Envy-Freeness with Limited Subsidies under Negative Dichotomous Valuations

Working draft — not for circulation

Mainak

What this document is. The complete state of the work, including results that are proved and machine-checked but not yet worth putting in the report. Sections 1 to 3 are shared verbatim with `main.tex`; everything after them is parked material. When a parked section earns its place, move its file from `working/` back to `sections/` and uncomment the matching line in `main.tex`. Nothing here is stale: every claim was re-verified against the scripts on the date of its section.

1 Introduction

Discrete fair division studies the allocation of resources that cannot be fractionally assigned among agents with individual preferences, and sits at the interface of mathematical economics and computer science [BCE+16, End17]. The classical fairness criterion is *envy-freeness* [Fol67, Var74]: no agent should prefer the bundle assigned to another agent over her own. For indivisible items an envy-free allocation need not exist, as the standard one-item two-agent instance shows, and a large body of work therefore studies relaxations such as envy-freeness up to one good (EF1) [Bud11, LMMS04].

A second and complementary line of work retains *exact* envy-freeness and buys off the residual envy with money. Each agent i receives, in addition to her bundle, a subsidy $p_i \geq 0$ supplied by a third party, and one asks for an allocation together with a subsidy vector under which every agent weakly prefers her own bundle-plus-subsidy to that of every other agent. The question is how much money this requires. Maskin [Mas87] answered it for unit-demand agents with $m = n$: under the scaling in which no agent values any single item above 1, a total subsidy of $n - 1$ suffices. Halpern and Shah [HS19] moved to the multi-demand setting with arbitrary m, gave the characterisation of *envy-freeable* allocations that the whole line now runs on, showed that a subsidy of $(n-1)m$ is necessary and sufficient in the worst case when the allocation is handed to us, and conjectured that $n - 1$ suffices when we may choose the allocation. Brustle et al. [BDN+20] proved that conjecture in the stronger per-agent form — one dollar to each agent suffices for additive valuations, with the allocation simultaneously EF1 and balanced — and showed that for general monotone valuations $2(n-1)$ per agent suffices, so that the total subsidy is $O(n^2)$ and, remarkably, *independent of the number of items*. Barman et al. [BKNS22] then improved the monotone bound by a factor of n on the dichotomous subclass: when every marginal

$v_i(S \cup \{g\}) - v_i(S)$ lies in $\{0,1\}$, there is an allocation whose accompanying subsidy is exactly 0 or 1 for each agent, hence at most $n-1$ in total, computable in polynomial time in the value-oracle model. Their bound assumes neither additivity nor submodularity nor even subadditivity, and it is tight: with n agents and a single unit-valued good, $n-1$ agents must each be paid 1.

Chores. Every result above is about *goods*. This paper asks the mirrored question, in which every item is a *chore*: an object that an agent would rather not hold, so that acquiring it weakly decreases her utility. Chores are not an exotic variant. Shift rotations, on-call duty, teaching and refereeing load, committee service, maintenance tasks and the liabilities attached to an estate are all allocation problems in which the items are burdens rather than prizes, and in all of them money is already the standard instrument of repair — as overtime pay, hardship allowance, or a course-release budget. If anything the subsidy model is more natural for chores than for goods, since compensating somebody for taking on unwanted work needs less justification than paying somebody for the privilege of not receiving a prize.

Concretely, we study *negative dichotomous* valuations: for every agent i, every $S \subseteq M$ and every $g \in M \setminus S$,

$$v_i(S) - v_i(S \cup \{g\}) \in \{0,1\}.$$

Equivalently, each marginal $v_i(S \cup \{g\}) - v_i(S)$ is 0 or -1: every item is a chore for every agent, and taking on one more chore costs an agent either nothing or exactly one unit. This is the exact mirror of the model of [BKNS22], and we impose no further structure — in particular the valuations need not be additive, submodular, or subadditive.

Binary marginals are the natural model whenever a task either lands on an agent as a fresh unit of burden or is absorbed by a commitment she has already made. An engineer already staffing an on-call window absorbs a second incident in that window at no extra cost; a referee already reading for a venue absorbs one more paper from the same batch; a driver already routed through a district absorbs an extra stop along the way. Whether a given chore is free depends in an arbitrary way on the rest of the agent's assignment, which is exactly why the model must tolerate non-additive and even non-subadditive valuations, and it is the counterpart of the scheduling example used by [BKNS22] to motivate dichotomous valuations for goods.

Why this is not a change of sign. It is tempting to expect that a chores result follows from the corresponding goods result by negating the valuations. It does not, for three reasons that we record here because they shape everything that follows.

First, money is one-signed. The subsidy vector is constrained to $p \in \mathbb{R}^n_+$, and negating an instance negates the valuations but leaves the subsidies where they are. The polyhedron of feasible subsidy vectors is therefore a genuinely different object in the two models, and no formal duality carries a bound across.

Second, the obvious transformation is available only for two agents. Writing $c_i := -v_i$ for the cost form of a negative dichotomous valuation, the complement $\tilde{v}_i(S) := c_i(M) - c_i(M \setminus S)$ is again dichotomous in the sense of [BKNS22], so the class is closed under complementation. But the complement transformation respects the allocation only when $M \setminus A_j = A_i$, that is, only when $n = 2$; see Remark 9. For $n \geq 3$ there is no such reduction, and that is where the content of the problem lies.

Third, and most tellingly, envy flows the other way. Under goods, envy concentrates *into* the agent holding the valuable items: many agents envy one, and the tight instance of [BKNS22] pays $n - 1$ agents one dollar each so that they stop envying the single agent who received the good. Under chores, envy emanates *out of* the agent carrying the load: one agent envies many. Since the minimum subsidy to an agent is the weight of the heaviest directed path leaving her in the envy graph [HS19, Theorem 2], the two situations are not interchangeable, and one should not assume in advance that the same bound is the right target.

What is already known, and what is open. Two facts delimit the problem before any work is done, and we state them explicitly so that the contribution can be measured against them.

On the one hand, existence of a bounded subsidy is not in question. Negative dichotomous valuations are doubly monotone — every item is a chore for every agent, which is a special case of every item being a good or a chore for each agent — and they satisfy the normalisation $|v_i(S \cup \{e\}) - v_i(S)| \leq 1$ that [KMS+24, §2] imposes. The reduction of Kawase et al., which converts any EF1 allocation into an envy-free one, therefore applies off the shelf and yields a subsidy of at most $n - 1$ per agent and $n(n-1)/2$ in total [KMS+24, Theorem 1]. Two details make the containment work and are easy to get wrong. Their normalisation is two-sided, so it bounds chore disutility as well as value; and the EF1 they require is the two-sided form of Theorem 5, which removes one item from *either* bundle, not the goods-only form of [HS19] — the two differ precisely when chores are present. An EF1 allocation is guaranteed to exist and to be computable in polynomial time for doubly monotone valuations, so the reduction has an input; for that step [KMS+24, §2] cites [LMMS04] and [BSV21], and the citation matters, because the envy-cycle argument of [LMMS04] does *not* port to chores unaltered and it is [BSV21] that supplies the repair.

On the other hand, the lower bound of the goods model survives intact.

Example 1 (The $n - 1$ lower bound transfers)**.** Take n agents and $m = n - 1$ chores, with $c_i(S) = |S|$ for every agent i — identical, binary, additive costs, which are in particular negative dichotomous. Every complete allocation leaves at least one agent empty-handed. Under the balanced allocation, in which $n - 1$ agents hold one chore each and one agent holds nothing, each of the $n - 1$ loaded agents envies the empty agent by exactly 1 and no other envy is present, so the minimum subsidy assigns 1 to each loaded agent and 0 to the empty one, for a total of $n - 1$. No allocation does better: any complete allocation gives some agent a bundle of size $k \geq 1$ while some other agent is empty, and that agent's heaviest outgoing path already has weight k.

So the target statement is precisely the analogue of [BKNS22, Theorem 4].

Conjecture 2. *For every instance with negative dichotomous valuations there is an envy-free solution $(\mathcal{A}, p)$ with $p \in \{0, 1\}^n$, hence with total subsidy at most $n - 1$; and it can be computed in polynomial time given value-oracle access.*

Example 1 shows Conjecture 2 would be tight. Between the $n - 1$ per agent that [KMS+24] already gives and the $\{0, 1\}$ per agent being asked for lies a factor of n in the total, exactly the gap that [BKNS22] closed on the goods side. Our objective is to close it on the chores side or to exhibit an instance showing that it cannot be closed.

Scope of this report. Conjecture 2 is open. This report sets up the problem and stops there: Section 2 fixes the model and notation and records what transfers from the goods theory unchanged, and Section 3 isolates three structural facts about the chore problem — an arc-update lemma (Lemma 14) reducing the entire goods/chore asymmetry to two lines, a two-tier characterisation of $\{0, 1\}$ solutions (Proposition 15), and a size-shift transform (Theorem 16) exhibiting the chore problem as exactly the goods problem of [BKNS22] together with a cardinality-balancing requirement. The last of these is the precise sense in which Conjecture 2 does not follow from [BKNS22] as a black box.

Conjecture 2 remains open. What stands in the way is recorded below: three techniques that settle the neighbouring corners of the square above, each of which fails here, and for each an explicit instance exhibiting the failure. Several special cases are settled; work towards the general statement is in progress and is not reported here. The remainder of this working draft, from Section 4 onwards, carries that material.

Related Work. The subsidy line for goods is described above. Within it, the dichotomous subclass has attracted separate attention: Halpern and Shah [HS19] settle binary additive valuations with a total subsidy of $n - 1$, Goko et al. [GIK+21] obtain the analogous bound for binary submodular valuations together with a strategy-proof mechanism, and Barman et al. [BKNS22] subsume both by dropping every structural assumption beyond binary marginals. Dichotomous preferences are themselves a well-studied restriction in social choice [BMS05, BEF21]. On the computational side, Caragiannis and Ioannidis [CI21] show that minimising the subsidy is NP-hard and give a fully polynomial-time approximation scheme for a constant number of agents, and Narayan et al. [NSV21] study the welfare cost of achieving envy-freeness with transfers.

The paper closest to the present setting is that of Kawase et al. [KMS+24], the only work in the envy-freeness-with-subsidy line that admits chores at all. Working with doubly monotone valuations under the two-sided normalisation, it decouples the two jobs that [BDN+20] performed together: rather than constructing a specific allocation and showing that it is cheap to subsidise, it takes *any* EF1 allocation as input and bounds the subsidy needed to convert it, obtaining $n - 1$ per agent and $n(n - 1)/2$ in total. Since EF1 allocations are known to exist for doubly monotone valuations, the wider class comes along at no cost. That reduction is what makes the present question well-posed rather than open-ended, and it is the baseline we are trying to beat.

The complementary repair — no money, weaker fairness notion — is pursued for binary valuations by Bu et al. [BSY23], who show that EFX allocations always exist and can be computed in polynomial time for general binary valuations, thus extending the matroid-rank result of Babaioff et al. [BEF21]. Under binary valuations one therefore has a clean dichotomy on the goods side: either pay 0 or 1 per agent and obtain exact envy-freeness [BKNS22], or pay nothing and obtain EFX [BSY23]. Whether either branch survives the passage to chores is open; this paper addresses the first.

The chores side. Work on indivisible chores has largely developed alongside rather than inside the subsidy line, and the two have met less often than one might expect. Where money has been brought to chores, it has been in the service of *proportionality* rather than envy-freeness: the question studied is the total subsidy needed to bring every agent down to her proportional share of the burden, for which the best bounds under additive disutilities normalised to at most 1 per chore are $n/4$ and then $n/3 - 1/6$, recently obtained together with Pareto optimality by Garg et al. [GSW25]. That is a different fairness notion and a different accounting; it does not bound the subsidy needed for envy-freeness.

The no-money branch, by contrast, has a direct chore analogue of [BSY23]. Tao et al. [TWYZ23] study chores with binary marginals and show that for binary *additive* costs an allocation that is simultaneously EFX and Pareto optimal always exists and is computable in polynomial time. So on the chores side the second half of the binary dichotomy — pay nothing, obtain EFX — is established at least for additive costs, while the first half, pay 0 or 1 per agent and obtain exact envy-freeness, is what Conjecture 2 asks for. That money is genuinely needed here is not obvious a priori but is known: Bhaskar et al. [BSV21] show that deciding whether an exactly envy-free allocation of chores exists is NP-complete already for binary additive costs, and give a polynomial-time EF1 algorithm for chores and for doubly monotone instances.

Where the two lines have met is on the *additive* side, and recently. Lu et al. [LMS26] prove the goods-and-chores analogue of [BDN$^+$20]: for additive utilities in which each item is a good for some agents and a chore for others, with every $\left|v_i(g)\right| \le 1$, a subsidy of one dollar per agent suffices, the bound is tight, and the allocation is computable in polynomial time — hence $n-1$ in total. This settles the additive chore problem completely.

It does not settle ours, and the reason is exactly the reason [BKNS22] was not a corollary of [BDN$^+$20]. Dichotomous valuations are not additive and additive valuations are not dichotomous; the two classes are incomparable. The four corners are

	goods	chores
additive	[BDN$^+$20]	[LMS26]
dichotomous	[BKNS22]	**Conjecture 2**

each of the three settled corners giving one dollar per agent and $n-1$ in total. The present question is the fourth. Note also that [LMS26] does not displace [KMS$^+$24] as the baseline of Section 1: for valuations that are dichotomous but not additive, $n-1$ per agent from [KMS$^+$24] remains the best published bound.

We are not aware of any treatment of envy-freeness with subsidy for chores under dichotomous costs. This is the outcome of a literature search rather than a proof of novelty — [LMS26] appeared in July 2026, which is a reminder of how quickly the surrounding corners are being filled in — and it should be repeated before the paper is circulated.

2 Notation and Preliminaries

We study the allocation of m indivisible items among n agents, with subsidies. Write $N = [n]$ for the set of agents and M for the set of items, $|M| = m$. Each agent $i \in N$ has a valuation

$v_i : 2^M \to \mathbb{R}$ with $v_i(\emptyset) = 0$, and we represent an instance by the tuple $\langle N, M, \{v_i\}_{i\in N}\rangle$. Algorithms operate in the standard *value-oracle* model: for each v_i we assume only an oracle that returns $v_i(S)$ for a queried set $S \subseteq M$, and running time is measured in n, m and the number of oracle calls.

An *allocation* $\mathcal{A} = (A_1, \dots, A_n)$ is an ordered tuple of pairwise disjoint *bundles*, with A_i assigned to agent i. The allocation is *complete* if $\bigcup_{i\in N} A_i = M$ and *partial* otherwise. The ordering matters throughout: much of the theory below concerns permuting a fixed family of bundles among the agents, and for a permutation $\sigma \in \mathbb{S}_n$ we write $\mathcal{A}_\sigma := (A_{\sigma(1)}, \dots, A_{\sigma(n)})$ for the allocation in which agent i receives $A_{\sigma(i)}$. The utilitarian welfare of $\mathcal{A}$ is $\mathrm{SW}(\mathcal{A}) = \sum_{i\in N} v_i(A_i)$.

Definition 3 (Envy-Freeness)**.** *An allocation $\mathcal{A} = (A_1, \dots, A_n)$ is* envy-free *(EF) iff $v_i(A_i) \geq v_i(A_j)$ for all agents $i, j \in N$.*

Envy-free allocations of indivisible items need not exist, so we allow a third party to supplement the allocation with money. Agents have quasi-linear utilities: agent i receiving bundle A_i and subsidy p_i enjoys utility $v_i(A_i) + p_i$, with money entering linearly and at the same exchange rate for every agent.

Definition 4 (Envy-Free Solution)**.** *An allocation $\mathcal{A} = (A_1, \dots, A_n)$ and a subsidy vector $p = (p_1, \dots, p_n) \in \mathbb{R}^n_+$ constitute an* envy-free solution $(\mathcal{A}, p)$ *iff $v_i(A_i) + p_i \geq v_i(A_j) + p_j$ for all $i, j \in N$.*

An allocation $\mathcal{A}$ is *envy-freeable* if some $p \in \mathbb{R}^n_+$ makes $(\mathcal{A}, p)$ an envy-free solution; this is a property of the allocation alone. Following [BKNS22] we write $\mathrm{M}(p) := \left\{i \in N \mid p_i \geq p_j \text{ for all } j \in N\right\}$ for the set of maximally subsidised agents.

We will occasionally need the standard relaxation of envy-freeness. Note that two inequivalent forms of it appear in this literature: the goods-only form of [HS19], which removes one item from the envied bundle, and the two-sided form used by [KMS+24], which removes one item from *either* bundle. They coincide when all items are goods and differ once chores are present, and only the two-sided form supports the doubly monotone results. We adopt the two-sided form.

Definition 5 (EF1, two-sided form)**.** *An allocation $\mathcal{A}$ is* envy-free up to one item *(EF1) iff for all $i, j \in N$ there exists $X \subseteq A_i \cup A_j$ with $|X| \leq 1$ such that $v_i(A_i \setminus X) \geq v_i(A_j \setminus X)$.*

2.1 Negative Dichotomous Valuations

Definition 6 (Negative Dichotomous Valuation)**.** *A valuation $v_i : 2^M \to \mathbb{R}$ with $v_i(\emptyset) = 0$ is* negative dichotomous *iff*

$$v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\} \qquad \text{for every } S \subseteq M \text{ and every } g \in M \setminus S,$$

equivalently iff every marginal $\Delta_i^+(S, g) := v_i(S \cup \{g\}) - v_i(S)$ lies in $\{-1, 0\}$.

Every item is thus a chore for every agent: v_i is non-increasing, so $v_i(S) \geq v_i(T)$ whenever $S \subseteq T$. We stress that nothing further is assumed — the valuations need not be additive, submodular, or subadditive — which is the generality at which [BKNS22] works on the goods side.

It is often more convenient to work with the non-negative *cost form*

$$c_i \;:=\; -v_i\,, \qquad \text{so} \qquad c_i(\emptyset)=0,\quad c_i(S)\le c_i(T)\text{ for }S\subseteq T,\quad c_i(S\cup\{g\})-c_i(S)\in\{0,1\}\,.$$

Definition 7 (Dichotomous cost function)**.** *A set function $c : 2^M \to \mathbb{R}$ is* dichotomous *if $c(\emptyset)=0$ and every marginal $c(S\cup\{g\})-c(S)$ lies in $\{0,1\}$.*

Monotonicity is not a separate hypothesis: marginals in $\{0,1\}$ are non-negative, so a dichotomous c is automatically non-decreasing. We nonetheless list it above because it is the property most often used directly.

Remark 8 (Three adjectives that are routinely confused)**.** *Monotone* means only $c(S)\le c(T)$ for $S\subseteq T$, with no bound whatever on how large an increment can be; it is far weaker than Theorem 7 and, on its own, supports almost none of the arguments below. *Dichotomous* adds the binary-marginal condition and is the class of this paper. *Binary additive* is the strictly smaller class $c_i(S)=|S\cap D_i|$ for a set D_i of disliked chores, i.e. additive with per-item costs in $\{0,1\}$; it is the case settled by [LMS26].

A warning about the literature: several papers say "binary valuations" for what is called dichotomous here — [BSY23] and [TWYZ23] both do — and reserve "binary additive" for the additive subclass. When a bound is quoted, check which is meant, since the classes are not the same and the gap between them is where this paper lives.

The correspondence $v_i \leftrightarrow c_i$ is a bijection between negative dichotomous valuations and dichotomous cost functions in the sense of [BKNS22]: the class we study is exactly the pointwise negation of the class they study. In cost form an envy-free solution is the requirement $c_i(A_i)-p_i \le c_i(A_j)-p_j$ for all i,j, read as "my own burden net of my compensation is no worse than what I perceive of yours". We use whichever of v_i and c_i is clearer locally and always state which.

The negation, however, does not carry results across, for the reasons given in Section 1. The following remark isolates the one case in which a transformation does exist, and shows why it is uninformative.

Remark 9 (Complementation, and why it stops at $n=2$)**.** For a negative dichotomous v_i with cost form c_i, define the *complement* $\tilde v_i(S) := c_i(M)-c_i(M\setminus S)$. Then $\tilde v_i(\emptyset)=0$ and, for $g\notin S$, writing $T := M\setminus(S\cup\{g\})$,

$$\tilde v_i(S\cup\{g\})-\tilde v_i(S) \;=\; c_i(M\setminus S)-c_i(T) \;=\; c_i(T\cup\{g\})-c_i(T) \;\in\; \{0,1\}\,,$$

so $\tilde v_i$ is dichotomous in the sense of [BKNS22]. The class is therefore closed under complementation, without any submodularity assumption. Now let $n=2$ and let $\mathcal{A}=(A_1,A_2)$ be complete, so that $M\setminus A_2 = A_1$ and $M\setminus A_1 = A_2$. Then $-c_i(A_i)=\tilde v_i(A_j)-c_i(M)$ and $-c_i(A_j)=\tilde v_i(A_i)-c_i(M)$, so $(\mathcal{A},p)$ is an envy-free solution for the chores instance $\{v_i\}$ if and only if $((A_2,A_1),p)$ is an envy-free solution for the goods instance $\{\tilde v_i\}$: swap the bundles and leave each subsidy attached to its agent. Two-agent chore division is thus exactly two-agent goods division, and [BKNS22] settles it. The step $M\setminus A_j = A_i$ has no analogue for $n\ge 3$, where the complement of one bundle is a union of several others, and no reduction of this kind is available.

2.2 The Envy Graph and the Halpern–Shah Characterisation

For an allocation $\mathcal{A}$, the *envy graph* $G_{\mathcal{A}}$ is the complete weighted directed graph on vertex set N in which the arc (i, j) carries weight

$$w_{\mathcal{A}}(i, j) \ := \ v_i(A_j) - v_i(A_i) \ = \ c_i(A_i) - c_i(A_j),$$

the envy that agent i has towards agent j. The weight of a directed path P is $w_{\mathcal{A}}(P) = \sum_{(i,j)\in P} w_{\mathcal{A}}(i, j)$, and $\ell_{\mathcal{A}}(i)$ denotes the maximum weight of any path in $G_{\mathcal{A}}$ starting at i, the empty path of weight 0 included.

The two results below are the engine of the entire subsidy line. Both are stated by Halpern and Shah [HS19] for additive valuations, but their proofs use only the arc weights and permutations of the bundles — neither additivity nor monotonicity nor any sign condition on v_i enters — and [BKNS22] invokes them in exactly this generality for (non-additive) dichotomous valuations. They therefore apply verbatim to negative dichotomous valuations, and we use them without further comment.

Theorem 10 ([HS19])**.** *For any allocation $\mathcal{A} = (A_1, \ldots, A_n)$, the following are equivalent.*

(i) $\mathcal{A}$ is envy-freeable.

(ii) $\mathcal{A}$ maximises utilitarian welfare across all reassignments of its own bundles among the agents: for every permutation σ over N, $\sum_{i\in N} v_i(A_i) \geq \sum_{i\in N} v_i(A_{\sigma(i)})$.

(iii) The envy graph $G_{\mathcal{A}}$ has no positive-weight directed cycle.

Theorem 11 ([HS19])**.** *For any envy-freeable allocation $\mathcal{A}$, the subsidy vector p^* given by $p^*_i := \ell_{\mathcal{A}}(i)$ realises an envy-free solution $(\mathcal{A}, p^*)$, and $p^*_i \leq p_i$ for every $i \in N$ and every p such that $(\mathcal{A}, p)$ is an envy-free solution. It can be computed in strongly polynomial time by running an all-pairs longest-path computation on $G_{\mathcal{A}}$.*

Condition (ii) of Theorem 10 has the standing consequence that an envy-freeable allocation can always be manufactured from an arbitrary one: fix the bundles and reassign them according to a maximum-weight perfect matching between agents and bundles. What is at issue is never the existence of an envy-freeable allocation, only the size of the subsidy that Theorem 11 then charges for it.

Specialising to our model gives the following, which we use throughout.

Observation 12 (Integrality)**.** *Let $\{v_i\}_{i\in N}$ be negative dichotomous. Then $v_i(S) \in \mathbb{Z}_{\leq 0}$ for every i and every $S \subseteq M$; every arc weight $w_{\mathcal{A}}(i, j)$ is an integer; and for every envy-freeable allocation $\mathcal{A}$ the minimum subsidy vector satisfies $p^* \in \mathbb{Z}^n_{\geq 0}$.*

Proof Fix i and $S = \{g_1, \ldots, g_k\}$. Telescoping from $v_i(\emptyset) = 0$ along any ordering of S writes $v_i(S)$ as a sum of k marginals, each of which lies in $\{-1, 0\}$ by Definition 6; hence $v_i(S) \in \mathbb{Z}$ and $-k \leq v_i(S) \leq 0$. Consequently each arc weight $w_{\mathcal{A}}(i, j) = v_i(A_j) - v_i(A_i)$ is a difference of integers, and the weight of any directed path, being a finite sum of arc weights, is an integer. By Theorem 11, $p^*_i = \ell_{\mathcal{A}}(i)$ is the maximum such weight over paths starting at i, and it is non-negative because the empty path has weight 0. ◻

Observation 12 is what makes the target statement — a subsidy vector in $\{0, 1\}^n$ — a statement about a *bound* rather than about rounding: once p^* is known to be integral, showing $p^*_i \le 1$ for every i is the same as showing $p^* \in \{0, 1\}^n$. The same observation underlies the corresponding step in [BKNS22], where the arc weights of a dichotomous instance are likewise integers.

A second normalisation costs nothing and removes a clause from the target.

Proposition 13 (Some agent is always unpaid)**.** *For every envy-freeable allocation $\mathcal{A}$ there is an agent h with $p^*_h = 0$. Consequently, if $p^* \in \{0, 1\}^n$ then $\sum_{i \in N} p^*_i \le n - 1$.*

Proof By Theorem 10(iii) the envy graph $G_{\mathcal{A}}$ has no positive-weight cycle, so a walk of maximum weight may be taken simple and $\ell_{\mathcal{A}}$ is finite; it is non-negative because the empty path qualifies. Choose a path P attaining $\max_i \ell_{\mathcal{A}}(i)$ and let h be its final vertex. If $\ell_{\mathcal{A}}(h) > 0$ there is a positive-weight walk Q leaving h, and P followed by Q is a walk from P's start of weight $w_{\mathcal{A}}(P) + w_{\mathcal{A}}(Q) > w_{\mathcal{A}}(P)$, contradicting the maximality of P among walks. Hence $\ell_{\mathcal{A}}(h) = 0$. The second claim is immediate: at most $n - 1$ coordinates of p^* can equal 1. □

Proposition 13 justifies the word "hence" in Conjecture 2: the bound on the *total* is not an independent requirement but a consequence of the per-agent one. The conjecture is therefore a single-clause statement,

$$\exists \mathcal{A} \ : \quad \max_{i \in N} \ell_{\mathcal{A}}(i) \ \le \ 1,$$

and it is this form that the rest of the report works with.

Two cautions are worth recording alongside it. First, the bound cannot be improved: the family of Example 1 attains $n - 1$ for every n, verified exhaustively for $n \le 7$. Second, the conjecture is *not* the assertion that the utilitarian-optimal subsidy is at most $n - 1$. Minimising the total over all allocations can do strictly better than any allocation with $p^* \in \{0, 1\}^n$, by concentrating the subsidy on a single agent: there are instances at $n = 4$ whose cheapest envy-free solution costs 2, borne entirely by one agent, while every allocation with $p^* \in \{0, 1\}^4$ costs 3. The $\{0, 1\}$ shape is a genuine restriction, not a normalisation of the total.

3 Structure of the Chore Problem

Throughout this section we work in cost form, so $w_{\mathcal{A}}(i, j) = c_i(A_i) - c_i(A_j)$.

3.1 The arc-update table, and the whole of the asymmetry

Every incremental algorithm in this literature adds one item to one bundle and then asks what happened to the envy graph. The answer is two lines long, and comparing those two lines against their goods counterparts accounts for the entire difference between the present problem and that of [BKNS22].

For an allocation $\mathcal{A}$, an agent x and an unallocated chore $g \notin \bigcup_i A_i$, write

$$Z^x \ := \ (A_1, \ldots, A_x \cup \{g\}, \ldots, A_n)$$

for the allocation obtained by giving g to x, and put

$$\delta_x \;:=\; c_x(A_x \cup \{g\}) - c_x(A_x), \qquad \delta_{i,x} \;:=\; c_i(A_x \cup \{g\}) - c_i(A_x),$$

both of which lie in $\{0,1\}$. So δ_x is what the chore costs x herself, and $\delta_{i,x}$ is what it costs in i's opinion.

Lemma 14 (Arc update)**.** *For all $i, j \in N$,*

$$w_{Z^x}(i,j) = w_{\mathcal{A}}(i,j) \quad (i,j \neq x), \qquad w_{Z^x}(x,j) = w_{\mathcal{A}}(x,j)+\delta_x, \qquad w_{Z^x}(i,x) = w_{\mathcal{A}}(i,x)-\delta_{i,x}.$$

Proof Only A_x changes, so an arc between two agents other than x has both endpoints untouched. For an arc leaving x, $w_{Z^x}(x,j) = c_x(A_x \cup \{g\}) - c_x(A_j) = w_{\mathcal{A}}(x,j) + \delta_x$. For an arc entering x, $w_{Z^x}(i,x) = c_i(A_i) - c_i(A_x \cup \{g\}) = w_{\mathcal{A}}(i,x) - \delta_{i,x}$. □

The mirror statement for goods is obtained by replacing costs with values: there the arcs *out of* the receiving agent fall and the arcs *into* her rise. The consequences are opposite in a way that matters.

	goods [BKNS22]	chores (here)
arcs out of the receiver	fall	**rise by** δ_x
arcs into the receiver	**rise**	fall by $\delta_{i,x}$
so the receiver must be	*maximally* subsidised	*minimally* subsidised
which is guaranteed by	$\mathrm{M}(p) \neq \emptyset$	$p_x = 0$ for some x

Both guarantees hold trivially — some agent is always maximally subsidised, and some agent always has $p_x = 0$ under the minimum subsidy vector. The asymmetry is subtler than the table suggests, and it is what Theorem 36 exploits. Under goods, the harm done by an insertion is confined to paths *ending* at the receiver, and requiring $x \in \mathrm{M}(p)$ caps every such path at $p_u - p_x \leq 0$ in one stroke. Under chores, the harm is done to paths *leaving* the receiver, and $p_x = 0$ caps only the paths that *start* at x. A path that merely passes through x — entering along an arc from some i with $\delta_{i,x} = 0$ and leaving along an arc that has just risen by $\delta_x = 1$ — is not controlled by any condition on p_x. That gap is not a defect of the analysis; Section 6 shows it is realised.

3.2 The shape of a $\{0,1\}$ solution

Proposition 15 (Two-tier characterisation)**.** *Let $p \in \{0,1\}^n$ and put $S := \{i \in N \mid p_i = 1\}$. Then $(\mathcal{A}, p)$ is an envy-free solution if and only if*

(a) $c_i(A_i) \leq c_i(A_j)$ whenever i, j lie both in S or both in $N \setminus S$;

(b) $c_i(A_i) \leq c_i(A_j) + 1$ for $i \in S$ and $j \in N \setminus S$;

(c) $c_i(A_i) + 1 \leq c_i(A_j)$ for $i \in N \setminus S$ and $j \in S$.

Proof In cost form Definition 4 reads $c_i(A_i) \leq c_i(A_j) + p_i - p_j$ for all i, j. Substituting the three possible values $0, 1, -1$ of $p_i - p_j$ gives (a), (b) and (c) respectively. □

The reading is worth keeping in mind when designing an algorithm. The unpaid agents are exactly envy-free among themselves and *strictly* prefer their own bundle to that of every paid agent, by a full unit; the paid agents are exactly envy-free among themselves and envy-free up to one unit towards the unpaid. This is a certificate format, and it also suggests choosing the partition $S \mid N \setminus S$ *first* and allocating afterwards, rather than building the allocation up one chore at a time. That is the one place where the incremental template killed in Section 6 is not obviously an obstacle. In the proof of Theorem 19 the set S turns out to be exactly the agents holding a $\lceil q/n \rceil$-share of the universally disliked chores.

3.3 The size-shift transform

Complementation relates the chore and goods models only for $n = 2$ (Remark 9). There is a second transform, available for every n, which does not solve the problem but says precisely how far it is from being a corollary of [BKNS22].

Theorem 16 (Size shift)**.** *For a negative dichotomous instance with cost functions c_i, define $\tilde{v}_i(S) := |S| - c_i(S)$. Then each $\tilde{v}_i$ is a dichotomous* goods *valuation in the sense of [BKNS22], and for every allocation $\mathcal{A}$ and all i, j,*

$$w_{\mathcal{A}}(i, j) \;=\; \tilde{w}_{\mathcal{A}}(i, j) \;+\; |A_i| - \left|A_j\right|,$$

where $\tilde{w}_{\mathcal{A}}$ denotes the envy graph of the goods instance $\{\tilde{v}_i\}$. Consequently every cycle *has the same weight in both graphs — so $\mathcal{A}$ is envy-freeable for the chores $\{c_i\}$ if and only if it is envy-freeable for the goods $\{\tilde{v}_i\}$ — while for a path P from u to v,*

$$w_{\mathcal{A}}(P) \;=\; \tilde{w}_{\mathcal{A}}(P) + |A_u| - |A_v|, \qquad \textit{hence} \qquad \ell_{\mathcal{A}}(u) \;=\; \max_{v \in N} \Big[\, \tilde{d}_{\mathcal{A}}(u, v) + |A_u| - |A_v| \,\Big],$$

with $\tilde{d}_{\mathcal{A}}(u, v)$ the maximum weight of a path from u to v in the goods graph and $\tilde{d}_{\mathcal{A}}(u, u) = 0$.

Proof $\tilde{v}_i(\emptyset) = 0$, and for $g \notin S$, $\tilde{v}_i(S \cup \{g\}) - \tilde{v}_i(S) = 1 - (c_i(S \cup \{g\}) - c_i(S)) \in \{0, 1\}$, so $\tilde{v}_i$ is dichotomous. For the identity,

$$\tilde{w}_{\mathcal{A}}(i, j) = \tilde{v}_i(A_j) - \tilde{v}_i(A_i) = \left(\left|A_j\right| - c_i(A_j)\right) - \left(|A_i| - c_i(A_i)\right) = w_{\mathcal{A}}(i, j) - |A_i| + \left|A_j\right|.$$

Summing along a path $(i_1, \ldots, i_r)$, the terms $|A_{i_t}| - \left|A_{i_{t+1}}\right|$ telescope to $\left|A_{i_1}\right| - |A_{i_r}|$; along a cycle they telescope to 0. The characterisation of envy-freeability by the absence of positive cycles (Theorem 10) then transfers, and the formula for $\ell_{\mathcal{A}}(u)$ follows from Theorem 11. ◻

Remark 17 (Why Conjecture 2 is not a corollary of [BKNS22])**.** Theorem 16 says the chore problem is the goods problem of [BKNS22] *plus a cardinality-balancing requirement*. A sufficient condition falls straight out: since $\tilde{d}_{\mathcal{A}}(u, v) \leq \tilde{p}_u - \tilde{p}_v$ for any envy-eliminating $\tilde{p}$, an allocation solving the goods instance in which the quantity $|A_i| + \tilde{p}_i$ has spread at most 1 across agents satisfies $\ell_{\mathcal{A}} \leq 1$. But [BKNS22] offers no control whatever on bundle cardinalities — its output need not even be EF1, by its own Appendix C — so the requirement is not met by invoking it as a black box. Nor can its algorithm simply be re-run with a balancing tie-break, because Theorem 36 rules out the entire item-by-item template on which it is built. The transform also shows the chore problem is at least as expressive as the goods problem, so no easier.

4 Our Results

The target is Conjecture 2, stated in Section 1: an envy-free solution with $p \in \{0,1\}^n$ and hence total subsidy at most $n-1$. Example 1 supplies the matching lower bound, so it would be tight. By Observation 12 the conjecture is equivalent to the purely graph-theoretic statement that some allocation $\mathcal{A}$ has no positive-weight cycle in $G_{\mathcal{A}}$ and satisfies $\ell_{\mathcal{A}}(i) \leq 1$ for every agent i.

We do not settle Conjecture 2. What we contribute is a map of the problem: two solved special cases, three structural tools, and — the substance of the paper — two theorems showing that the two most natural proof strategies both fail, one of them the strategy that succeeds on the goods side.

Solved cases. Conjecture 2 holds, constructively and in polynomial time, for **binary additive** costs (Theorem 19) and for **identical** costs (Theorem 21). The case $n = 2$ follows from [BKNS22] through the complementation of Remark 9. The binary additive proof is the one to imitate: dump every chore on somebody who finds it free, then balance whatever nobody wants. Its bound telescopes along paths in the envy graph, which is the mechanism all three tools below try to reproduce.

Structural tools (Section 3). The arc-update table of Lemma 14 isolates the entire difference between the goods and chore problems in two lines. Proposition 15 characterises $\{0,1\}$ solutions as a two-tier structure, which is the natural certificate format and suggests choosing the tiers before allocating. Theorem 16 exhibits the chore problem as *exactly* the goods problem of [BKNS22] plus a cardinality-balancing requirement, which is the precise sense in which Conjecture 2 is not a corollary of [BKNS22].

First negative result (Section 6). Item-by-item insertion — allocate one chore at a time, never move an already-allocated chore, and repair only by permuting whole bundles — is the template of [BKNS22]. Theorem 36 exhibits a three-agent instance on which that template provably cannot be completed: a legal partial state with $p \in \{0,1\}^3$ and an unallocated chore such that *every* agent who receives it forces a subsidy of 2, while the full instance admits a zero-subsidy allocation. Any proof of Conjecture 2 must therefore be allowed to re-open bundles that have already been allocated. We state this bluntly because the template is seductive and patching the insertion invariant is a dead end, not an unsolved exercise.

Second negative result (Section 7). The natural repair is to stop building the allocation incrementally and instead select a good allocation globally, the obvious candidate being one that minimises total cost. This suggests the following strengthening.

Conjecture 18. *The allocation in Conjecture 2 may in addition be taken* utilitarian-optimal, *i.e. minimising $\sum_{i \in N} c_i(A_i)$ over all allocations.*

Proposition 38 refutes Conjecture 18: there is a three-agent, four-chore instance whose *unique* utilitarian-optimal allocation requires a subsidy of 2, although another allocation — of strictly higher total cost — requires no subsidy at all. Utilitarian optimality is therefore not merely an unhelpful restriction but an actively wrong one, and the programme of searching for a tie-breaking rule inside the utilitarian-optimal set is unsound as posed.

A third approach, still open (Section 8). Both failures above are failures of *timing*: insertion commits a chore to an owner and cannot take it back, and utilitarian optimality commits to the wrong allocation outright. Section 8 changes coordinates so that ownership is decided only by the last move that touches a chore. Theorem 43 re-expresses the chore instance as a *goods* instance on an item set carrying $n - 1$ copies of each chore, with identical envy graphs, so the dichotomous class is closed under the duality and [BKNS22] applies to the replica verbatim. Corollary 45 then reduces Conjecture 2 to a single residual constraint — *coverage*, that no agent receive two copies of one type. Read dynamically this is the *peel* process, in which the orientation of [BKNS22] is restored because the quantity handed out is relief rather than burden, and the witness of Theorem 36 is handled with the invariant intact. The approach is not known to work: Proposition 52 exhibits legal states from which no legal continuation exists, so what remains is to find a peel rule that never enters one.

A third negative result (Section 9). Given Corollary 45, the natural move is to change the numbers so that a coverage violation becomes unattractive, and thereby reuse [BKNS22] as a black box. Section 9 closes that at both levels. No dichotomous reweighting of the replica instance separates coverage from non-coverage (Theorems 93 and 94, the latter exhaustive over all 990^3 candidate triples), and no inflation of item weights can steer the algorithm of [BKNS22], because its repair subroutine provably runs where every candidate's marginal is 0 and a duplicate moves no arc at all (Theorem 95, Corollary 96). Both failures have one cause, named in Section 9.1: a rule fixed before the allocation is chosen cannot act on a quantity the allocation determines. Coverage must be enforced by the procedure that builds the allocation, not by the objective it is scored against.

A working algorithm (Section 11). The size-shift transform of Theorem 16 is an involution (Lemma 110), so the chore problem restates as a pure *goods* question, Target G, with no chores or scheduling in it (Proposition 112). Running [BKNS22]'s own algorithm on that goods instance fails for a structural reason — item-by-item insertion can lock two items into the same bundle permanently, verified by an exhaustive backtracking search that finds a 0-spread allocation *unreachable* by any legal execution of the algorithm (Section 11.3). Abandoning insertion for a construct-and-repair procedure — one pass of iterated maximum-weight perfect matching, repaired by cardinality-preserving single-item swaps, restarted on failure — has **zero failures across** 389,215 **test instances**, including every hard instance that defeated Approaches 1 through 5.

Four properties of that algorithm are now *theorems*: its outputs are certified (Theorem 116), it terminates in polynomial time $O(n^3m^3\tau + n^4m^3)$ (Theorem 117), its search space is genuinely the unordered balanced partitions (Lemma 115), and — the main theoretical result of this line — **it is complete for two agents** (Theorem 118): every two-agent dichotomous instance admits a balanced split of spread at most 1, found constructively in $O(m\tau)$ time with no restarts, by a discrete intermediate-value argument on the aggregate $u = \tilde{v}_1 + \tilde{v}_2$ along swap paths. This strengthens even the known $n = 2$ chore result, adding balancedness that the complementation route does not provide. For $n \geq 3$, completeness remains the one open statement (Section 11.8).

Evidence (Section 19). Conjecture 2 survives exhaustive verification for $n = m = 3$ over all 9,880 instances up to agent symmetry, and randomised search over some 3.5×10^4 further instances. Section 19 also explains why that evidence is weaker than the counts suggest, and what would strengthen it.

5 Solved Cases

Conjecture 2 holds in three restrictions of the model, by explicit polynomial-time constructions, and on one further class of instances cut out by a certificate rather than by a restriction on the valuations (Section 5.4). The first three proofs run the same way: bound $w_{\mathcal{A}}(i, j)$ by a difference $\phi_i - \phi_j$ of a per-agent potential, so that the bound telescopes to $\phi_{i_1} - \phi_{i_r}$ along a path and to 0 around a cycle. Envy-freeability and the $\{0, 1\}$ bound then both follow from controlling the spread of ϕ. Reproducing that mechanism in general is the open problem.

What is and is not new here. This section should be read with its overlap against the literature stated plainly, because two of the three cases are largely or wholly not ours.

- **Binary additive costs** (Theorem 19) are *subsumed* by [LMS26], which settles all additive goods-and-chores instances. See Remark 20. The statement is not new; only the potential-and-telescoping mechanism is retained, because that is what the general case has to reproduce.
- **Identical costs** (Theorem 21) are *not* subsumed, for the reason given in Remark 22 — but the result is elementary and is best read as a warm-up.
- **Two agents** is [BKNS22, Theorem 4]. Only the reduction carrying it to chores (Remark 9) is ours.
- **The uniformly balanced class** (Section 5.4) is new, and is the only case here not implied by a cited paper. It is a certificate rather than an algorithm, and it is incomplete.

Nothing in this section, in other words, is load-bearing. The substance of the paper is in Sections 6, 7, 9 and 10, which are negative, and in Sections 3 and 8, which are structural.

5.1 Binary additive costs

Let $c_i(S) = |S \cap D_i|$ for a set $D_i \subseteq M$ of chores that agent i dislikes; every other chore is free for her. Call a chore *universally bad* if it lies in D_i for every i, write U for the set of universally bad chores and $q := |U|$.

Theorem 19. *For binary additive costs, the following allocation is envy-freeable with minimum subsidy $p^* \in \{0, 1\}^n$, hence total subsidy at most $n - 1$, and it is computable in time $O(nm)$.*

Give each chore $g \notin U$ to some agent i with $g \notin D_i$; at least one exists, by definition of U. Distribute U as evenly as possible, so that $u_i := |A_i \cap U| \in \{\lfloor q/n \rfloor, \lceil q/n \rceil\}$ for every i.

The bound is tight, by Example 1.

Proof By construction every chore in $A_i \setminus U$ is free for i, so $c_i(A_i) = |A_i \cap D_i| = u_i$. For any j we have $A_j \cap U \subseteq A_j \cap D_i$, since $U \subseteq D_i$, and therefore $c_i(A_j) = |A_j \cap D_i| \geq u_j$. Hence

$$w_{\mathcal{A}}(i,j) \;=\; c_i(A_i) - c_i(A_j) \;\leq\; u_i - u_j.$$

Summing along a path $P = (i_1, \ldots, i_r)$ the right-hand side telescopes:

$$w_{\mathcal{A}}(P) \;\leq\; u_{i_1} - u_{i_r} \;\leq\; \lceil q/n \rceil - \lfloor q/n \rfloor \;\leq\; 1.$$

Around a cycle the same computation gives $w_{\mathcal{A}}(C) \leq 0$, so $G_{\mathcal{A}}$ has no positive-weight cycle and $\mathcal{A}$ is envy-freeable by Theorem 10. By Theorem 11, $p_i^* = \ell_{\mathcal{A}}(i) \leq 1$, and Observation 12 forces $p^* \in \{0,1\}^n$. Since some agent receives $\lfloor q/n \rfloor$ universally bad chores and hence requires no subsidy, the total is at most $n-1$. Example 1 is binary additive with $D_i = M$ for every i, so the bound cannot be improved. □

Remark 20 (Superseded, and by how much)**.** Theorem 19 is subsumed by [LMS26], which appeared in July 2026 and gives one dollar per agent for *all* additive goods-and-chores instances with $|v_i(g)| \leq 1$; binary additive costs are the special case $v_i(g) \in \{-1, 0\}$. We keep the proof because the mechanism, not the statement, is what the general case needs: the potential $\phi_i = u_i$ and its telescoping along paths is the only device in this paper that produces a $\{0,1\}$ bound from scratch, and the general conjecture is precisely the problem of finding the right ϕ. Theorem 21 and the $n = 2$ case are *not* subsumed, since dichotomous costs need not be additive.

The construction is a two-phase version of the shape the general case must reproduce: *dump every chore on somebody who finds it free, then balance whatever nobody wants*. The obstruction to running the same argument for general negative dichotomous costs is that "free for i" becomes context dependent. A chore may be free on top of A_j and cost a full unit on top of A_i, so there is no fixed set D_i to compute with and no fixed potential u_i to telescope against. Theorem 36 shows that this is a genuine obstacle rather than a bookkeeping nuisance.

5.2 Identical costs, and two agents

Theorem 21. *Suppose $c_1 = \cdots = c_n = c$. Then every allocation is envy-freeable, and greedily adding each chore to a currently cheapest bundle yields $p^* \in \{0,1\}^n$ in time $O(m(n+\tau))$, where τ is the cost of one oracle call.*

Proof For any permutation σ we have $\sum_i c(A_{\sigma(i)}) = \sum_i c(A_i)$, since both sides sum the same function over the same multiset of bundles. So condition (ii) of Theorem 10 holds with equality for every allocation, and every allocation is envy-freeable. Moreover $w_{\mathcal{A}}(i,j) = c(A_i) - c(A_j)$ telescopes *exactly* along a path $P = (i_1, \ldots, i_r)$, giving $w_{\mathcal{A}}(P) = c(A_{i_1}) - c(A_{i_r})$ and hence

$$\ell_{\mathcal{A}}(u) \;=\; c(A_u) - \min_{j \in N} c(A_j).$$

It therefore suffices to keep $\max_i c(A_i) - \min_i c(A_i) \le 1$. This is an invariant of the greedy rule. It holds vacuously at the empty allocation. Suppose it holds before a step, so that $c(A_i) \in \{\mu, \mu+1\}$ for every i, where μ is the current minimum, and let the next chore go to a bundle attaining μ. Its cost becomes μ or $\mu + 1$, because marginals lie in $\{0, 1\}$; every other bundle is unchanged. So all costs still lie in $\{\mu, \mu+1\}$ and the invariant survives. The bound $p^* \in \{0, 1\}^n$ follows with Observation 12. □

Remark 22 (Why this one survives the literature)**.** Theorem 21 is not covered by [LMS26], because identical dichotomous costs need not be additive. The instance $c_i(S) = \max(0, |S| - 1)$ for every i is identical and dichotomous, with marginals $0, 1, 1, \ldots$, and is supermodular rather than additive — it is the very cost function that drives the obstruction of Theorem 36. Nor does [BKNS22] cover it, that being a goods result, and [KMS+24] gives only $n-1$ per agent on this class. So the statement is new, in the narrow sense that no cited paper implies it.

It is also easy, and should not be oversold. With identical costs every agent agrees on the cost of every bundle, the problem degenerates to load balancing, and the potential $\phi_i = c(A_i)$ telescopes *exactly* rather than merely bounding the arcs. That exactness is what makes the proof three lines long, and it is precisely what fails as soon as two agents disagree.

For $n = 2$ the conjecture holds as well, with total subsidy at most 1, but here the theorem is somebody else's: by Remark 9 two-agent chore division *is* two-agent goods division with the bundles swapped, so the content is [BKNS22, Theorem 4] and ours is only the reduction. As noted there, that reduction consumes the identity $M \setminus A_j = A_i$ and so says nothing for $n \ge 3$.

5.3 Small bundles, and fewer chores than agents

Every other case in this section, and every approach in the paper, eventually needs an *exchange* step: given a bad allocation, produce a better one by moving items. That is exactly what dichotomous costs refuse to supply, since a bundle of positive cost need not contain any single item whose removal lowers it — for instance $c(S) = \min(\max(0, |S| - 1), 1)$ on a three-element set has $c = 1$ while every single removal leaves it at 1. The following theorem needs no exchange at all; it reads the conclusion straight off the cycle-closing bound.

Theorem 23 (Small-bundle theorem)**.** *Let $\mathcal{A}$ be envy-freeable and suppose every bundle costs at most one unit to every agent: $c_v(A_i) \le 1$ for all $v, i \in N$. Then $p^* \in \{0, 1\}^n$ and the total subsidy is at most $n - 1$.*

Proof By Theorem 33, $\ell_{\mathcal{A}}(u) \le \max\left(0, \max_{v \ne u}[c_v(A_u) - c_v(A_v)]\right) \le \max(0, 1 - 0) = 1$, using $c_v(A_u) \le 1$ and $c_v(A_v) \ge 0$. Observation 12 then gives $p^* \in \{0, 1\}^n$, and the endpoint of a heaviest path has $\ell = 0$, so the total is at most $n - 1$. □

Corollary 24 (Fewer chores than agents)**.** *If $m \le n$ then Conjecture 2 holds, and a witness is computable by a single minimum-cost bipartite matching.*

Proof Among all allocations giving each agent at most one chore, take one minimising $\sum_i c_i(A_i)$; such an allocation exists because $m \le n$, and it is a minimum-cost matching between agents and chores. Every permutation of its own bundles is again an allocation of this form, so the minimisation was over a set containing all of them, and condition (ii) of Theorem 10 holds: $\mathcal{A}$ is envy-freeable. Each bundle has $|A_i| \le 1$, so $c_v(A_i) \le |A_i| \le 1$ for every v — because marginals lie in $\{0, 1\}$ — and Theorem 23 applies. The bound is tight by Example 1, which has $m = n - 1$. □

The hypothesis of Theorem 23 is genuinely weaker than $m \le n$: it also covers instances with many chores in which every agent finds all but a few of them free, so that no bundle is expensive to anybody. Both statements were checked semantically against the true longest-path computation — exhaustively for $m \le 3$, $n \le 3$, on 3,000 random instances for the theorem, and on 111,232 instances for the corollary — with no violations (`structural.py`).

Remark 25 (Why this is worth stating despite being easy)**.** The proof is three lines and the class is small, so the content is not the theorem but the *mechanism*: it is the only argument in this paper that reaches $\ell \le 1$ without an exchange step, a balance hypothesis, a schedule, or a search. It does so by making the cycle-closing bound tight from the far side — bounding $c_v(A_u)$ from above rather than $c_v(A_v)$ from below. Whether that trick has a wider reach is open; the obvious next question is what happens when bundles cost at most 2, where the same computation gives only $\ell \le 2$ and the argument stops.

5.4 Instances admitting a uniformly balanced family

The fourth case is not a restriction on the valuation class but on the instance, and it is the only one of the four that is genuinely new here.

Theorem 26. *Conjecture 2 holds for every instance admitting a uniformly balanced family, i.e. a partition of M into n bundles that every agent values within one unit of each other (Theorem 98). Given such a family, an envy-free solution with $p^* \in \{0, 1\}^n$ and total subsidy at most $n - 1$ is obtained by assigning the bundles by a single maximum-weight matching.*

Proof Proposition 99. □

Three things should be said about the scope of this, none of which the proof makes evident.

- **It is a certificate, not an algorithm.** Verifying a proposed family and assigning it are polynomial; *finding* one is a search over partitions, and no polynomial procedure for that is known. So Theorem 26 decides an instance only once somebody supplies the family.
- **The class is large but not everything.** It contains every instance at $n = m = 3$ — all 9,880, checked exhaustively — and every hard instance the project had accumulated before it was tested. It does *not* contain the discrepancy instance of Proposition 100, at $n = 3$ and $m = 4$, nor the certified-hard set-splitting instance.
- **Membership is not the same as difficulty.** The instances outside the class are not the instances that need large subsidies; on the counterexample of Proposition 100, 19 allocations are good and several need no subsidy at all. Uniform balance is a property of the certificate, not of the instance's hardness.

So this is a solved case in the same sense as the others in this section: real, checkable, and not load-bearing. What it contributes is the counterexample and Remark 102, which say that any complete method must reason about paths rather than arcs.

5.5 The residual at $n = 3$, and who gets paid

Every approach in this report has attacked the general case, and each died on a general obstruction. The one gap in that coverage is a dedicated attack on a bounded case beyond $n = 2$. At $n = 3$ the target is finite in a useful way: by Proposition 13 Conjecture 2 is the single clause $\max_i \ell_{\mathcal{A}}(i) \le 1$, and by the two-tier characterisation with Theorem 77 the paid set lies in a lattice inside $2^{\{1,2,3\}}$ in which $\emptyset$ and N impose the same condition. So there are exactly three cases, $|S| = 0, 1, 2$, and Example 1 shows the last is sometimes forced.

Observation 27 (The residual is small and its paid set is extremal)**.** *Over* 2,460 *instances at* $n = 3$ *drawn from six families — uniform, nested, mixed, capped, threshold and disjoint-interest — searched exhaustively over all* 3^m *allocations:*

- *every instance admits a good allocation; the minimal paid-set size is* 0 *on* 2,304, 1 *on* 135 *and* 2 *on* 21*, so the residual beyond exact envy-freeness is* 156 *instances,* 6.2%*;*
- *in* every *minimal witness with* $|S| = 1$*, the paid agent attains* $\max_i c_i(A_i)$ *—* 135 *of* 135*;*
- *in* every *minimal witness with* $|S| = 2$*, the unpaid agent attains* $\min_i c_i(A_i)$ *—* 21 *of* 21*.*

Bundle size does not *characterise it: the paid agent holds a largest bundle in only* 91 *of* 135 *witnesses, and the unpaid agent a smallest in* 2 *of* 21 *(*`update_44/n3_cases.py`*).*

So at a minimal witness the singled-out agent sits at an extreme of *own cost*, which removes the choice of *which* agent is paid: one may assume it is an $\arg\max_i c_i(A_i)$ when $|S| = 1$, and that the unpaid one is an $\arg\min$ when $|S| = 2$. That is a necessary condition, not a sufficient one, and it is what a case analysis at $n = 3$ would start from.

Remark 28 (The tempting unification is false)**.** Both clauses look like instances of $S = \{i : c_i(A_i) > \min_j c_j(A_j)\}$ — the paid agents being exactly those not attaining the minimum own cost. They are not. Over the residual instances that equality holds at *some* minimal witness on 59 of 69 at $n = 3$, and at *every* one on only 37; the failure is visible already when two agents tie at the maximum while $|S| = 1$. At $n = 4$ and $n = 5$ it fares better (48/53 and 67/69 for every witness), which is a warning rather than a comfort: the equality is not a theorem and should not be assumed.

Remark 29 (No counterexample, and the limits of looking)**.** Conjecture 2 was also attacked directly, by searching for an instance admitting *no* good allocation. Over 683 instances at $n \le 6$, $m \le 11$, each searched exhaustively over all n^m allocations and drawn from the adversarial families above, none was found. The power of that search is limited and should be read as such: only 32 of the 683 were hard at all, in the sense of admitting no exactly envy-free allocation, so the conjecture was under strain on under 5% of the sample (`update_44/counterexample_hunt.py`).

Remark 30 (Why the two-agent theorem does not bootstrap)**.** In the $|S| = 1$ case the two unpaid agents are exactly envy-free between themselves, which invites using Theorem 118 on the sub-instance they share. It does not apply: that theorem delivers a subsidy of at most one for two agents, not exact envy-freeness, and Example 1 at $n = 2$ already forces a dollar. The sub-problem at $n = 3$ is strictly harder than the one solved at $n = 2$, and the case must be attacked directly.

Remark 31 (No explicit rule solves the $n = 3$ residual either)**.** Section 13 found no global objective whose optima are all good at general n, but it never tested one on the $n = 3$ residual, which is the only place the question has content. Seven lexicographic rules were tried there — minimise the maximum own cost, with tie-breaks by total cost, by the number of agents at that maximum, by $\sum_i |A_i|^2$, and the sorted own-cost vector with and without a balance tie-break.

On a first sweep of 227 residual instances, the rule "minimise the sorted own-cost vector, then $\sum_i |A_i|^2$" had *every* optimum good, with no failures — the first rule anywhere in this report to do so on a residual. It does not survive. On 368 residual instances it fails on 4, and on one of them *every* optimum is bad, so no tie-break can rescue it; the smallest failure is at $m = 4$ with singleton costs $(1, 1, 1, 0)$, $(1, 0, 0, 1)$, $(0, 0, 0, 1)$. The earlier clean sweep was small-sample luck, of exactly the kind Remark 41 records elsewhere: there a refinement survived 1,050 instances and failed at a rate near 1 in 2.6×10^4.

So the explicit-rule route is closed at $n = 3$ as well as in general. What survives from this section is Observation 27: the residual is 6.2% of instances, and at a minimal witness the singled-out agent is extremal in own cost.

6 Approach 1: Item-by-Item Insertion

The template. The algorithm of [BKNS22] maintains an envy-free solution $(\mathcal{A}^t, p^t)$ over a partial allocation with $p^t \in \{0, 1\}^n$, allocates one further good at each step, and repairs the invariant by permuting whole bundles among the agents. Nothing already allocated is ever moved between bundles. This section follows that template on the chore side.

Why it looks promising. Two of its steps come out *easier* than for goods.

Theorem 32 (Free insertion)**.** *Let $\mathcal{A}$ be envy-freeable and suppose $c_x(A_x \cup \{g\}) = c_x(A_x)$ for some agent x. Then Z^x — the allocation with g added to A_x — is envy-freeable and $\ell_{Z^x}(i) \le \ell_{\mathcal{A}}(i)$ for every i.*

Proof By Lemma 14, arcs between agents other than x are unchanged, arcs out of x are unchanged since $\delta_x = 0$, and arcs into x satisfy $w_{Z^x}(i, x) = w_{\mathcal{A}}(i, x) - \delta_{i,x} \le w_{\mathcal{A}}(i, x)$. Every arc weight weakly decreases, so no positive cycle appears and no path weight rises. □

The corresponding step of [BKNS22] needs the receiver to be maximally subsidised, a bundle permutation, and extendability machinery, because for goods the insertion *raises* the arcs into the receiver. Here it lowers them for free. So the only case needing work is a chore whose marginal cost is 1 for every agent on her own bundle — the *hard case*.

The second tool survives the approach and is used throughout the report.

Theorem 33 (Cycle-closing bound)**.** *For any envy-freeable allocation $\mathcal{A}$ and any agent u,*

$$\ell_{\mathcal{A}}(u) \;\leq\; \max\Big(0,\ \max_{v\neq u}\big[\,c_v(A_u) - c_v(A_v)\,\big]\Big).$$

Proof Since $\mathcal{A}$ is envy-freeable, $G_{\mathcal{A}}$ contains no positive-weight directed cycle (Theorem 10(iii)). Every closed walk decomposes into simple cycles, so every closed walk also has weight at most 0; deleting a closed sub-walk from a walk therefore never decreases its weight, and every walk starting at u has weight at most that of some *simple* path starting at u. As there are finitely many simple paths, the supremum defining $\ell_{\mathcal{A}}(u)$ is attained, by a simple path P^*.

If P^* is empty then $\ell_{\mathcal{A}}(u) = 0$ and the claim holds, the right-hand side being non-negative. Otherwise P^* is simple and so does not revisit u; its terminal vertex v therefore satisfies $v \neq u$, and appending the arc (v, u) yields a genuine simple cycle C. By Theorem 10(iii), $w_{\mathcal{A}}(C) \leq 0$, so

$$\ell_{\mathcal{A}}(u) \;=\; w_{\mathcal{A}}(P^*) \;=\; w_{\mathcal{A}}(C) - w_{\mathcal{A}}(v,u) \;\leq\; -w_{\mathcal{A}}(v,u) \;=\; c_v(A_u) - c_v(A_v),$$

which is at most the maximum over $v \neq u$. □

Remark 34 (The bound is not an evaluation of $\ell_{\mathcal{A}}$)**.** $\ell_{\mathcal{A}}(u)$ *is* the maximum path weight out of u; the bound is a one-directional estimate of it in terms of a single arc, discarding everything about the heaviest path except where it ends. The two are not equivalent and the inequality is routinely strict: a path $u \to a \to b$ of weight 2 with $w_{\mathcal{A}}(b,u) = -3$ closes a cycle of weight $-1 \leq 0$, so $\ell_{\mathcal{A}}(u) = 2$ while the bound returns 3. Checked against the true longest-path computation over 172,484 (agent, allocation) pairs, the bound holds without exception but is exact on only 70.9% of them, with slack up to 5 (`update_46/cyclebound_check.py`). Evaluating $\ell_{\mathcal{A}}$ requires the longest-path computation; the bound costs $O(n)$ arc lookups and is used only where an upper estimate suffices — Corollary 35 and Theorem 23. Where it was pressed into service as an equality it failed, which is the content of Proposition 144.

Corollary 35. *If $c_v(A_u) \leq c_v(A_v) + 1$ for all u, v — no agent would suffer more than one extra unit by taking anybody else's bundle — then $p^* \in \{0,1\}^n$.*

Where it fails. One instance kills the template.

Theorem 36 (The insertion template cannot be completed)**.** *There is a three-agent instance with negative dichotomous valuations, an envy-freeable partial allocation $\mathcal{A}$ with $p^* \in \{0,1\}^3$, and an unallocated chore g, such that for* every *agent x the allocation Z^x requires a subsidy of* 2 *— even though the full instance admits an allocation with subsidy* 0.

Proof Take $M = \{a_1, a_2, g\}$ with

$$c_1(S) = \max\big(0, |S| - 1\big), \qquad c_2(S) = c_3(S) = |S|,$$

all dichotomous. Let $\mathcal{A} = (\{a_1, a_2\}, \emptyset, \emptyset)$, whose envy matrix

$$\big(w_{\mathcal{A}}(i,j)\big) = \begin{pmatrix} 0 & 1 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{pmatrix}$$

has no positive cycle and gives $p^* = (1, 0, 0)$: a legal state. The marginal cost of g is 1 for every agent on her own bundle, so this is the hard case, and inserting g into each bundle in turn gives minimum subsidy vectors

$$Z^1 \mapsto (2, 0, 0), \qquad Z^2 \mapsto (2, 1, 0), \qquad Z^3 \mapsto (2, 0, 1),$$

each with an entry equal to 2. Yet the complete allocation $(\{a_1\}, \{a_2\}, \{g\})$ has an identically zero envy matrix and subsidy $(0, 0, 0)$. □

Note that c_1 is *supermodular*, which is legal — neither [BKNS22] nor this report restricts to submodular valuations — and the instance genuinely needs it.

Remark 37 (Verdict)**.** Any proof of Conjecture 2 must be allowed to **re-open already-allocated bundles**. Allocating one item at a time, never moving an allocated item, and repairing only by permuting whole bundles is provably insufficient. This is a genuine asymmetry between the goods and chore problems, not a gap in the analysis, so effort spent strengthening the insertion invariant is wasted. The same failure recurs in two further coordinate systems (Section 8, Section 11.3).

7 Approach 2: Selecting a Utilitarian-Optimal Allocation

The idea. Theorem 36 rules out building the allocation one chore at a time. The natural response is to stop building it and instead *select* one globally, from a set small enough to reason about and rich enough to contain a witness. Utilitarian optimality is the obvious candidate.

Why it looks promising. Two reasons, both real. First, by Theorem 10 an allocation minimising $\sum_i c_i(A_i)$ over *all* allocations — not merely over reassignments of its own bundles — is automatically envy-freeable, so half of what is needed comes for free. Second, it carries a strong exchange property: if $g \in A_i$ is *active* for i, meaning $c_i(A_i) - c_i(A_i \setminus \{g\}) = 1$, then $c_j(A_j \cup \{g\}) = c_j(A_j) + 1$ for every j — nobody can absorb it for free, or the allocation would not have been optimal. With Theorem 33, a path of weight at least 2 from u to v forces $c_v(A_u) \geq c_v(A_v) + 2$, so A_u carries two units live for v, and one would like to transfer one along the path and watch a potential drop.

Where it fails.

Proposition 38. *There is an instance with $n = 3$ and $m = 4$ whose* unique *utilitarian-optimal allocation requires a subsidy of* 2*, although another allocation — of strictly greater total cost — admits a subsidy of* 0*.*

Proof Take $M = \{g_1, g_2, g_3, g_4\}$ with the costs below.

S	c_1	c_2	c_3	S	c_1	c_2	c_3
$\emptyset$	0	0	0	g_3g_4	2	1	1
g_1	1	1	1	$g_1g_2g_3$	2	3	2
g_2	1	1	1	$g_1g_2g_4$	3	2	2
g_3	1	1	0	$g_1g_3g_4$	2	2	2
g_4	1	1	1	$g_2g_3g_4$	2	2	2
g_1g_2	2	2	2	$g_1g_2g_3g_4$	3	3	3
g_1g_3	1	2	1				
g_1g_4	2	2	2				
g_2g_3	1	2	1				
g_2g_4	2	2	2				

Each column is 0 at $\emptyset$ with every marginal in $\{0, 1\}$, so all three are negative dichotomous in cost form.

Agent 1 pays 1 for every singleton, so $c_1(A_1) = 0$ only if $A_1 = \emptyset$; agent 3 pays 0 for $\{g_3\}$ but 1 for every superset. Any allocation of total cost below 2 would need two agents at cost 0, forcing $A_2 \supseteq \{g_1, g_2, g_4\}$, which costs agent 2 at least 2. So the minimum total cost is 2, attained only by

$$\mathcal{A}^\star = \left(\emptyset,\ \{g_1, g_2, g_4\},\ \{g_3\}\right), \qquad 0 + 2 + 0 = 2,$$

whose envy graph has $w_{\mathcal{A}^\star}(2, 1) = 2$, giving $p^* = (0, 2, 0)$. But $\mathcal{B} = (\{g_1, g_3\}, \{g_2\}, \{g_4\})$, of total cost 3, has envy matrix

$$\left(w_{\mathcal{B}}(i, j)\right) = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

every entry at most 0, hence $p^* = (0, 0, 0)$. □

Two features make this sharper than a bare counterexample: the optimum is *unique*, so there is no tie to break, and the gap is the largest possible in one step — subsidy 2 against 0, for one extra unit of total burden.

Corollary 39 (No refinement of the utilitarian-optimal set can work)**.** *Let $\mathcal{R}$ be any rule selecting a nonempty subset of the utilitarian-optimal allocations. Then $\mathcal{R}$ fails on the instance of Proposition 38: the optimal set is the singleton $\{\mathcal{A}^\star\}$, so $\mathcal{R}$ returns it, and $\ell_{\mathcal{A}^\star}(2) = 2$ while a subsidy-free allocation exists outside.*

Remark 40 (Verdict)**.** Conjecture 2, if true, is *not* witnessed by welfare-maximising allocations, and looking for a tie-breaking rule inside the utilitarian-optimal set is unsound as posed. The exchange property quoted above remains true and usable; what fails is the assumption that a witness may be sought among cost-minimising allocations at all. **Any search procedure must be free to give up total cost.** The slip underneath is a misreading of Theorem 10: it says an envy-freeable allocation maximises welfare across reassignments of *its own bundles*, not that it is globally welfare-maximising — and $\mathcal{B}$ above, envy-freeable at cost 3 against the optimum 2, is a direct witness that the two notions differ.

Remark 41 (The refinement that looked safe)**.** Refining the optimal set lexicographically by cardinality balance and then by own-cost — suggested independently by Theorem 16, by [BDN+20]'s balancedness guarantee, and by Remark 53 — survives randomised search over roughly 1050 instances and is nevertheless false by Corollary 39, both refinements being no-ops on a singleton. The failure rate is about one instance in 2.6×10^4, so the sweep was two orders of magnitude short of seeing it (Section 19.1).

7.1 Selection rules that fail

Three tie-breaking rules inside the utilitarian-optimal set were tested before Proposition 38 was found — minimising total perceived load, leximin on the perceived-load vector, and local search by single transfers — and all three fail. The informative one is the last: single transfers are too coarse, because the utilitarian-optimal allocations of a dichotomous instance have natural moves that are exchange *paths and cycles*, as Yankee Swap and matroid union are in the matroid-rank literature [BEF21]. Exchange-path local search over the full allocation lattice — not the optimal set, per Remark 40 — remains untried.

8 Approach 3: The Replica Transform and the Peel Reformulation

Both preceding approaches fail because of *when* decisions are made. Item-by-item insertion (Section 6) commits a chore to an owner and cannot take it back; selecting a utilitarian optimum (Section 7) commits to the wrong allocation outright. This section changes coordinates so that the commitment is deferred. The chore problem is re-expressed as a *goods* problem on a replicated item set, and then read dynamically as a process that relieves agents of chores one at a time. Ownership of a chore is decided only by the last move that touches it.

The payoff is that the proof orientation of [BKNS22] becomes correct again and the obstruction of Theorem 36 ceases to bind. The price is a new constraint — coverage — which is then shown to be non-vacuous.

8.1 The replica instance

Fix a negative dichotomous instance with cost functions c_i on chores $M = \{a_1, \dots, a_m\}$. Build a *replica instance* $\hat{I}$ whose item set carries $n-1$ copies of a good b_j for each chore a_j:

$$\hat{M} \;:=\; \left\{ b_j^{(1)}, \dots, b_j^{(n-1)} \;:\; j \in M \right\}.$$

For $S \subseteq \hat{M}$ let $\tau(S) \subseteq M$ be the set of *types* occurring in S, and define

$$\hat{v}_i(S) \;:=\; c_i(M) \;-\; c_i\big(M \setminus \tau(S)\big).$$

The reading is that *handing agent i a copy of b_j relieves i of chore a_j*. Each chore ends with exactly one owner, so exactly $n-1$ agents must be relieved of it — hence $n-1$ copies — and no agent may be relieved of the same chore twice, so no agent may hold two copies of one type. That last clause is load-bearing rather than cosmetic: it is what makes $\hat{v}_i$ well defined as a dual, since a second copy of a type has marginal 0 and never +1 again.

Remark 42 (The naive reading is false)**.** If one instead declares a copy of b_j to be worth $+1$ to whoever receives it, the construction collapses. An agent indifferent to a_j would gain from b_j, every $\hat{v}_i$ would reduce to $|\tau(S)|$, and the induced envy graph would be $\hat{w}(i,k) = |B_i| - |B_k|$ — the pure cardinality term already isolated in Theorem 16, carrying none of the cost information. The correct reading is *marginal*: a copy of b_j is worth to agent i exactly i's marginal cost of a_j on what remains of i's workload.

This also completes Remark 9. There the dual $\tilde{v}_i(S) = c_i(M) - c_i(M \setminus S)$ settled $n = 2$ and broke at $n \geq 3$ because $M \setminus A_i$ stops being a bundle of the partition. Replication is precisely the repair: with $n-1$ copies of every type, the n complements $M \setminus A_1, \ldots, M \setminus A_n$ become *simultaneously* realisable as disjoint bundles.

8.2 The transform

Call an allocation $B = (B_1, \ldots, B_n)$ of all of $\hat{M}$ a *coverage allocation* if no B_i contains two copies of the same type.

Theorem 43 (Replica transform)**.** *Let $c_1, \ldots, c_n$ be* dichotomous *in the sense of Theorem 7 — that is, every marginal $c_i(S \cup \{g\}) - c_i(S)$ lies in $\{0, 1\}$, which already forces c_i to be monotone. Then*

(i) each $\hat{v}_i$ is a dichotomous valuation on $\hat{M}$ in the sense of [BKNS22];

(ii) $\Phi : \mathcal{A} \mapsto B$, where B_i receives one copy of each type in $M \setminus A_i$, is a bijection between partitions of M and coverage allocations of $\hat{M}$, up to relabelling copies of a type;

(iii) for corresponding $\mathcal{A}$ and B the envy graphs agree arc for arc: $\hat{w}_B(i,k) = w_{\mathcal{A}}(i,k)$ for all i, k.

Proof *(i)* Normalisation: $\hat{v}_i(\emptyset) = c_i(M) - c_i(M) = 0$. Monotonicity: $S \subseteq S'$ gives $\tau(S) \subseteq \tau(S')$, hence $M \setminus \tau(S) \supseteq M \setminus \tau(S')$, and c_i is monotone, so $\hat{v}_i(S) \leq \hat{v}_i(S')$. Marginals: take $b \notin S$ of type j. If $j \in \tau(S)$ then $\tau(S \cup \{b\}) = \tau(S)$ and the marginal is 0. Otherwise, writing $T := M \setminus \tau(S)$, we have $M \setminus \tau(S \cup \{b\}) = T \setminus \{j\}$ with $j \in T$, so

$$\hat{v}_i(S \cup \{b\}) - \hat{v}_i(S) \; = \; c_i(T) - c_i(T \setminus \{j\}) \; \in \; \{0, 1\},$$

because c_i has marginals in $\{0, 1\}$.

(ii) Forward: type j is absent from exactly one bundle of $\mathcal{A}$, its owner's, so exactly $n-1$ agents require a copy of b_j and the $n-1$ available copies are routed injectively. Which physical copy goes where is immaterial, since $\hat{v}_i$ depends on S only through $\tau(S)$. By construction no B_i holds a duplicate and $\bigcup_i B_i = \hat{M}$. Backward: in a coverage allocation the $n-1$ copies of type j lie in $n-1$ *distinct* bundles, so exactly one agent's bundle contains no copy of j; setting $\Psi(B)_i := M \setminus \tau(B_i)$ therefore puts j in exactly one $\Psi(B)_i$, making $\Psi(B)$ a partition. Finally $\Psi(\Phi(\mathcal{A}))_i = M \setminus (M \setminus A_i) = A_i$, and $\Phi(\Psi(B))$ agrees with B up to which copy of a type sits where.

(iii) For corresponding $\mathcal{A}, B$ we have $\tau(B_k) = M \setminus A_k$, so $\hat{v}_i(B_k) = c_i(M) - c_i(A_k)$ and

$$\hat{w}_B(i,k) = \hat{v}_i(B_k) - \hat{v}_i(B_i) = \big(c_i(M) - c_i(A_k)\big) - \big(c_i(M) - c_i(A_i)\big) = c_i(A_i) - c_i(A_k) = w_{\mathcal{A}}(i,k). \quad \square$$

Remark 44 (Monotonicity alone is not enough, and which clause does what)**.** The two halves of the hypothesis do different jobs in the proof of (i), and it is worth separating them because the statement is easy to misread. Monotonicity of c_i is what makes $\hat{v}_i$ monotone, and *nothing else*. The binary-marginal condition is what makes $\hat{v}_i$'s marginals binary, because those marginals are *downward* marginals of c_i:

$$\hat{v}_i(S \cup \{b\}) - \hat{v}_i(S) \;=\; c_i(T) - c_i(T \setminus \{j\}), \qquad T = M \setminus \tau(S),$$

so they inherit exactly the range of c_i's own marginals. Monotonicity alone bounds this only by ≥ 0.

The failure is real rather than an artefact of the proof. Take $m = 2$ with $c(\emptyset) = c(\{a\}) = c(\{b\}) = 0$ and $c(\{a, b\}) = 2$. This c is monotone and normalised but has a marginal of 2, and the resulting dual has $\hat{v}(\{b_a\}) - \hat{v}(\emptyset) = 2 - 0 = 2$, so $\hat{v}$ is not dichotomous and [BKNS22] does not apply to $\hat{I}$ at all. Conversely, the normalisation $c_i(\emptyset) = 0$ is *not* used in Theorem 43: $\hat{v}_i(\emptyset) = c_i(M) - c_i(M) = 0$ regardless, and the constant $c_i(M)$ cancels in part (iii). It is retained only because it is part of the model. Verified by `monotone_check.py`, which also confirms by exhaustive enumeration that binary marginals imply monotonicity, so the operative hypothesis is a single clause.

Part (i) is worth stating on its own: *the dual of a dichotomous cost function is a dichotomous value function*, so the class is closed under this duality and the hypotheses of [BKNS22] apply to $\hat{I}$ verbatim, with no additivity, submodularity or subadditivity assumed anywhere. Because the graphs of part (iii) are equal, envy-freeability, every $\ell(i)$, the whole set of feasible subsidy vectors and the $\{0, 1\}$ target all transfer in both directions without loss.

Corollary 45 (Coverage is the entire gap)**.** *Conjecture 2 holds if and only if every replica instance $\hat{I}$ admits a* coverage *allocation B with $\ell_B(i) \leq 1$ for all i.*

Applying [BKNS22] to $\hat{I}$ as a black box is legal by Theorem 43(i) and returns some B with $\hat{p} \in \{0, 1\}^n$ and total at most $n - 1$. But it is free to place two copies of one type in one bundle. If it does, that type is absent from at least two bundles, the chore has at least two owners, and $\Psi(B)$ is not a partition. Coverage is the sole residual difficulty — and, by Section 8.5, a real one.

Lemma 46 (Duplicate extraction is envy-neutral)**.** *Removing from B_x a copy whose type occurs at least twice in B_x leaves $\tau(B_x)$ unchanged, hence leaves every $\hat{v}_k(B_x)$ unchanged, hence leaves the entire envy graph unchanged.*

Proof Immediate, since $\hat{v}_k(S)$ depends on S only through $\tau(S)$. □

So any output of [BKNS22] may be stripped to a duplicate-free partial allocation for free. The difficulty is never in removing duplicates; it is in *re-inserting* the stripped copies into agents that do not already hold that type.

8.3 The peel process

Read a coverage allocation as being built one copy at a time and translate back into chores.

Definition 47 (Peel process)**.** *A* state *is a* workload profile $W = (W_1, \ldots, W_n)$ *with* $W_i \subseteq M$*, such that the* owner-candidate set $S_j := \{i : j \in W_i\}$ *is nonempty for every* $j \in M$*. Read* W_i *as "agent* i *is still on the hook for* W_i*".*

- ***Root:*** $W_i = M$ *for every* i*.*
- ***Peel*** peel(x, j)*: legal iff* $j \in W_x$ *and* $|S_j| \geq 2$*; it sets* $W_x \leftarrow W_x \setminus \{j\}$*.*
- ***Permutation:*** *replace* W *by* $(W_{\sigma(1)}, \ldots, W_{\sigma(n)})$*.*
- ***Terminal:*** $|S_j| = 1$ *for every* j*; the unique element of* S_j *is the* survivor*, i.e. the owner of* j*, and the induced allocation is* $A_i = \{j : S_j = \{i\}\}$*.*
- ***Graph:*** $w_W(i, k) = c_i(W_i) - c_i(W_k)$*. The root graph is identically zero.*

A state is legal *if its graph has no positive-weight cycle and* $\ell_W(i) \leq 1$ *for every* i*.*

By Theorem 43 a peel state is exactly a duplicate-free partial allocation of $\hat{M}$, a peel is exactly the insertion of one good into one bundle, a permutation is exactly a bundle reassignment, and terminal states are exactly coverage allocations. Nothing is added or lost in the translation.

Lemma 48 (Peel dynamics)**.** *Write* $\mu_k := c_k(W_x) - c_k(W_x \setminus \{j\})$ *for* k*'s marginal cost of* j *on* x*'s residual workload. Then* peel(x, j) *changes only the arcs incident to* x*, by*

$$\Delta w(k, x) \;=\; +\mu_k \;\in\; \{0, 1\} \qquad \textit{and} \qquad \Delta w(x, k) \;=\; -\mu_x \;\in\; \{-1, 0\} \qquad (k \neq x).$$

Proof Only W_x changes. For $k \neq x$, $w(k, x) = c_k(W_k) - c_k(W_x)$ and $c_k(W_x)$ drops by μ_k, so the arc rises by μ_k. For arcs out of x, $w(x, k) = c_x(W_x) - c_x(W_k)$ and $c_x(W_x)$ drops by μ_x. Arcs between two agents other than x involve neither W_x nor c_x. □

The orientation reversal. This is the point of the whole construction. Compare Lemma 48 against Lemma 14:

frame	move	in-arcs at x	x should be
goods [BKNS22]	insert a good	rise	maximally subsidised
chores, insertion (§6)	insert a chore	fall	unsubsidised, $p_x = 0$
chores, **peel**	relieve of a chore	**rise**	maximally subsidised

The obstruction of Theorem 36 was exactly this sign clash. The repair subroutine of [BKNS22] navigates to an agent whose tentative subsidy reached 2; in the goods world that agent is automatically maximally subsidised, which is the set the argument needs, whereas in the chore-insertion world she automatically has $p_u = 1$ and is precisely the agent who must *not* receive the chore. In the peel frame the quantity handed out is *relief*, relief correctly flows to the most envious agents, and the orientation points the right way again.

Lemma 49 (Spectator-free peels are safe)**.** *If* $\mu_k = 0$ *for every* $k \neq x$*, then* peel(x, j) *weakly decreases every arc weight, hence* ℓ *pointwise, and preserves envy-freeability — whatever* μ_x *may be.*

Proof By Lemma 48 the arcs into x are unchanged, the arcs out of x change by $-\mu_x \leq 0$, and all other arcs are untouched. Path and cycle weights are sums of arc weights. □

Lemma 49 is the peel-frame replacement for Theorem 32, and note where the safety condition has moved. In the insertion frame a chore that was free *for the receiver* was unconditionally safe; here what matters is that the chore be free *for everyone else*. That is the goods-side safety structure of [BKNS22], as the identification predicts. The hard case is now a pair (x, j) for which some other agent k has $\mu_k = 1$.

What does *not* transfer is the coverage constraint, which has no analogue in [BKNS22]: any good may go in any bundle there, whereas a peel demands $x \in S_j$ and a type is peelable only while $|S_j| \geq 2$.

8.4 The insertion obstruction dissolves

Theorem 36 remains true — it is a theorem about the insertion template, and the peel frame is a different template, not a repair of that one. What follows is that its witness no longer obstructs.

Recall the witness: $M = \{a_1, a_2, g\}$ with $c_1(S) = \max(0, |S| - 1)$ and $c_2 = c_3 = |S|$.

The trap state is refused before it can form. The peel-frame counterpart of the trap is $W = (\{a_1, a_2, g\}, \{g\}, \{g\})$, and

$$w_W(1,2) \;=\; c_1(\{a_1, a_2, g\}) - c_1(\{g\}) \;=\; 2 - 0 \;=\; 2,$$

so $\ell_W(1) = 2$ and the state is already illegal. No legal peel schedule passes through it.

Remark 50 (The conditioned-remainder principle)**.** The two frames disagree because they compare different things. At a stage where the finished types form a partial assignment $\mathcal{A}^{(t)}$ and the untouched types form R, every agent is still on the hook for R, so the peel-frame arc is

$$c_i\big(\mathcal{A}_i^{(t)} \cup R\big) - c_i\big(\mathcal{A}_k^{(t)} \cup R\big), \qquad \text{not} \qquad c_i\big(\mathcal{A}_i^{(t)}\big) - c_i\big(\mathcal{A}_k^{(t)}\big).$$

Every comparison is conditioned on the common unfinished remainder. For additive costs the R terms cancel and the two frames coincide — consistent with the additive case already being closed by Theorem 19. For non-additive costs they differ, and on the witness it is exactly the supermodularity of c_1 that the conditioning catches: with g still pending, loading agent 1 with both of a_1, a_2 already costs 2, and the peel invariant sees it. *The insertion invariant was blind to pending load; the peel invariant has hindsight built in.* This is the substantive content of the change of coordinates.

A legal schedule reaches the zero-subsidy allocation. Six peels, every one of them the hard case ($\mu_k = 1$ for all k — this instance never offers a spectator-free move), with the invariant holding throughout. Verified by `peel.py`.

step	move	resulting W	ℓ	$\sum_i \ell(i)$
0	root	$(a_1a_2g,\ a_1a_2g,\ a_1a_2g)$	$(0,0,0)$	0
1	relieve 1 of g	$(a_1a_2,\ a_1a_2g,\ a_1a_2g)$	$(0,1,1)$	2
2	relieve 2 of g [owner(g) = 3]	$(a_1a_2,\ a_1a_2,\ a_1a_2g)$	$(0,0,1)$	1
3	relieve 3 of a_1	$(a_1a_2,\ a_1a_2,\ a_2g)$	$(0,0,0)$	0
4	relieve 1 of a_2	$(a_1,\ a_1a_2,\ a_2g)$	$(0,1,1)$	2
5	relieve 2 of a_1 [owner(a_1) = 1]	$(a_1,\ a_2,\ a_2g)$	$(0,0,1)$	1
6	relieve 3 of a_2 [owner(a_2) = 2]	$(a_1,\ a_2,\ g)$	$(0,0,0)$	0

The terminal allocation is $(\{a_1\},\{a_2\},\{g\})$, the known zero-subsidy solution. This establishes that the peel template is not killed by the existing obstruction; it does not establish that the template always works, which is Section 8.5. Note also how it meets the requirement of Remark 37 that a correct proof be free to re-open allocated bundles: *peeling never commits a chore's owner until the last peel of that type, so ownership is a deferred decision and there is nothing to re-open.*

8.5 The new obstruction: peel dead ends

The natural conjecture is local.

Conjecture 51 (Local extendability — **refuted**). *From every legal non-terminal state, some legal move exists.*

Proposition 52. *Conjecture 51 is false. There are legal, non-terminal peel states from which no sequence of peels and permutations reaches a terminal state without violating the invariant.*

Proof A complete backward fixed-point computation over the entire state space of the witness, with both move types and no heuristics, is carried out by `peel3.py`; it reports the following.

quantity	value
states in total (7^3)	343
legal states	175
terminal (partition) states that are legal	6 of 27
legal states from which a terminal is reachable within the invariant	166
legal dead ends	**9**
root is good	yes

The smallest dead end is $W = (\{a_1, a_2\},\{g\},\{g\})$, which is legal with $\ell = (1,0,0)$. The only peelable type is g, and both continuations fail: relieving agent 2 or agent 3 of g creates a weight-2 path out of agent 1. Permutations do not help — the profile $(\{g\},\{a_1,a_2\},\{g\})$ is legal but its two continuations fail identically. □

What the dead ends are, in the language of [BKNS22]. This is the sharpest statement available. At $W = (\{a_1, a_2\}, \{g\}, \{g\})$ the remaining unallocated copy is a copy of b_g, and it has three possible recipients. Giving it to agent 2 or 3 is a genuine peel and violates the invariant. Giving it to agent 1 — who already holds a copy of b_g — creates a *duplicate*, which is perfectly legal in $\hat{I}$ and, by Lemma 46, changes nothing whatever in the graph. So the algorithm of [BKNS22] never gets stuck on $\hat{I}$: it gets stuck only in the sense that its one legal move is to spend a copy on a duplicate, and spending a copy on a duplicate is exactly the loss of coverage. If agent 1 takes the second b_g then g ends with owner-candidate set $\{2, 3\}$ — two owners, not a partition. Corollary 45 is therefore not a formal restatement; the gap it names is realised.

Remark 53 (The balance signal — **observation only).** All nine dead ends of the witness have the same shape: *one agent is already committed to a two-element terminal bundle*. They are exactly the three choices of which pair times the three choices of which agent. Correspondingly all six reachable terminals are the six perfect matchings, so on this instance the only legal partitions are the matchings. This is the same signal as Theorem 16 — chores are goods plus cardinality balancing — and as the balancedness guarantee of [BDN+20], arriving from a third direction, and it suggests the missing ingredient is a *balance* discipline rather than a subsidy-based one. Consistent with that, restricting recipients to $\arg\max_i \ell(i)$, the direct mirror of $\mathrm{M}(p)$, shrinks the reachable state space but does *not* avoid the dead ends: a subsidy-based rule alone is not enough. This is a single instance and should not be generalised without more data.

8.6 The reduced open problem

Conjecture 54 (Reachability — **open).** *From the root, some sequence of peels and permutations reaches a terminal state with the invariant holding at every intermediate state.*

Conjecture 54 implies Conjecture 2. The root graph is identically zero, so the invariant holds there; if a legal schedule reaches a terminal state W^{end} then the induced allocation $\mathcal{A}$ has $G_{\mathcal{A}} = G_{W^{\text{end}}}$, with no positive cycle and $\ell_{\mathcal{A}}(i) \le 1$ for all i; Observation 12 then forces $p^* \in \{0, 1\}^n$, and the endpoint of a maximum-weight path has $\ell = 0$, giving a total of at most $n - 1$.

Remark 55 (The converse is open, and it matters for what the experiments test)**.** Conjecture 54 implies Conjecture 2, but the implication has not been established in the other direction and should not be assumed. Conjecture 2 asserts that a good terminal state *exists*; Conjecture 54 asserts that one is *reachable along a path every state of which is legal*. These can come apart: a good terminal state may exist while every schedule leading to it passes through an illegal intermediate state, and Proposition 52 shows that illegal-looking intermediate states are not hypothetical. The practical consequence is that an instance with a *bad root* would refute Conjecture 54, and hence kill this approach, but would say nothing directly about Conjecture 2, which would have to be attacked by exhaustive search over allocations as in Section 19. Establishing the converse — that a good terminal state, when one exists, is always reachable legally — would make the peel frame a complete reformulation rather than a sufficient one, and is worth attempting on its own.

It is not automatic. Conjecture 54 is a reachability statement about a state space with genuine dead ends, so a proof must supply a *rule* for choosing the next peel that provably never enters the bad region, or else avoid the state space altogether by a potential or exchange argument. Two resources are available that [BKNS22] never had:

1. **Scheduling freedom.** Many types are unfinished at once, and we need only *some* type to admit a legal peel, not a prescribed one.
2. **$|S_j| \geq 2$.** A peelable type always has at least two candidate recipients, so the forbidden recipient set is never all of N.

Call a legal non-terminal state a *deadlock* if for every unfinished type j and every permutation, no recipient in S_j preserves the invariant. Proposition 52 shows deadlocks exist; Conjecture 54 asks only that the *root* avoid them. So the target theorem has the shape *there is a peel rule under which no deadlock is ever entered*, and Remark 53 is the first candidate for what that rule must enforce.

8.7 The sweep, and the rule it produces

The experiment this section has been calling for — run the reachability computation over the full $n = m = 3$ family — was finally run. At $n = m = 3$ a state is a workload profile, so there are at most $(2^m)^n = 512$ of them and reachability is decidable exactly.

Observation 56 (The sweep)**.** *Over the complete exhaustive $n = m = 3$ dichotomous family — all 9,880 instances — there is* no bad root*: from every root some sequence of peels and permutations reaches a terminal state with the invariant holding throughout. Extending to random instances at $n = 3$ with $m \leq 6$, $n = 4$ with $m \leq 4$ and $n = 5$ with $m = 3$ gives 0 bad roots as well (`update_32/peel_sweep.py`, `peel_general.py`).*

So Conjecture 54 is not refuted, and Approach 3 is alive. But reachability from the root is not a rule: dead ends are plentiful — 278, 1,319 and 1,633 of them at $n = 3$, $m = 4, 5, 6$ — and all three natural greedy rules walk into them (peel from an $\arg\max_i \ell(i)$ agent, peel the chore with the largest candidate set, peel from the most burdened residual workload, failing 81, 64 and 66 times respectively). What was missing was the rule. Remark 53 guessed it should enforce balance, and that guess is now confirmed in a sharp form.

Observation 57 (Balance characterises the dead ends)**.** *Say a state W admits a* balanced terminal *if some choice of owner $f(j) \in S_j$ for each chore induces bundle sizes differing by at most 1. Over 1,192,108 reachable legal states, the count of states that are dead yet admit a balanced terminal is* zero*. That is,*

$$W \text{ admits a balanced terminal} \implies W \text{ is live.}$$

The converse fails — 132,958 live states admit no balanced terminal — so the condition is sufficient, not necessary (`update_32/deadend_char.py`).

Admitting a balanced terminal is a degree-constrained bipartite feasibility question: with $m = qn + r$, it asks for an assignment of each chore j to an agent of S_j with r agents receiving $q + 1$ chores and $n - r$ receiving q. That is a transportation problem, decidable by one max-flow computation. So Observation 57 supplies something the earlier sections never had: a *checkable* sufficient condition for not being stuck.

The balance rule. Never peel in a way that destroys the existence of a balanced terminal.

The root admits a balanced terminal trivially, every S_j being all of N, so the rule can start. Whether it can be *followed* is the remaining question, and the answer so far is yes.

Observation 58 (The rule is complete, so far)**.** *Restricting the state graph to states that are both legal and balance-admitting, a terminal is still reachable from the root on every instance tested:* 0 *failures over* 305 *instances at* $n = 3$ *(*$m \leq 6$*),* $n = 4$ *(*$m \leq 4$*) and* $n = 5$ *(*$m = 3$*) (*`update_32/balance_rule.py`*).*

Conjecture 59 (Balance-rule reachability)**.** *Restricting the state graph to states that are both legal and balance-admitting, a terminal is reachable from the root.*

Remark 60 (The pointwise form is false)**.** The stronger and more convenient statement — *from every* legal, non-terminal, balance-admitting state some legal move leads to another such state — is refuted. Over 537,541 legal balance-admitting non-terminal states, 5 admit no legal balance-preserving peel and no legal permutation followed by one; the smallest is $n = 3$, $m = 4$ with $W = (\{g_1, g_2\}, \{g_3, g_4\}, \{g_3, g_4\})$. So the rule is not a local invariant that holds everywhere: those states are simply not reached from the root along the restricted graph, which is why Observation 58 sees no failure. Any proof must therefore argue about reachability from the root, not about arbitrary states.

Proposition 61. *Conjecture 59 implies Conjecture 54, hence Conjecture 2; and it yields a polynomial-time algorithm, each step choosing a peel by one max-flow feasibility test.*

Proof The root is legal, its graph being identically zero, and admits a balanced terminal, so the restricted graph is non-empty. Conjecture 59 supplies a path in it from the root to a terminal; every state on that path is legal, which is Conjecture 54, and that implies Conjecture 2 as already shown. Each step tests $O(nm)$ candidate peels, each by a transportation feasibility check on a bipartite graph with $n + m$ nodes. □

What a proof must supply, and what it need not. Two reductions narrow the target considerably.

Lemma 62 (Balance-preservation is free)**.** *Let W be non-terminal and admit a balanced terminal f. Then for every chore j with $|S_j| \geq 2$ and every $i \in S_j \setminus \{f(j)\}$, the state* $\mathrm{peel}(i, j)$ *still admits f.*

Proof Peeling removes i from S_j only, and $f(j) \neq i$ lies in S_j still, so f remains a legal choice function; the induced sizes are unchanged. □

So the balance half of the rule costs nothing: the entire content is *legality*. A legal balance-preserving peel exists at 529,579 of 537,541 states, and at 342,048 of them one such peel is *free* — no arc into the peeled agent rises at all, so legality is preserved automatically by Lemma 48. Free peels are thus the common case, and the open question is what to do at the remainder.

Second, the earlier guess at *why* balance certifies liveness is wrong and should not be reused. One would expect a balance-admitting state to admit a balanced terminal that is itself legal, reached by a legal schedule. Both halves fail: over 349,188 balance-admitting legal states, 76,220 (21.8%) admit *no* legal balanced terminal at all, and of those that do, 3,669 admit no legal ordering of the peels down to one (`update_33/why_balance_works.py`). Those states are live nonetheless — the legal terminal they reach is *unbalanced*. Balance is a certificate of liveness, not a description of the destination.

What the dead states do look like is visible directly. Every dead state has $\min_j \left|S_j\right| = 1$, and no state with $\min_j \left|S_j\right| \geq 2$ was ever dead; the smallest witnesses are

$$W = (\left\{g_2\right\}, \left\{g_2\right\}, \left\{g_1, g_3, g_4\right\}), \qquad c_i(W_i) = (0, 0, 2),$$

in which g_1, g_3, g_4 are already committed to agent 3, so every terminal below forces it to hold three chores at cost 2 while the others hold at most one — and that same over-commitment is exactly what makes a balanced terminal impossible. Death is over-commitment to one agent, and balance-admissibility is the checkable shadow of it.

8.8 When is a peel legal? A potential calculus

The legality invariant has a potential form, and stating it that way turns "which peels are safe" from a search into algebra.

Lemma 63 (Potential form of legality)**.** *A state W is legal if and only if there is $p \in \{0,1\}^n$ with*

$$w_W(i,k) \;\leq\; p_i - p_k \qquad \textit{for all } i \neq k.$$

Proof ($\Leftarrow$) Summing the inequality along a path from i_0 to i_r telescopes to $w(P) \leq p_{i_0} - p_{i_r} \leq 1$, and around a cycle to $w(C) \leq 0$. So there is no positive cycle and $\ell_W \leq 1$. ($\Rightarrow$) Take $p = \ell_W$, which lies in $\{0,1\}^n$ by legality and Observation 12. By definition of a heaviest path, $\ell(i) \geq w(i,k) + \ell(k)$, which is the inequality. □

Write $\mu_k := c_k(W_x) - c_k(W_x \setminus \left\{j\right\}) \in \{0,1\}$ as in Lemma 48, and call an arc (k,x) *tight* for p if $w_W(k,x) = p_k - p_x$. Since Lemma 48 changes only the arcs incident to x — raising those into x by μ_k and lowering those out of x by μ_x — the question of whether the *same* potential still certifies the result is immediate.

Proposition 64 (Tight-arc criterion)**.** *Let W be legal with potential p. Then p certifies* peel(x,j) *if and only if $\mu_k = 0$ for every $k \neq x$ whose arc (k,x) is tight for p.*

Proof The constraints not involving x are unchanged. Out of x, $w(x,k) - \mu_x \leq w(x,k) \leq p_x - p_k$ holds automatically. Into x, the requirement is $w(k,x) + \mu_k \leq p_k - p_x$; the slack $p_k - p_x - w(k,x)$ is a non-negative integer, so this holds exactly when $\mu_k = 0$ or the slack is at least 1, i.e. the arc is not tight. □

Two sufficient conditions follow, and they cover different ground. The first is the peel-side counterpart of Theorem 32.

Definition 65. *Call* peel(x,j) free *if $\mu_k := c_k(W_x) - c_k(W_x \setminus \left\{j\right\}) = 0$ for every $k \neq x$, i.e. chore j has zero marginal for every other agent on x's residual workload.*

Lemma 66 (Free peels are always legal)**.** *A free peel applied to a legal state yields a legal state.*

Proof By Lemma 48 the only arcs that change are those incident to x: $\Delta w(k,x) = +\mu_k = 0$ for $k \neq x$, and $\Delta w(x,k) = -\mu_x \leq 0$. So every arc weight weakly decreases. No positive-weight cycle can appear and no path weight can rise, so the invariant survives. □

Immediate from Proposition 64, since $\mu_k = 0$ for every $k \neq x$. The second condition is new and is proved by *moving* the potential rather than keeping it.

Lemma 67 (Paid-agent peel)**.** *Let W be legal with potential p. If $p_x = 1$ and $\mu_x = 1$ — the peeling agent is paid, and the chore is costly to it on its own residual workload — then* $\mathrm{peel}(x, j)$ *is legal, whatever the other marginals.*

Proof Take $p'_x := 0$ and $p'_i := p_i$ for $i \neq x$; then $p' \in \{0,1\}^n$. The constraints not involving x are unchanged and p' agrees with p off x. Into x: $w(k,x) \leq p_k - p_x = p_k - 1$, so $w(k,x) + \mu_k \leq p_k - 1 + 1 = p_k = p'_k - p'_x$. Out of x: $w(x,k) \leq p_x - p_k = 1 - p_k$, so $w(x,k) - \mu_x \leq 1 - p_k - 1 = -p_k = p'_x - p'_k$. So p' certifies the new state, which is legal by Lemma 63. □

The two conditions are genuinely complementary: a free peel constrains the *other* agents' marginals and says nothing about x, while Lemma 67 constrains only x and leaves the others free. Both were checked against the implementation — 670,927 free peels and 693,276 paid-agent peels, with 0 illegal outcomes between them — and together with Lemma 62 they give a peel that is *provably* safe and balance-preserving at

409,003 of 477,922 reachable non-terminal states (85.6%),

against 58.7% for free peels alone. The residual is 68,919 states at which neither condition fires; by Proposition 64 those are exactly the states where every peelable chore has a tight incoming arc carrying marginal 1 into every unpaid candidate owner.

8.9 What the two lemmas cannot do

Corollary 68 (Structure of a stuck state)**.** *If W is stuck then for every peelable chore j and every admissible $x \in S_j$, either $\ell_W(x) = 0$ or $\mu_x = 0$.*

Proof Contrapositive of Lemma 67. □

That is the whole of what the two safety lemmas say about stuck states, and it is not enough: they cannot certify a schedule at all.

Proposition 69 (The two lemmas do not suffice)**.** *Restricting the peel process to steps certified by Lemma 66 or Lemma 67, a terminal is reachable from the root on only* 48 *of* 191 *test instances. Soundness is not at issue: over* 1,558,435 *certified peels,* 0 *were illegal.*

Remark 70 (A correction: the potential is a free choice)**.** An earlier count reported 0 of 191. That test evaluated Lemma 67 at the single potential $p = \ell_W$, whereas the lemma quantifies over *any* $S \in \mathcal{P}(W)$. The difference is not cosmetic: at the root $\ell \equiv 0$, so the lemma appears never to fire, yet $N \in \mathcal{P}(\mathrm{root})$ by Observation 73 and with $S = N$ every agent is paid, so the lemma fires whenever some $\mu_x = 1$. That is exactly consistent with Proposition 71, since $\mu_x = 1$ forces $\mu_x \geq \max_k \mu_k$. Reading a lemma at one witness rather than at its existential is the error; the corrected figure is 48.

The root is nevertheless settled exactly, and the answer is a genuine constraint on every schedule.

Proposition 71 (The first peel)**.** *From the root,* peel(x, j) *is legal if and only if* $\mu_x \geq \mu_k$ *for every* $k \neq x$*: the chore must be taken from an agent who finds it at least as costly as everybody else does.*

Proof After the peel, $w(k, x) = \mu_k$, $w(x, k) = -\mu_x$, and $w(i, k) = 0$ for $i, k \neq x$. A potential p' must satisfy $0 \leq p'_i - p'_k$ both ways for $i, k \neq x$, forcing $p'_i = c$ off x; then the remaining constraints are $\mu_k \leq c - p'_x$ and $-\mu_x \leq p'_x - c$, i.e. $c - \mu_x \leq p'_x \leq c - \max_{k \neq x} \mu_k$. This interval contains an integer iff $\mu_x \geq \max_{k \neq x} \mu_k$, and when it does, $c = 1$ with $p'_x = 1 - \max_k \mu_k$ works. Conversely if $\mu_x < \max_k \mu_k$ the interval is empty, so no potential certifies the state and Lemma 63 makes it illegal. □

Remark 72 (The generalisation fails)**.** Proposition 71 is exact only at the root, where all arcs vanish. The natural extension — always peel a chore from an owner of maximal marginal — is neither a lemma nor a usable rule: of 2,912,968 dominant-marginal peels, 272,757 (9.4%) are illegal, and greedy descent under the rule reaches a terminal on only 10 of 191 instances.

8.10 Peel dynamics on the admissible paid sets

Remark 70 shows the right object is not a single potential but the whole admissible set, so it is worth naming and studying in its own right. Identify $p \in \{0, 1\}^n$ with the paid set $S = \{i : p_i = 1\}$ and put

$$\mathcal{P}(W) \ := \ \{ S \subseteq N \ : \ w_W(i, k) \leq \lambda_S(i, k) \ \text{ for all } i \neq k \},$$

where

$$\lambda_S(i, k) := \begin{cases} 0 & i, k \text{ on the same side of } S, \\ 1 & i \in S, \ k \notin S, \\ -1 & i \notin S, \ k \in S, \end{cases}$$

so that W is legal iff $\mathcal{P}(W) \neq \emptyset$, by Lemma 63. This is the two-tier data of Proposition 15 carried along the peel process.

Observation 73. *At the root every arc weight is* 0*, so* $\lambda_S(i, k) = -1$ *is violated for every pair with* $i \notin S$*,* $k \in S$*; hence* $\mathcal{P}(\mathit{root}) = \{\emptyset, N\}$ *— only the two constant potentials. Verified on* 169 *instances without exception.*

The framework reproduces what is already proved. After peel(x, j) from the root the arcs are $w(k, x) = \mu_k$, $w(x, k) = -\mu_x$ and 0 elsewhere, and $S = N \setminus \{x\}$ is admissible exactly when $\mu_x \geq 1$, which with the free case is Proposition 71.

What the framework adds is the general safety criterion, of which Proposition 64 is the special case $S = \{i : \ell(i) = 1\}$.

Proposition 74 (Peel criterion)**.** *For any* $S \subseteq N$*,* $S \in \mathcal{P}(W')$ *if and only if*

$$\begin{aligned} &\text{(i)} \quad w(i, k) \leq \lambda_S(i, k) && \text{for } i, k \neq x, \\ &\text{(ii)} \quad w(i, x) + \mu_i \leq \lambda_S(i, x) && \text{for } i \neq x, \\ &\text{(iii)} \quad w(x, k) - \mu_x \leq \lambda_S(x, k) && \text{for } k \neq x. \end{aligned}$$

If moreover $S \in \mathcal{P}(W)$*, then (i) and (iii) hold automatically, and only the in-arc condition (ii) remains.*

Proof The three groups are exactly the constraints on the arcs of W', which by Lemma 48 are the old arcs off x, the old in-arcs raised by μ_i, and the old out-arcs lowered by μ_x. If $S \in \mathcal{P}(W)$ then (i) is one of its defining constraints and (iii) follows from $w(x,k) - \mu_x \leq w(x,k) \leq \lambda_S(x,k)$. □

Remark 75 (A correction)**.** An earlier version of Proposition 74 asserted the biconditional with (iii) omitted, for arbitrary S. That is false: (iii) is automatic only for S already admissible at W, and the whole point of working with $\mathcal{P}$ is that new sets appear. The corrected form was checked on 21,451,744 instances of the criterion with 0 violations.

The correction is what identifies where a further criterion can come from.

Lemma 76 (New paid sets require a costly chore)**.** *If $\mu_x = 0$ then $\mathcal{P}(W') \subseteq \mathcal{P}(W)$. If $\mu_x = 1$, every $S \in \mathcal{P}(W') \setminus \mathcal{P}(W)$ violates, at W, exactly one kind of constraint: an out-arc of x, and by exactly one unit.*

Proof Let $S \in \mathcal{P}(W')$. By (ii), $w(i,x) \leq \lambda_S(i,x) - \mu_i \leq \lambda_S(i,x)$, so the in-arc constraints already held at W; by (i) so did the constraints off x. Hence the only constraint S can violate at W is an out-arc $w(x,k) \leq \lambda_S(x,k)$, and (iii) bounds the violation by μ_x. If $\mu_x = 0$ there is no room for one. □

Verified: over 418,662 peels at which a new paid set appears, $\mu_x = 1$ in every case.

Two consequences worth having. First, legality of a peel is decided entirely by the arcs *into* the peeled agent: peeling is safe exactly to the extent that the agents who envy x can absorb μ_i. Second — and this is why $\mathcal{P}$ is the right invariant rather than ℓ — $\mathcal{P}(W') \nsubseteq \mathcal{P}(W)$ in general: a set S failing an out-arc constraint at W may satisfy it at W', since that arc dropped by μ_x. New paid sets can appear. Lemma 67 is exactly an instance: it produces the set $S \setminus \{x\}$, which need not lie in $\mathcal{P}(W)$.

8.11 The admissible paid sets form a lattice

So far $\mathcal{P}(W)$ has been used only as a feasibility object — non-empty or not. It has more structure than that, and the structure removes a search.

Theorem 77 (Lattice)**.** *$\mathcal{P}(W)$ is closed under union and intersection. Hence, when non-empty, it has a unique maximum element $S_{\max}$ and a unique minimum element $S_{\min}$.*

Proof Work with potentials. Let $p, q \in \mathcal{P}(W)$ and let $r := \max(p,q)$ coordinatewise; we must show $w(i,k) \leq r_i - r_k$. If r agrees with p at both i and k, or with q at both, this is one of the hypotheses. Otherwise, say $r_i = p_i \geq q_i$ and $r_k = q_k \geq p_k$; then

$$w(i,k) \;\leq\; q_i - q_k \;\leq\; p_i - q_k \;=\; r_i - r_k,$$

using the q-constraint and $q_i \leq p_i$. The remaining mixed case is symmetric. For $\min(p,q)$ the same argument applies with the roles reversed: if the minimum is p_i at i and q_k at k then $p_k \geq q_k$ gives $w(i,k) \leq p_i - p_k \leq p_i - q_k$. □

This is the standard fact that the feasible set of a difference-constraint system is a lattice, but it has two consequences here that are worth naming.

Corollary 78. $S_{\min} = \{i : \ell_W(i) = 1\}$*: the longest-path potential is the* smallest *admissible paid set.*

Corollary 79 (The canonical potential for peeling)**.** *Some admissible paid set contains x if and only if $x \in S_{\max}$. Consequently Lemma 67 fires for x exactly when $\mu_x = 1$ and $x \in S_{\max}$, and Lemma 85 need only ever be run at $S = S_{\max}$, no larger admissible set being available to absorb blockers.*

Verified on 140,785 legal states and 492,100 pairs: closure under union and intersection with 0 violations, and both corollaries with 0 violations.

Corollary 78 also explains the error recorded in Remark 70. Evaluating Lemma 67 at $p = \ell_W$ is evaluating it at $S_{\min}$ — the admissible set with the *fewest* paid agents, hence the worst possible choice for a lemma whose hypothesis is $x \in S$. The right representative is $S_{\max}$, and by Corollary 79 it is the only one that ever needs to be computed.

8.12 The dynamics of $S_{\max}$

Theorem 77 supplies a canonical element, so one may ask how it evolves rather than merely whether it exists. Write $F(W) := S_{\max}(W)$.

Proposition 80. *If $\mu_x = 0$ then $F(W') \subseteq F(W)$.*

Proof By Lemma 76, $\mu_x = 0$ gives $\mathcal{P}(W') \subseteq \mathcal{P}(W)$, and the maximum of a subfamily of a lattice is contained in the maximum of the family. □

The maximum of a difference-constraint system has a closed form, dual to the one for the minimum.

Proposition 81 (What $S_{\max}$ is)**.** *Let $\ell^*(k)$ denote the heaviest weight of a directed path* ending *at k, the empty path counting as 0. Then*

$$S_{\max}(W) \;=\; \{\, k \;:\; \ell^*(k) \le 0 \,\}\,.$$

Proof The constraints $w(i,k) \le p_i - p_k$ read $p_k \le p_i - w(i,k)$, so iterating along a path ending at k gives $p_k \le p_{\text{start}} - \ell^*(k) \le 1 - \ell^*(k)$. Hence $\ell^*(k) \ge 1$ forces $p_k = 0$. Conversely the vector $p_k := 1$ if $\ell^*(k) \le 0$ and 0 otherwise is admissible: for an arc (i,k), if $p_k = 1$ then $\ell^*(k) \le 0$, so $\ell^*(i) + w(i,k) \le \ell^*(k) \le 0$ and, as $\ell^*(i) \ge 0$, $w(i,k) \le 0 = p_i - p_k$ when $p_i = 1$; the case $p_i = 0$ gives $\ell^*(i) \ge 1$, whence $w(i,k) \le \ell^*(k) - \ell^*(i) \le -1$ as required. □

Compare Corollary 78: $S_{\min}$ is read off the paths *leaving* each agent, $S_{\max}$ off the paths *entering* it. That duality is what makes the $\mu_x = 1$ case tractable.

Lemma 82 (Monotonicity off the peeled agent)**.** *If $\mu_x = 1$ then $S_{\max}(W) \setminus \{x\} \subseteq S_{\max}(W')$.*

Proof Fix $k \ne x$ with $\ell^*(k) \le 0$, and let P be any path ending at k in W'. If P avoids x its weight is unchanged. If P passes through x, it uses one arc into x, which rose by some $\mu_i \le 1$, and one arc out of x, which fell by $\mu_x = 1$; the net change is $\mu_i - 1 \le 0$. If P starts at x it uses only a lowered arc. In every case the weight does not increase, so the heaviest path ending at k in W' still has weight at most $\ell^*(k) \le 0$, and Proposition 81 places k in $S_{\max}(W')$. □

Verified: the closed form on 93,819 states and the monotonicity lemma on 264,139 peels with $\mu_x = 1$, both with 0 violations. Together with Proposition 80 this accounts for all of the observed behaviour except one case: $\mu_x = 1$ with $x \in S_{\max}(W)$ leaving, where $S_{\max}(W')$ contains $S_{\max}(W) \setminus \{x\}$ and might in principle also acquire a new element, which would make the two incomparable. It cannot.

Theorem 83 (F never moves sideways)**.** *Let W be legal and let $W' = \mathrm{peel}(x, j)$ also be legal. Then $S_{\max}(W)$ and $S_{\max}(W')$ are comparable under inclusion. More precisely, in the case $\mu_x = 1$ with $x \in S_{\max}(W)$ and $x \notin S_{\max}(W')$ one has $S_{\max}(W') = S_{\max}(W) \setminus \{x\}$ exactly.*

Proof If $\mu_x = 0$, Proposition 80 gives $S_{\max}(W') \subseteq S_{\max}(W)$. So assume $\mu_x = 1$; Lemma 82 gives $S_{\max}(W) \setminus \{x\} \subseteq S_{\max}(W')$. If $x \in S_{\max}(W')$, or if $x \notin S_{\max}(W)$, that containment already reads $S_{\max}(W) \subseteq S_{\max}(W')$ and we are done. There remains the case $x \in S_{\max}(W)$, $x \notin S_{\max}(W')$, where it suffices to show no new element enters.

Suppose some $k \notin S_{\max}(W)$ had $k \in S_{\max}(W')$; note $k \neq x$. By Proposition 81 there is a simple path Q ending at k with $w_W(Q) \geq 1$, while $w_{W'}(Q) \leq 0$. Only arcs incident to x changed, so Q meets x, and being simple it meets x once. Write $Q = Q_1 \cdot Q_2$, split at x, with Q_1 arriving from some i' (possibly empty, if Q starts at x). By Lemma 48,

$$w_{W'}(Q) \;=\; w_W(Q) + \mu_{i'} - \mu_x,$$

so $w_{W'}(Q) \leq 0$ and $w_W(Q) \geq 1$ force $\mu_{i'} = 0$ and $w_W(Q) = 1$. Since $x \in S_{\max}(W)$, Proposition 81 gives $w_W(Q_1) \leq 0$, whence $w_W(Q_2) \geq 1$ and, the first arc of Q_2 having dropped by $\mu_x = 1$, $w_{W'}(Q_2) \geq 0$.

Since $x \notin S_{\max}(W')$ there is a path P ending at x with $w_{W'}(P) \geq 1$. Concatenating, $P \cdot Q_2$ is a walk ending at k of weight at least 1 in W'. As W' is legal its envy graph has no positive cycle, so a heaviest walk may be taken simple; hence there is a simple path ending at k of positive weight, contradicting $k \in S_{\max}(W')$ by Proposition 81. ◻

Verified on 755,051 peels with 0 violations of the conclusion or of any step of the case split; the critical case — $\mu_x = 1$ with x leaving $S_{\max}$ — arose 212,930 times, so the theorem is not vacuous there (`update_41/comparability.py`).

So a peel moves the canonical paid set along a *chain* in the lattice. Two cautions. The jump is not bounded by one: over the same sweeps the symmetric difference reached 4, and $F(W') \subseteq F(W) \cup \{x\}$ held only 87.9% of the time, so peels are not single lattice flips. And F is *not* a monovariant: it grows on roughly one peel in eight, so it supplies no monotone progress. Termination was never the difficulty in any case, since each peel deletes an element and $\sum_i |W_i|$ already decreases; the difficulty is legality.

Remark 84 (The canonical-support conjecture is false)**.** It is tempting to look for a bridge to the transportation side of the form: *every reachable legal balance-admitting state admits a balanced terminal f with $f(j) \in S_{\max}$ whenever $S_j \cap S_{\max} \neq \emptyset$*, so that every chore whose candidates include a paid agent could be given to one. This is false, and not marginally: over 117,141 balance-admitting states it fails on 59,623, a little over half. The mixed case is common enough for the statement to have content — $\left|S_j \cap S_{\max}\right|$ was 0 on 14,082 chores and strictly between 0 and $\left|S_j\right|$ on most of the rest — so the failure is not a vacuity. A bridge between balanced terminals and the admissible potentials, if one exists, does not take this form.

Lemma 85 (Slack-transfer peel)**.** *Let $S \in \mathcal{P}(W)$ with $x \notin S$, and let*

$$T := \left\{ i \neq x \;:\; w(i,x) = \lambda_S(i,x) \text{ and } \mu_i = 1 \right\}$$

be the agents whose in-arc blocks the peel. Suppose

(1) $w(k,i) \le \lambda_S(k,i) - 1$ for every $i \in T$ and every $k \notin T \cup \{i\}$, and

(2) $w(x,k) - \mu_x \le -1$ for every $k \in T$.

Then $S \cup T$ certifies peel(x, j)*, which is therefore legal.*

Proof Write $S' := S \cup T$; note $x \notin S'$. Moving i from outside S to inside raises $\lambda(i,\cdot)$ by one and lowers $\lambda(\cdot,i)$ by one, and leaves all pairs within T and all pairs avoiding T unchanged. Condition (1) is exactly the slack needed for the lowered constraints, so $S' \in \mathcal{P}(W)$ apart from the arcs at x, giving (i) of Proposition 74. For (ii): if $i \in T$ then $\lambda_{S'}(i,x) = \lambda_S(i,x) + 1$ absorbs $\mu_i = 1$; if $i \notin T$ then either the arc is not tight, leaving slack for $\mu_i \le 1$, or $\mu_i = 0$. For (iii): out-arcs into $S' \setminus S = T$ have $\lambda_{S'}(x,k) = -1$, which is condition (2); out-arcs elsewhere are unchanged and drop by μ_x. □

Verified: 46,723 peels certified this way, none illegal. The three sufficient conditions together — free, paid-agent, slack-transfer, each with S ranging over all of $\mathcal{P}(W)$ — give a provably safe balance-preserving peel at

$$166{,}480 \text{ of } 174{,}630 \text{ reachable non-terminal states} \quad (95.3\%),$$

with the free and paid-agent pair supplying 94.8% and slack transfer the remaining 0.5%. The residual is 8,150 states, 4.7%.

Remark 86 (Random testing has run out of power)**.** Greedy rules driven by $|\mathcal{P}|$ were tried, maximising it at each step and — as a control — minimising it. Both terminate on all 169 instances tested, so the criterion is not what makes them work; on random instances almost any legal balance-preserving greedy succeeds, because stuck states are of order 10^{-4} of the reachable set. Further progress on Conjecture 59 has to be mathematical, not experimental: what is needed is a proof that some choice keeps $\mathcal{P}$ non-empty, and Proposition 74 says such a proof only ever has to control the in-arcs at the peeled agent.

So closing Conjecture 59 needs a safety criterion strictly stronger than Lemmas 66 and 67. Proposition 64 says what such a criterion must engage with: it characterises legality for the potential $p = \ell_W$ only, and both lemmas are the two ways of exploiting it that keep the potential fixed or shift one coordinate. A criterion that closes the residual will have to reason about *which potentials are available* at a state — the set $\left\{p \in \{0,1\}^n : w(i,k) \le p_i - p_k\right\}$, equivalently the set of admissible paid sets — rather than about ℓ alone. That set is exactly the two-tier data of Proposition 15, which is the one piece of machinery in this report not yet brought to bear on the peel frame.

Observation 87 (The residual, and a correction)**.** *Of* 522,117 *reachable non-terminal states,* 215,416 *(*41.3%*) have no free balance-preserving peel. Of those,* 207,543 *still admit a non-free legal* *one and* 7,836 *admit one after a permutation — but* 37 *admit* none at all. *So progress-stuck states are reachable, and Conjecture 59 does not follow from a greedy-safety argument.*

An earlier count of 0 *stuck reachable states was wrong: it treated a permutation as a move out of a stuck state, but permutations remove nothing, so a state whose only legal moves are permutations makes no progress and is stuck.*

Two things survive this. The free-first strategy — take a free balance-preserving peel when one exists, otherwise any legal one, permuting if necessary — reached a terminal on every instance tested, so the 37 stuck states are reachable but *avoided* in practice. And they are rare, at 7×10^{-5} of reachable states. What is missing is a proof that some rule provably avoids them, and that is now the whole of the gap: the free case is settled by Lemma 66, and the non-free case is settled empirically but not in principle.

Remark 88 (Dropping balance does not help, and the commitment predicate narrowly fails)**.** Since Conjecture 59 is strictly stronger than Conjecture 2, it is worth asking whether Conjecture 54 is easier without the balance restriction. It is not: Conjecture 51 — from every legal non-terminal state some legal move exists — is already refuted by Proposition 52, so the wider graph carries the same dead ends and the same obstruction.

What Remark 53 suggests is more promising, because it is a predicate on the *state* alone, which is what Remark 89 shows a rule must be. Writing $C_i(W) := \{j : S_j = \{i\}\}$ for the chores whose owner is already decided, consider

$$\text{(COMMIT)} \qquad c_i\big(C_i(W)\big) \le 1 \quad \text{for every agent } i.$$

It holds at the root, where every C_i is empty, and it very nearly characterises liveness: over 151,374 legal reachable states, 239 of the 256 dead states violate it. But it is not sufficient — 17 dead states satisfy it, the smallest at $n = 3$, $m = 4$ with $W = (\{g_2\}, \{g_2\}, \{g_1, g_3, g_4\})$ — so it cannot serve as Φ (`update_43/commitment.py`).

The figures above are from a sweep restricted to instances whose *entire* reachable set was enumerated, 88 of them with none truncated. An earlier run capped the search at 6,000 states per instance and reported 154,019 dead states satisfying (COMMIT); that was an artifact of the cap, since the backward liveness pass then runs on a truncated graph and misclassifies states whose route to a terminal leaves the explored set.

Remark 89 (The avoiding choice is not locally legible)**.** Theorem 83 suggests looking for the rule in the dynamics of $S_{\max}$: if a peel moves it along a chain, a stuck state might be entered only by a distinguished kind of step. It is not. Taking every predecessor of a stuck state — a state admitting a legal balance-preserving peel into one, of which there are 171 — and comparing the peels that enter a stuck state against those that do not, no feature available to a rule separates them:

feature	separates bad from good, per predecessor
whether $S_{\max}$ shrinks, stays or grows	2/171
whether the peeled agent lies in $S_{\max}$	0/171
μ_x	2/171
the change in $\lvert S_{\max}\rvert$	7/171
the number of chores still peelable	0/171

Indeed the same signature occurs on both sides: the profile (shrinks, $x \in S_{\max}$, $\mu_x = 1$) arises 102 times on peels into a stuck state and 35 times on peels away from one. Whatever distinguishes the schedules that avoid the stuck states is not visible in the state and the move (`update_42/predecessors.py`).

Qualifications. First, Conjecture 59 is *stronger* than Conjecture 2, as Remark 55 already warns for Conjecture 54: a good allocation may exist while no legal schedule reaches one. Second, the sufficient condition is not tight — 132,958 live states fail it — so the rule discards live states, and its completeness is a genuine claim rather than a bookkeeping identity. What makes it worth more than the earlier strengthenings is that it comes with an *explicit, polynomial-time checkable rule*: unlike the descent lemma of Section 12, whose witnessing move was shown not to be local, here the move to make is decided by a flow computation, and the open statement is only that a legal such move always exists.

8.13 The type-ordered restriction

One restriction of the peel process survives every test so far and shrinks the search substantially. Present the chores in *blocks*: decide the owner of a_1 completely, then a_2, and so on. Coverage is automatic in the peel frame — a peel deletes j from W_x, so the same pair (x, j) cannot be peeled twice — and after r complete blocks the state is $W_i = \mathcal{A}_i^{(r)} \cup R$ with R the undecided chores, which is literally the conditioned-remainder principle of Remark 50. A state is then just a triple (partial allocation, current chore, set of agents already relieved of it), which is a far smaller space than the general peel process.

Conjecture 90 (Type-ordered reachability — **open**). *From the root, some chore order π, together with some choice of owner and within-block sequence for each chore, reaches a terminal state with the invariant holding at every intermediate state.*

Conjecture 90 implies Conjecture 54 implies Conjecture 2, and it is a strictly smaller search problem than either. The evidence, from `typeorder.py` and `stress.py`, is that the restriction costs nothing: across the witness of Theorem 36 and 890 random dichotomous instances at $n \in \{3, 4\}$, $m \leq 4$, *every* instance that admits a legal schedule at all admits a type-ordered one. The chore *order* is not free — in 8/120 instances at $n = m = 3$, 16/60 at $n = 3, m = 4$ and 8/60 at $n = 4, m = 3$ only some orders work — but in every instance tested some order does. Note that the schedule exhibited in Section 8.4 is *not* type-ordered, since it interleaves a_1 and a_2; the table says a type-ordered one exists for that instance anyway, which was not obvious.

What does not survive is the natural recipient rule to pair with it.

Proposition 91 (arg max recipient selection is insufficient — **refuted**). *Giving each copy to an agent of maximum current subsidy does not suffice, even under the type-ordering.*

Proof Take $n = 3$, $m = 2$ with the costs

$$c_1(\emptyset) = 0,\ c_1(\{a_1\}) = c_1(\{a_2\}) = 0,\ c_1(M) = 1; \qquad c_2(S) = \min(|S|, 1); \qquad c_3 = c_1.$$

All three are dichotomous (Theorem 7). Exhaustive search over both chore orders, both roles and all within-block sequences finds no legal type-ordered schedule in which every recipient lies in $\arg\max_i \ell_W(i)$. Dropping only the arg max restriction, the schedule that relieves agent 1 of a_1, then agent 3 of a_1, then agent 1 of a_2, then agent 3 of a_2 is legal and terminates at $\mathcal{A} = (\emptyset, \{a_1, a_2\}, \emptyset)$ with $\ell = (0, 1, 0)$. □

This is the signal of Remark 53 again — a subsidy-based recipient rule alone is not enough — now confirmed *inside* the type-ordered restriction, which is the sharper statement. Combined with Theorem 95, which says the recipient in the hard case has zero marginal, the rule provably cannot be read off the subsidy vector: it must see the residual workload, which is exactly what the conditioned-remainder arcs already encode.

8.14 A subclass that may fall first — conjectural

On type space the dual of a *supermodular* dichotomous cost is submodular with binary marginals, that is, a matroid rank function; lifting through τ to the copies is the parallel extension of that matroid, still a matroid rank function. Additive costs dualise to partition-matroid ranks. If that is right — and the parallel-extension step in particular deserves a full proof before being relied on — then supermodular dichotomous chores are exactly matroid-rank goods coverage problems, and the witness of Theorem 36 lies entirely inside this class, since c_1 dualises to the rank of $U_{2,3}$, parallel extended.

Matroid rank is where the machinery of the surrounding literature is strongest, through the exchange properties used in [BSY23, BEF21]. That makes Conjecture 2 restricted to supermodular dichotomous costs, attacked by basis exchange in the parallel-copy matroid, the natural intermediate target — sitting between Theorem 19, which is closed, and the full conjecture.

9 Approach 4: Staged Allocation with Escalating Type Costs

The idea. Corollary 45 reduces Conjecture 2 to a single constraint on the replica instance $\hat{I}$: find an allocation in which no agent holds two copies of one type. Each type b_j has $n - 1$ copies, so a coverage allocation is exactly one that, for every type, spreads its $n - 1$ copies over $n - 1$ *distinct* agents.

That suggests allocating in *stages*. Deal out all $n - 1$ copies of b_1, then all $n - 1$ copies of b_2, and so on. Within a stage the copies must land on distinct agents; across stages nothing is required. To make a black-box algorithm respect that, arrange the costs so that each type is far costlier than the one before,

$$b_1 \lll b_2 \lll \cdots \lll b_m,$$

the intention being that an agent who has just taken a copy of the current type is thereby so heavily loaded — in the goods reading of the replica, so well-supplied — that the algorithm's selection rule will not return to her while copies of that same type remain. Coverage would then be enforced by the *schedule* rather than checked afterwards.

Why it looks promising. The staging really does make coverage automatic: if each stage places its $n - 1$ copies on distinct agents, the result is a coverage allocation by construction, with no constraint to verify at the end. And the selection rule of [BKNS22] does have the shape the intuition wants — it hands the next item to a maximally subsidised agent, and an agent who has just received something valuable is not maximally subsidised. So the mechanism being appealed to exists.

Why it fails. The escalation separates *types*, but the duplicate problem lives *within* a type. Inside a stage all $n-1$ copies of b_j are the same object, so no cost assigned to types can distinguish the copy an agent already holds from the copy she is about to be given. The formal statement of that is already available:

Theorem 92. *In $\hat{I}$ the marginal value of a second copy of a type is exactly* 0*, for every agent and every set: if b has type j and $j \in \tau(S)$ then $\hat{v}_i(S \cup \{b\}) - \hat{v}_i(S) = 0$.*

Proof $\tau(S \cup \{b\}) = \tau(S)$, and $\hat{v}_i$ factors through τ. ◻

So an agent holding a copy of b_j is *indifferent* to a second one, not averse to it. Indifference is exactly the wrong sign for the intuition: the selection rule is not repelled by the duplicate, it simply gains nothing from avoiding it, and nothing in the invariant of [BKNS22] forbids handing it over. Escalating b_j against b_{j-1} does not touch this, because both copies in question are copies of b_j.

Two further obstacles stand behind that one, and each is fatal on its own.

Leaving the class. Making one type "much costlier" than another requires weights growing faster than 1 per item, which leaves the dichotomous class — every marginal must lie in $\{0, 1\}$ — and so forfeits the very theorem the staging was meant to invoke. There is no room inside the hypothesis for an escalation.

No weighting works even if the class is abandoned. Suppose we allow arbitrary reweightings u_i of the replica and ask only that they separate the coverage allocations from the rest. They cannot.

Theorem 93 (No-go)**.** *There is a negative dichotomous instance on which* no *faithful reweighting separates: $n = 3$, $m = 2$, $c_1 = c_2 = c_3 = |\cdot|$.*

Theorem 94. *On the same instance separation fails even if faithfulness is dropped entirely. Among all* 990 *dichotomous valuations on the four replica copies, none of the* 990^3 *triples* (u_1, u_2, u_3) *excludes all twelve non-coverage allocations of the shape "one duplicated pair plus two singletons". Exhaustive, by* `dupsep.py`*: collapsing the* 990 *valuations by their behaviour on the critical family leaves* 30 *classes, and* 0 *class-triples survive.*

Steering the algorithm instead of the numbers. One might try to keep the valuations and instead bias the selection rule, so that the staging is enforced procedurally. That fails for a matching reason.

Theorem 95. *Suppose the algorithm of [BKNS22] reaches its* FindSink *branch, i.e.* $(\mathcal{A}, p)$ *is not extendable with g. Then every $\ell \in \mathrm{M}(p)$ satisfies $v_\ell(A_\ell \cup \{g\}) - v_\ell(A_\ell) = 0$.*

Corollary 96. *Let s be the agent returned by* FindSink *and* $\mathcal{A}' = (A_1, \ldots, A_s \cup \{g\}, \ldots, A_n)$*. Then* $\ell_{\mathcal{A}'}(s) = \ell_{\mathcal{A}}(s)$*.*

In the branch that matters the recipient's marginal is already 0 — the very weights the staging wants to inflate are the ones the algorithm has already driven to zero — so no inflation can change which agent is selected. The intuition that "receiving a copy makes an agent ineligible" is precisely what Theorem 95 denies.

9.1 Why both fail: holder-blindness

Both failures have one cause. A price on duplicates bites only through the arcs of the envy graph, and there it cancels: whatever an agent is charged for the duplicates she holds lowers every arc *out of* her by the same amount that the charge raises the arcs *into* her from everyone else. Around a cycle the two contributions annihilate, so a price can disqualify an allocation only if the agent who *holds* the duplicate does not feel it while somebody else does. But the prices are fixed before the allocation is chosen, and it is the allocation that decides who holds the duplicate. Requiring the holder to be blind for every possible holder forces the price to zero.

Remark 97 (Verdict)**. Ruled out:** enforcing coverage by pricing — whether by escalating types across stages, or by any reweighting of the replica (Theorem 94) — and enforcing it by biasing the selection rule (Theorem 95). The staging intuition is sound about *what* a coverage allocation looks like, and wrong about *how* to obtain one: the distinction it needs is between two identical objects, and prices cannot make that distinction.

Not ruled out: enforcing coverage *structurally*. If the copies of a type are simply not allowed to land on the same agent, coverage holds by construction and no pricing is needed. That is exactly what the peel process of Section 8.3 does — a peel deletes j from W_x, so the same pair (x, j) can never be peeled twice — and it is why Section 8 inherits the staging intuition without inheriting its failure.

The general shape of the failure is the same as in Section 6: *a rule fixed ex ante against a quantity determined ex post*. There the owner was fixed before its consequences were visible; here the price must be levied on an agent who cannot be identified in advance.

10 Approach 5: Agent Induction and the Balance Invariant

Every approach so far inducts on *items*. Inducting on *agents* instead is attractive for chores because a newcomer starts with an empty bundle, which is the cheapest bundle there is, so the newcomer is the natural sink and the orientation problem of Section 6 does not arise. This section sets up what such an induction would have to carry, and shows that the natural candidate fails.

10.1 What the induction would have to carry

Corollary 35 says that if $\mathcal{A}$ is envy-freeable and every arc satisfies $w_{\mathcal{A}}(i, j) \geq -1$ — equivalently $c_i(A_j) \leq c_i(A_i) + 1$ — then $p^* \in \{0, 1\}^n$. The useful feature of that condition for an induction on agents is that it is *assignment-invariant*: it constrains only the family of bundles, not which agent holds which. That suggests carrying the family rather than the allocation.

Definition 98 (Uniformly balanced family)**.** *A family of n disjoint bundles covering M is* uniformly balanced *if every agent values all n bundles within one unit of each other:* $\max_t c_i(B_t) - \min_t c_i(B_t) \le 1$ *for every $i \in N$.*

Proposition 99. *If an instance admits a uniformly balanced family, then Conjecture 2 holds for it.*

Proof Assign the bundles to agents by a maximum-weight matching. By Theorem 10 the resulting allocation is envy-freeable, and by uniform balance every arc satisfies $w_{\mathcal{A}}(i, j) = c_i(A_i) - c_i(A_j) \ge -1$ — the bound holds for *any* assignment, so in particular for this one. Corollary 35 then gives $p^* \in \{0, 1\}^n$, and Observation 12 plus the vanishing of ℓ at the endpoint of a heaviest path give a total of at most $n - 1$. □

This is exactly the shape an agent induction wants: a purely combinatorial, schedule-free, coverage-free statement, with the step from n to $n+1$ agents being a refinement of the family. It also holds on every hard instance the project has accumulated, and on the full exhaustive family at $n = m = 3$ — all 9,880 instances, with the implication chain of Proposition 99 verified end to end on each (`routeA.py`). It is nevertheless false.

10.2 The invariant is unreachable

Proposition 100. *There is a negative dichotomous instance with $n = 3$ and $m = 4$, with binary additive costs, admitting no uniformly balanced family.*

Proof Take $c_i(S) = |S \cap D_i|$ with

$$D_1 = \{g_1, g_2\}, \qquad D_2 = \{g_1, g_3, g_4\}, \qquad D_3 = \{g_2, g_3, g_4\}.$$

Uniform balance for agent i means the elements of D_i are spread across the three bundles with counts of range at most 1. Three elements over three bundles have range at most 1 only in the pattern $(1, 1, 1)$, so D_2 forces g_1, g_3, g_4 into three *different* bundles, one each, and D_3 likewise forces g_2, g_3, g_4 into three different bundles. Hence g_2 must lie in the one bundle containing neither g_3 nor g_4, which is the bundle containing g_1. But two elements over three bundles have range at most 1 only in the pattern $(1, 1, 0)$, so D_1 forces g_1 and g_2 into *different* bundles. Contradiction. □

The instance is binary additive and therefore already settled by [LMS26]; what it refutes is the method, on a class the literature closes. Indeed 19 of its 81 allocations are good, several with $\ell = (0, 0, 0)$ — it is an easy instance for the conjecture and an impossible one for the invariant.

The failure is sharper than mere unreachability.

Proposition 101. *On the instance of Proposition 100,* every *good allocation has an arc of weight -2 or less. Hence no sufficient condition of the form "all arcs ≥ -1" can certify any of them.*

Proof Exhaustive over all $3^4 = 81$ allocations (`routeA_cex.py`): 19 are good, and the minimum arc weight over them is -3, attained at $(\{g_1, g_3, g_4\}, \{g_2\}, \emptyset)$, with every other good allocation reaching -2. □

Remark 102 (What this rules out)**.** Corollary 35 is a bound on *arcs*; the target is a bound on *paths*. Propositions 100 and 101 show the gap between the two is not slack that a cleverer construction can close — it is where all the surviving witnesses live. **Any correct invariant must be path-based.** That is precisely what makes agent induction hard, because path weights depend on which agent holds which bundle, so the invariant cannot be assignment-invariant and the newcomer's arrival re-opens the whole graph. Agent induction is not refuted here; the only invariant that would have made it easy is.

Remark 103 (The obstruction is hypergraph discrepancy)**.** For binary additive costs, uniform balance says every D_i is split evenly across the n bundles, which is a discrepancy condition on the set system $\{D_1, \dots, D_n\}$. Proposition 100 is a small discrepancy obstruction, and it is the same combinatorial phenomenon that drives the reduction by which [BSV21] proves deciding exact envy-freeness for binary additive chores NP-complete. The certified-hard set-splitting instance recorded from that reduction also admits no uniformly balanced family, checked by `routeA_adv.py`. So the balance signal of Remark 53, which three independent routes had pointed at, cannot be imposed exactly — only approximately, and the approximation is exactly the path slack.

10.3 The $n = 3$ specialisation

Since the attack was aimed at $n = 3$ first, it is worth recording exactly what the target reduces to there. The point is that with three agents the path structure is finite and small, so "$\ell \le 1$" becomes a fixed list of inequalities.

Proposition 104 (Characterisation at $n = 3$)**.** *Let $n = 3$. An allocation $\mathcal{A}$ is good — envy-freeable with $\ell_{\mathcal{A}} \le 1$ — if and only if*

(i) every directed cycle of $G_{\mathcal{A}}$ has weight at most 0*;*

(ii) every arc has weight at most 1*;*

(iii) every directed path of length two has weight at most 1*.*

Proof Condition (i) is envy-freeability, by Theorem 10. Given it, no positive cycle exists, so $\ell_{\mathcal{A}}(i)$ is the maximum weight of a *simple* path leaving i; on three vertices such a path has length 0, 1 or 2, contributing weight 0, an arc, or a two-path respectively. So $\ell_{\mathcal{A}} \le 1$ is exactly (ii) together with (iii). ◻

Corollary 105 (Only six inequalities, and most are free)**.** *Let $n = 3$ and let $\mathcal{A}$ satisfy (i) and (ii). For distinct i, j, k the three-cycle bound gives $w_{\mathcal{A}}(i, j) + w_{\mathcal{A}}(j, k) \le -w_{\mathcal{A}}(k, i)$, so the two-path $i \longrightarrow j \longrightarrow k$ has weight at most 1 whenever $w_{\mathcal{A}}(k, i) \ge -1$. As (i, j, k) runs over the six orderings of N, the closing arc (k, i) runs over all six ordered pairs exactly once. Hence exactly those two-paths whose closing arc lies below -1 require separate checking, one per offending arc. In particular, if every arc lies in $[-1, 1]$ then (iii) is automatic and $\mathcal{A}$ is good.*

Both statements are machine-checked against the true longest-path computation over 5.8×10^5 (instance, allocation) pairs — exhaustively at $n = m = 3$ and randomised to $m = 6$ — by `n3check.py`.

Corollary 105 is the sharpest handle available at $n = 3$, and it also explains precisely why Section 10 failed. The clause "if every arc lies in $[-1, 1]$" is the uniform-balance hypothesis; the residual work is the two-paths whose closing arc is very negative; and Proposition 101 says that on the counterexample *every* good allocation has such an arc, so the residual work is not a corner case but the whole problem.

10.4 What survives: the two-tier restatement

One reformulation does survive, and it is worth recording because it is equivalent rather than merely sufficient.

Proposition 106. *Let $n = 3$ and let $\mathcal{A}$ be envy-freeable with every arc at most 1. Then $\mathcal{A}$ is good if and only if the digraph of weight-1 arcs contains no directed path of length two.*

Proof If no such path exists then every directed path carries at most one positive arc, so has weight at most 1, and $\ell_{\mathcal{A}} \le 1$. Conversely if $\ell_{\mathcal{A}} \le 1$ then $p^* \in \{0,1\}^n$ by Observation 12; writing $S = \left\{i : p_i^* = 1\right\}$, Proposition 15 puts every arc within S, within $N \setminus S$, or from $N \setminus S$ into S at weight at most 0, so weight-1 arcs run only from S to $N \setminus S$ and cannot be composed. □

Remark 107 (The hypothesis $n = 3$ is needed, and only in one direction)**.** The converse direction above is general: for any n, a good allocation has its weight-1 arcs running one way across a bipartition, so none compose. The forward direction is not. It reads "every directed path carries at most one positive arc", which needs the two positive arcs of a heavy path to be *consecutive* — true at $n = 3$, where a simple path has length at most two, and false beyond it: at $n = 4$ the path $i_1 \to i_2 \to i_3 \to i_4$ with arc weights $1, 0, 1$ has weight 2 while its two weight-1 arcs are not composable, so the digraph of weight-1 arcs contains no path of length two. For general n the correct statement is Proposition 15 itself — a bipartition across which all strict envy runs one way — and the composability form is a genuinely three-agent convenience.

So the target is exactly: *find an envy-freeable allocation together with a bipartition of the agents such that all strict envy runs from one side to the other*. This is Proposition 15 again, now as a criterion rather than a certificate format. It is not assignment-invariant, consistent with Remark 102, but it is a condition on a bipartition rather than on a schedule, so it inherits none of the dead ends of Section 8 and none of the ex-ante failures of Section 9.

Two facts about the paid set are worth recording before that route is attempted. The first is a theorem and constrains what the set can be; the second is an observation from Section 19 and suggests where to look.

Proposition 108 (Paid agents are those somebody finds expensive)**.** *Let $\mathcal{A}$ be envy-freeable. If $p_i^* \ge 1$ then there is an agent $v \ne i$ with $c_v(A_i) \ge c_v(A_v) + 1$.*

Proof Immediate from Theorem 33: $\ell_{\mathcal{A}}(i) \le \max(0, \max_{v \ne i}[c_v(A_i) - c_v(A_v)])$, so $\ell_{\mathcal{A}}(i) \ge 1$ forces the inner maximum to be at least 1. □

So the paid set is contained in the set of agents whose bundle looks strictly more expensive to somebody than that somebody's own — which is exactly condition (c) of

Proposition 15 read backwards. An agent cannot be paid merely for holding a bundle she herself dislikes; somebody else has to corroborate.

Remark 109 (Where the paid set sits, empirically)**.** Across the 164 adversarial instances of Remark 296 that genuinely require a subsidy, the minimum-size paid set of the witness found is in every case contained in $\arg\max_i c_i(A_i)$ — the paid agents are among those carrying the largest own cost.

That lead is now withdrawn. The 164 instances were not filtered for additivity, and the containment turns out to be an artefact of additive instances. Re-tested on 553 instances that are both envy-free-free *and* non-additive, it fails on 141 of them (`update_15/nonadditive_residual.py`). Own costs of different agents are measured by different functions, so there was never a reason to expect the comparison to mean anything; the earlier 164/164 simply reflected a sample that was effectively additive. What survives is Proposition 108, which is proved: a paid agent's bundle must look at least one unit more expensive to *somebody* than that somebody's own.

11 Approach 6: A Pure Goods Reformulation, and a Working Algorithm

Every approach so far has worked inside the chore model or the replica model. This one leaves both, and it ends in a concrete algorithm with no observed failures: a construct-and-repair procedure that has solved every one of 389,215 test instances tried against it, including every adversarial instance that defeated Approaches 1 through 5.

11.1 The transform is an involution

The size-shift transform of Theorem 16 is not merely a comparison device: it is an involution of the dichotomous class onto itself, so the chore problem can be restated as a question about *goods* with no chores, no replica, no coverage constraint and no schedule anywhere in it.

Lemma 110. *The map $c \mapsto \tilde{v}$, $\tilde{v}(S) := |S| - c(S)$, is an involution of the set of dichotomous set functions (Theorem 7) onto itself.*

Proof $\tilde{v}(\emptyset) = 0$, and for $g \notin S$ the marginal is $\tilde{v}(S \cup \{g\}) - \tilde{v}(S) = 1 - \big(c(S \cup \{g\}) - c(S)\big) \in \{0, 1\}$, so $\tilde{v}$ is dichotomous. Applying the map twice returns $|S| - (|S| - c(S)) = c(S)$. □

11.2 The target

Fix an allocation $\mathcal{A}$ and write $\tilde{p}$ for the minimum subsidy vector of the goods instance $\{\tilde{v}_i\}$ at $\mathcal{A}$, defined exactly when the chore instance is envy-freeable at $\mathcal{A}$, since by Theorem 16 the two envy graphs have the same cycle weights. Put $q_i := \tilde{p}_i + |A_i|$ — goods subsidy plus bundle size.

Conjecture 111 (Target G)**.** *Every dichotomous goods instance admits an allocation whose vector q has spread at most* 1.

Proposition 112. *Conjecture 111 implies Conjecture 2.*

Proof By Theorem 16, $\ell_{\mathcal{A}}(u) = \max_v[\tilde{d}_{\mathcal{A}}(u,v) + |A_u| - |A_v|]$, where $\tilde{d}_{\mathcal{A}}(u,v)$ is the heaviest $u \to v$ path in the goods envy graph. Appending to any $u \to v$ path a heaviest path leaving v gives $\tilde{d}_{\mathcal{A}}(u,v) + \tilde{p}_v \le \tilde{p}_u$, so $\tilde{d}_{\mathcal{A}}(u,v) \le \tilde{p}_u - \tilde{p}_v$ and therefore $\ell_{\mathcal{A}}(u) \le q_u - \min_v q_v$. A spread of at most 1 therefore gives $\ell_{\mathcal{A}} \le 1$, and Observation 12 upgrades this to $p^* \in \{0,1\}^n$ with total at most $n-1$. □

Target G quantifies over *all* allocations, and $\tilde{p}$ is itself a longest-path weight, so Target G is a condition on paths rather than arcs — exactly what killed Approach 5 (Remark 102). It also survives extensive testing on its own: 0 failures on the full exhaustive $n = m = 3$ family and on the adversarial families that refuted the invariant of Approach 5, detailed in `targetG.py` and `targetG_adv.py`. But Target G alone does not say how to *find* the allocation. The rest of this section does.

11.3 Item-by-item insertion is the wrong template here too

The obvious next move is to run the algorithm of [BKNS22] directly on the size-shifted goods instance $\{\tilde{v}_i\}$, since it is a legitimate dichotomous instance and the algorithm's correctness proofs (Lemmas 3, 7, 8, 9, 10, 11 of that paper) are generic over *which* valid choice Extend and FindSink make at each step: any valid (ρ, κ) pair, any starting agent in $\mathrm{M}(p)$. That freedom can be spent trying to steer bundle sizes toward balance while $\tilde{p} \in \{0,1\}^n$ is preserved automatically.

It does not work. Greedily steering the choice that locally minimises q-spread fails on 1,635 of the 9,880 exhaustive $n = m = 3$ instances with a fixed item order, and on 440 of them even after trying every item order. Worse than a weak heuristic: on the smallest such instance, a full backtracking search over *every* legal execution of the algorithm — every item order, every Extend choice, every FindSink starting point, 474 states in total — never reaches spread ≤ 1, although spread 0 is achievable (the perfectly balanced allocation $\{0\}, \{1\}, \{2\}$) and is confirmed reachable by direct, algorithm-free enumeration. This is a genuine reachability obstruction, verified by `guidedR3_full.py` and `verify_reach_gap.py`, in the same family as the peel dead ends of Section 8.5: item-by-item insertion fixes which items end up sharing a bundle the moment they are grouped, and no later permutation can split a bundle back apart, so an early bad grouping can be permanent.

The lesson is to stop building the allocation one item at a time. The rest of this section instead *constructs* a candidate directly and *repairs* it if needed — an operation insertion cannot perform, since repair means moving an item out of a bundle it is already in.

11.4 Construct and repair

Definition 113 (Target G-bal)**.** *A partition of the items into n groups is* cardinality-balanced *if the group sizes differ by at most* 1*.* Target G-bal *is the statement that every dichotomous goods instance admits a cardinality-balanced partition whose optimal (maximum-welfare) assignment of groups to agents gives q-spread at most* 1*.*

Observation 114. *Target G-bal implies Target G, since a cardinality-balanced partition is in particular a partition.*

Cardinality-balanced partitions are exactly what the balanced-matching machinery of [BDN+20, LMS26] is built to produce, so this restriction moves onto ground the literature already occupies. The algorithm below constructs one directly, using no chores, no replica, and no item-by-item insertion order.

Algorithm (construct-and-repair).

1. *Construct.* Run one pass of R11's iterated maximum-weight perfect matching (IMWPM): pad the items with inert dummies to a multiple of n, and in each of $\lceil m/n \rceil$ rounds assign every agent exactly one remaining item (real or dummy), by a matching maximising the total *marginal* value $\tilde{v}_i(A_i \cup \{g\}) - \tilde{v}_i(A_i)$ over the current partial bundles — the non-additive generalisation of the item-value weights of [BDN+20], needed because $\tilde{v}_i$ need not be additive. Discard the dummies.
2. *Repair.* While the current partition's optimal assignment has q-spread greater than 1: among all single-item swaps between two groups that keep the sizes cardinality-balanced, take the one that most reduces the (q-spread, −welfare) pair, re-optimise the assignment, and repeat.
3. *Restart on failure.* If repair reaches a local optimum with spread still above 1, restart step 1 from a freshly shuffled round-robin partition instead of IMWPM's output, and repeat step 2. Try up to a small constant number of restarts.

Every step preserves cardinality balance, so termination at spread ≤ 1 directly witnesses Target G-bal — and hence, by Proposition 112 and Observation 114, produces an envy-free chore allocation with $p^* \in \{0,1\}^n$ once translated back through Theorem 16.

11.5 Evidence: zero failures across 389,215 instances

Test set	Instances	Failures
Exhaustive, $n = m = 3$, all instances	9,880	0
Exhaustive, structured non-additive triples, $n = 3$, $m \in \{3,4,5\}$	378,176	0
Every named hard instance in this project (Theorem 36's witness, Theorem 93's no-go instance, Proposition 100's discrepancy counterexample, Proposition 38's instance)	4	0
Randomised, $n \in \{3,\ldots,8\}$, $m \in \{3,\ldots,10\}$, endpoint-constant biased	1,155	0
Total	**389,215**	**0**

Two components of this deserve to be separated out, since the construction step and the repair step have different track records on their own.

IMWPM alone, with no repair, already succeeds on all 9,880 exhaustive $n = m = 3$ instances and on all four hard instances, but fails on roughly 18% of larger randomised instances (132 of 720 tested) — it is greedy and short-sighted, so it can lock in a bad grouping in an early round exactly as insertion does, just less often.

Repair from a bare round-robin start, with no IMWPM warm start, reaches spread ≤ 1 on all 9,880 exhaustive instances and on 357,746 of the 357,760 structured triples at $m = 5$; the remaining 14 are genuine local optima of the swap neighbourhood — confirmed by independent exhaustive search over all cardinality-balanced partitions, which finds spread 0 achievable on every one of them (`classify_failures.py`) — and every one of the 14 is recovered by restarting from a freshly shuffled partition, using at most 4 restarts (`restart_check.py`).

Combining both, as in Section 11.4, removed every failure found by either component alone, across every test run against it.

11.6 Complexity, and what is proved unconditionally

Three properties of the algorithm are theorems, not conjectures: its outputs are certified correct, it terminates in polynomial time, and its search space is genuinely the set of *unordered* balanced partitions. We state them with proofs, in the value-oracle model with per-query cost τ.

Lemma 115 (Assignment invariance)**.** *Fix an unordered family of n bundles. The assignments of bundles to agents under which the allocation is envy-freeable are exactly the welfare-maximising ones, and every welfare-maximising assignment induces the same minimum subsidy on each* bundle *[KMS+24, Lemma 2]. Hence the multiset $\{\tilde{p}(B) + |B|\}$, and with it the q-spread, depends only on the unordered partition.*

Proof The first claim is condition (ii) of Theorem 10 applied to the goods instance: an assignment is envy-freeable iff no reassignment of its own bundles raises welfare, which is exactly welfare maximality over assignments. The second is Lemma 2 of [KMS+24], which proves that the minimum subsidy vectors of any two maximum-weight assignments coincide bundle-by-bundle. Adding $|B|$, a bundle attribute, preserves this. $\square$

Two consequences. First, the $n!$ assignment loop in the implementation is mathematically equivalent to a single maximum-weight matching computation, so the complexity below is stated with the Hungarian algorithm, $O(n^3)$. Second, the algorithm's true search space is unordered balanced partitions — the object Conjecture 123 should be read as quantifying over.

Theorem 116 (Soundness, certified)**.** *Suppose the algorithm outputs a partition and assignment with q-spread at most* 1*. Then the* same *allocation, read in the original chore instance — the size-shift acts on valuations only, so the allocation needs no translation — together with $p := \ell_{\mathcal{A}}$ is an envy-free solution with $p \in \{0,1\}^n$ and total subsidy at most $n-1$. Moreover the output is verifiable in $O(n^2\tau + n^3)$ time independently of how it was found.*

Proof Proposition 112 gives $\ell_{\mathcal{A}} \leq 1$ on the chore side; Observation 12 upgrades to $p^* \in \{0,1\}^n$; the endpoint of a heaviest path has $\ell = 0$, giving total at most $n-1$. Verification recomputes the n^2 bundle values (n^2 oracle calls), one maximum-weight matching and one longest-path computation ($O(n^3)$ each). $\square$

Theorem 117 (Termination and complexity)**.** *On any dichotomous instance:*

(i) evaluating one partition costs $O(n^2\tau + n^3)$*;*

(ii) the construction pass (IMWPM) costs $O(m^2\tau + nm^2)$*;*

(iii) each repair iteration examines $O(nm)$ *moves and costs* $O(n^3m\tau + n^4m)$*;*

(iv) the repair loop performs at most $O(m^2)$ *iterations, so with a constant number of restarts the whole algorithm runs in time* $O(n^3m^3\tau + n^4m^3)$*.*

In particular the algorithm always terminates, in polynomial time, regardless of whether it succeeds.

Proof (i) is the count in Theorem 116, using Lemma 115 to replace the assignment loop by one Hungarian computation. (ii): there are $\lceil m/n \rceil$ rounds; each builds an $n \times O(m)$ marginal table ($O(nm)$ oracle calls, two per entry) and solves one rectangular assignment problem in $O(n^2m)$; summing gives the bound. (iii): a move sends one of m items to one of $n-1$ other groups, and each candidate is evaluated by (i). (iv) is the substantive claim. Because marginals lie in $\{0,1\}$, every bundle value satisfies $\tilde{v}_i(B) \le |B|$; hence total welfare is an integer in $[0,m]$, and every subsidy $\tilde{p}_i$, being a path weight, is an integer in $[0,m]$, so the spread is an integer in $[0, m+1]$. The repair loop moves only when the pair (spread, −welfare) strictly decreases lexicographically, and that pair takes at most $(m+2)(m+1)$ values, so at most $O(m^2)$ moves occur before the loop halts — either at spread ≤ 1 or at a local optimum, where a restart or a failure report follows. □

The gap between these theorems and a full correctness proof is exactly Conjecture 123: soundness says the algorithm never lies, termination says it always answers, and what is missing is that the answer is always "success." For two agents, that missing piece can be supplied.

11.7 A complete proof for two agents

Theorem 118 (Completeness at $n = 2$)**.** *Every two-agent dichotomous goods instance admits a cardinality-balanced split with q-spread at most* 1*. Consequently Target G-bal, Target G, and Conjecture 2 all hold at* $n = 2$ *— the last with the additional guarantee, not provided by the complementation route of Remark 9, that the two bundles differ in size by at most one.*

Fix dichotomous $\tilde{v}_1, \tilde{v}_2$ on M, $|M| = m$. For an *ordered* balanced split (A, B) with $|A| = \lceil m/2 \rceil \ge |B|$, write

$$\alpha := \tilde{v}_1(A) - \tilde{v}_1(B), \qquad \beta := \tilde{v}_2(A) - \tilde{v}_2(B), \qquad u := \tilde{v}_1 + \tilde{v}_2, \qquad h := u(A) - u(B) = \alpha + \beta.$$

Call the split *good* if some envy-freeable orientation (assignment of A, B to the two agents) has q-spread at most 1.

Lemma 119 (Swap step)**.** *Exchanging one item of A with one item of B changes each of* $\tilde{v}_i(A)$ *and* $\tilde{v}_i(B)$ *by at most* 1*, hence each of* α, β *by at most* 2 *and h by at most* 4*.*

Proof Removing an item changes a dichotomous value by 0 or −1 and adding one by 0 or +1, so the composite changes each side's value by at most 1 in absolute value. □

Lemma 120 (Window)**.** *If m is even and $|h| \le 3$, or m is odd and $-1 \le h \le 3$, then the split is good.*

Proof *Even m.* Sizes are equal, so q-spread equals $|\tilde{p}_1 - \tilde{p}_2|$, and it suffices to find an envy-freeable orientation in which both envies are at most 1. Giving A to agent 1 yields envies $(-\alpha, \beta)$, so that orientation is satisfactory iff $\alpha \ge -1$ and $\beta \le 1$; the flipped orientation iff $\alpha \le 1$ and $\beta \ge -1$. Both fail only when $\alpha, \beta \le -2$ or $\alpha, \beta \ge 2$, and in either case $|h| \ge 4$. It remains to check envy-freeability. The two-cycle of the first orientation has weight $\beta - \alpha$; if it is positive while that orientation is satisfactory ($\alpha \ge -1$, $\beta \le 1$, $\beta > \alpha$), then $\alpha < \beta \le 1$ and $\beta > \alpha \ge -1$ show the flipped orientation is satisfactory as well, and its cycle $\alpha - \beta$ is negative. So a satisfactory envy-freeable orientation exists, its subsidies lie in $\{0, 1\}$, and the spread is at most 1.

Odd m. Now $|A| = |B| + 1$, and with subsidies in $\{0, 1\}$ the spread is at most 1 iff the holder of the large bundle needs no subsidy — both agents needing 1 is impossible, since then the two-cycle ≥ 2 would be positive. Giving A to agent 1 is satisfactory iff $\alpha \ge 0$ and $\beta \le 1$; flipped, iff $\beta \ge 0$ and $\alpha \le 1$. Both fail only when $\alpha, \beta \le -1$ or $\alpha, \beta \ge 2$, i.e. when $h \le -2$ or $h \ge 4$. Envy-freeability is repaired exactly as in the even case: if the first satisfactory orientation has $\beta > \alpha$, then $\alpha < \beta \le 1$ gives $\alpha \le 0$ and $\beta > \alpha \ge 0$ gives $\beta \ge 0$, so the flipped orientation is satisfactory with a negative cycle. □

Proof of Theorem 118 *Even m.* Take any equal split, ordered so that $h \ge 0$. If $h \le 3$, Lemma 120 finishes. Otherwise $h \ge 4$; walk to the complementary split (B, A) by exchanging one paired item at a time ($m/2$ swaps). The final value is $-h \le -4$, and by Lemma 119 each step changes h by at most 4, so at the first step where the value drops below 4 it lies in $[0, 3]$ — inside the window.

Odd m. The big sides are the $(k+1)$-subsets, $m = 2k + 1$. Suppose no split is good; then by Lemma 120 every big side A has $h(A) \le -2$ or $h(A) \ge 4$. Any two big sides are connected by single exchanges (repeatedly swap an element of the difference), and a step from $h \ge 4$ to $h \le -2$ would change h by at least $6 > 4$, so *all* big sides lie on one side. If all have $h \ge 4$: let A^* minimise u over big sides, and pick any $x \in A^*$. Then $B := (M \setminus A^*) \cup \{x\}$ is a big side with $u(B) \le u(M \setminus A^*) + 2 \le u(A^*) - 2$, contradicting minimality — the first inequality because u has marginals in $\{0, 1, 2\}$, the second because $h(A^*) \ge 4$. If all have $h \le -2$: let A° maximise u over big sides and pick any $x \in A^\circ$. Then $u((M \setminus A^\circ) \cup \{x\}) \ge u(M \setminus A^\circ) \ge u(A^\circ) + 2$, by monotonicity and $h(A^\circ) \le -2$, contradicting maximality. □

Corollary 121 (An exact $O(m\tau)$ algorithm for two agents)**.** *For $n = 2$ the proof is constructive and needs no restarts: start from any balanced split; if it is not good, then for even m walk toward the complement, and for odd m first apply the jump $A \mapsto (M \setminus A) \cup \{x\}$ — which maps $h \ge 4$ to $h \le 0$ and $h \le -2$ to $h \ge 2$ — and then, if the jump target is still outside the window, walk between the two splits. Some split within $m/2 + 2$ evaluations lies in the window and is good. Each evaluation uses $O(1)$ oracle calls, so the whole procedure runs in $O(m\tau)$ time.*

Proof The even case is the walk in the proof above. For the odd jump, write $B = (M \setminus A) \cup \{x\}$ with $x \in A$, so $M \setminus B = A \setminus \{x\}$ and $h(B) = u(B) - u(A \setminus \{x\})$. Recall that $u = \tilde{v}_1 + \tilde{v}_2$ has marginals in $\{0, 1, 2\}$, so adding or removing a single item changes u by at most 2. If $h(A) \ge 4$ then $u(B) \le u(M \setminus A) + 2 \le u(A) - 2$ while $u(A \setminus \{x\}) \ge u(A) - 2$, giving $h(B) \le 0$. If $h(A) \le -2$ then $u(B) \ge u(M \setminus A) \ge u(A) + 2$ while $u(A \setminus \{x\}) \le u(A)$, giving $h(B) \ge 2$.

Both bounds are attained, so neither can be sharpened. In either case the jump target is in the window $[-1, 3]$ or strictly on the far side of it, and in the latter case the swap walk between A and B crosses the window as in the theorem: descending from ≥ 4, the first value below 4 is at least 0; ascending from ≤ -2, the first value above -2 is at most 2. $\square$

The theorem and the walk are machine-verified by `n2proof_check.py`: exhaustively over all instances at $m \leq 4$ (490,545 pairs at $m = 4$ alone) and on 30,000 random instances at $m \in \{5, 6, 7\}$ — 0 failures of either, with the walk never needing more than 2 evaluations in practice against its $m/2 + 2$ bound.

Remark 122 (What the proof teaches the algorithm)**.** Two findings from the proof effort feed back into the implementation. First, the moves that the $n = 2$ proof walks are *pairwise exchanges*, which preserve group sizes exactly — but the implemented repair neighbourhood contains only single-item *moves*, which change two group sizes by one. When n divides m, every group has size m/n and every single move is rejected by the balance filter, so the repair loop is inert there and the implementation succeeds purely through IMWPM plus restarts; the zero failures at $n = m$ and at $(n, m) \in \{(3, 6), (6, 6), (7, 7), (8, 8)\}$ were all achieved in that degenerate mode. Adding exchanges to the neighbourhood is therefore both principled — they are the proof's own moves — and necessary for the repair step to act at all when $n \mid m$. Second, the h-window technique suggests the right potential for a general proof: a one-dimensional aggregate whose swap-Lipschitz constant is smaller than the width of the region it must cross. At $n = 2$ the aggregate $u = \tilde{v}_1 + \tilde{v}_2$ works because the goodness condition is two-sided in a single difference; for $n \geq 3$ goodness involves $\binom{n}{2}$ pairwise differences, and no single aggregate is yet known to control them.

11.8 What remains

After Sections 11.6 and 11.7, the standing of the approach is: soundness, certification, termination and polynomial complexity are theorems for every n; completeness is a theorem for $n = 2$; and for $n \geq 3$ completeness is the one remaining open statement, sharp and almost entirely combinatorial, with no chores, no replica and no scheduling left in it:

Conjecture 123 (Construct-and-repair succeeds, $n \geq 3$)**.** *For every dichotomous goods instance, the algorithm of Section 11.4 — with pairwise exchanges added to the repair neighbourhood, per Remark 122 — reaches a cardinality-balanced partition with q-spread at most* 1*, using a number of restarts bounded by a constant independent of n and m.*

The 14 local optima found at $n = 3$ show that the move neighbourhood does contain bad local optima, so a proof cannot simply be "every non-optimal balanced partition has an improving move." What was observed instead is that bad local optima are escapable from other starting partitions, always within a handful of restarts. The $n = 2$ proof suggests what to look for: a one-dimensional aggregate — there, $u = \tilde{v}_1 + \tilde{v}_2$ with the window $|h| \leq 3$ — whose per-move variation is smaller than the forbidden gap it must cross, together with an extremal argument at the ends. Finding the analogous aggregate for $n \geq 3$, where goodness involves $\binom{n}{2}$ pairwise comparisons rather than one, is the whole remaining problem; solving it would upgrade Conjecture 123, and with it Conjecture 2, from extensively tested to proved.

11.9 The canonical aggregate, tested and refuted

The aggregate the previous paragraph asks for has an obvious candidate. At $n = 2$ the quantity is $h = u(A) - u(B)$ with $u = \tilde{v}_1 + \tilde{v}_2$, which is the *spread of u across the bundles*; for general n that reads

$$\mathrm{uspread}(A) \;:=\; \max_j u(A_j) - \min_j u(A_j), \qquad u := \textstyle\sum_i c_i,$$

and at $n = 2$ it is $|h|$ exactly, so this is a genuine generalisation rather than an analogy. Splitting Lemma 120 and the walk of Theorem 118 along the same seam gives two statements:

$$\begin{aligned} \textsc{Window}(K): &\quad \text{every balanced partition with uspread} \le K \text{ is good;} \\ \textsc{Exists}(K): &\quad \text{every instance has a balanced partition with uspread} \le K. \end{aligned}$$

Together they give Conjecture 2, and at $n = 2$ both hold with $K = 3$ — that is precisely the content of the two-agent proof.

The measurement (`update_22/window_general.py`) first reproduces the proved case: at $n = 2$ the smallest uspread carried by a *bad* balanced split is 4, i.e. $|h| \le 3$ suffices, matching Lemma 120 exactly. That agreement is what licenses reading the rest of the table as evidence about the same quantity rather than a different one.

For $n \ge 3$ the two requirements separate and the route closes:

	smallest uspread of a bad balanced partition		largest uspread forced on some instance	
$n = 2$	4	(so $\textsc{Window}(3)$)	3	(so $\textsc{Exists}(3)$)
$n \ge 3$	2	(so $K \le 1$)	3	(so $K \ge 3$)

At $n = 3$, $m = 7$ a bad balanced partition occurs with uspread = 2, forcing $K \le 1$; and 45 of the instances tested admit no balanced partition with uspread below 3, forcing $K \ge 3$. No K serves both. *The window shrinks below the achievable spread*, and this is exactly why $n = 2$ is tractable and $n \ge 3$ is not: there, both numbers are 3 and the interval is non-empty.

So the canonical aggregate is refuted. Any surviving aggregate must be sensitive to how welfare is distributed among agents, not only to its total — u discards precisely that information, and at $n = 2$ it could afford to, because with two agents a single difference determines the split.

11.10 Sharpening what needs proving

Conjecture 123 allows a constant number of restarts, and the 14 local optima at $n = 3$ show the repair neighbourhood really does have bad local optima, so the allowance is not vacuous. But a local optimum matters only if the algorithm can *reach* it, and the warm start is not an arbitrary partition.

Observation 124 (Restarts are never consumed)**.** *Over 1,580 random instances with $n \le 7$ and $m \le 9$, the IMWPM warm start followed by repair alone succeeded every time; the restart branch was never entered (`update_22/restart_necessity.py`).*

This matters for what a proof must establish, because the two readings of Conjecture 123 are of quite different difficulty. Were restarts genuinely needed, completeness would be a *reachability* claim over starting partitions — which presupposes that a good partition exists and then adds that a shuffled round-robin start finds one, making it strictly harder than Conjecture 2 and useless as a route to it. Observation 124 indicates the weight sits elsewhere:

Conjecture 125 (Warm-start descent)**.** *For every dichotomous goods instance, repair from the IMWPM warm start reaches q-spread at most* 1 *without restarts.*

Conjecture 125 implies Conjecture 123 with zero restarts, hence Conjecture 2, and it is the better proof target for a specific reason: unlike the descent lemma of Section 12, whose witnessing move proved not to be local (Section 12.6), it does not quantify over all partitions. It concerns a *single, structured* starting point — the output of an iterated maximum-weight matching — and the properties of that output are available to the proof. Whether the bad local optima are reachable from it is the question; the data say they are not.

11.11 What the warm start guarantees, and the two claims that remain

Asking what the IMWPM output looks like, rather than only whether repair eventually succeeds, turns out to answer the question almost completely.

Observation 126 (The warm start never exceeds spread two)**.** *Across every family tested — the full exhaustive $n = m = 3$ family, all* 357,760 *triples of the structured non-additive pool at $m = 5$, sampled triples at $m = 6$, random instances up to $n = 7$, $m = 10$, disjoint-interest instances built for maximal disagreement, and the named hard instances of Section 11.5 — the q-spread of the raw IMWPM partition, before any repair, never exceeded* 2 *(`update_23/warmstart.py`, `imwpm_bound.py`).*

Two refinements make this sharper than it first appears. First, the bound is rarely attained: on the $m = 5$ pool, 352,468 of 357,760 instances start at spread at most 1, so IMWPM *alone* already produces a witness for Target G-bal and repair is never invoked. The discrepancy counterexample, the insertion witness and the W4 no-go instance — the three named obstructions of this section — all start at spread at most 1. Second, when spread 2 does occur, repair is immediate:

Observation 127 (One move suffices)**.** *Among* 3,000 *instances whose IMWPM start has q-spread exactly* 2*, repair reached spread at most* 1 *in exactly one improving move in every case, with no stalls.*

Together these decompose Conjecture 125 into two claims, neither of which is a descent along an unbounded chain:

Conjecture 128 (Warm-start bound)**.** *For every dichotomous goods instance, the IMWPM partition has q-spread at most* 2*.*

Conjecture 129 (Last rung)**.** *Every IMWPM partition with q-spread exactly* 2 *admits a balance-preserving single-item move to q-spread at most* 1*.*

Proposition 130. *Conjectures 128 and 129 together imply Conjecture 125, hence Conjecture 123 with zero restarts, hence Conjecture 2 together with its polynomial-time clause.*

Proof By Conjecture 128 the warm start has spread at most 2. If it is at most 1 the partition is already a Target G-bal witness. Otherwise it is exactly 2, and Conjecture 129 supplies a move to spread at most 1; the move preserves cardinality balance by hypothesis. Either way a cardinality-balanced partition with q-spread at most 1 is produced, which is Target G-bal; Observation 114 and Proposition 112 give Conjecture 2. The running time is that of Theorem 117 with the repair loop executing at most one iteration. □

The reason this is a better position than anything in Sections 12 and 13 is that Conjecture 128 is not a statement about local search at all. It is a statement about what one round-based matching construction produces, and constructions of exactly this kind are what [BDN+20] analyses: for *additive* goods, iterated maximum-weight matching is shown there to need a subsidy of at most one per agent, which combined with cardinality balance is precisely a q-spread bound of 2. So Conjecture 128 asks for that analysis to survive the passage from additive to dichotomous values, where the matching weights are marginals $\tilde{v}_i(A_i \cup \{g\}) - \tilde{v}_i(A_i)$ rather than fixed item values. This is the first point in this report at which the remaining gap abuts a theorem that is already proved in the literature rather than a fresh combinatorial obstruction, and it is where effort should go.

Conjecture 129 is the smaller of the two and the more likely to yield to the two-tier machinery of Section 10: spread exactly 2 means the agents split into a top and a bottom tier differing by 2, and what must be shown is that one item can always be moved across.

11.12 Reading [BDN+20] line by line: what transfers and what does not

Conjecture 128 asks whether [BDN+20, Theorem 1.3] — iterated maximum-weight matching needs a subsidy of at most one per agent — survives the passage from additive to dichotomous valuations. The two classes are *incomparable*: $v(S) = \min(|S|, 1)$ is dichotomous and not additive, an additive function with an item of value 5 is not dichotomous, and they intersect exactly in the binary additive functions. So nothing is inherited for free, and the question is which parts of the proof survive.

Where additivity is used. In [BDN+20] it appears in five places. Lemma 3.1 (envy-freeability), Lemma 3.2 (EF1), and equations (1) and (7) all rest on the per-round decomposition $v_i(A_k) = \sum_t v_i(\mu_k^t)$; Claim 4.4, equation (9), invokes additivity by name. These are serious but conceivably routable. The fatal use is the *engine* of their Section 4: the modified profile $\bar{v}_i$ is *defined by assigning item values*,

$$\bar{v}_i(\mu_k^t) \;=\; \max\bigl(v_i(\mu_k^t),\, v_i(\mu_i^{t+1})\bigr),$$

and extended additively. A non-additive dichotomous function admits no such extension, so the construction has no direct analogue at all.

What transfers. Their Lemma 4.1 — *if $\mathcal{A}$ is envy-freeable and every arc of its envy graph has weight at least* -1, *the subsidy is at most one per agent* — is proved by closing a maximum-weight path into a cycle and using that the cycle has non-positive weight. It is pure envy-graph combinatorics and uses no additivity whatever, so it holds verbatim in our setting. Moreover

$\bar{v}$ enters their argument only through the n^2 numbers $\bar{v}_i(A_k)$. That suggests performing the modification on the *matrix* rather than on a valuation: set

$$a_{ik} := \tilde{v}_i(A_k), \qquad \bar{a}_{ik} := \max\bigl(a_{ik},\, a_{ii} - 1\bigr),$$

so that $\bar{a}_{ii} = a_{ii}$, every modified arc is at least -1, and every modified arc dominates the true one, whence $\bar{\ell} \geq \ell$ pointwise. If the modified graph has no positive cycle, Lemma 4.1 gives $\bar{\ell} \leq 1$ and hence $\ell \leq 1$ — with no additivity used anywhere and no valuation function to construct.

Why this does not settle it. The condition "the modified matrix has no positive cycle" implies $\ell \leq 1$, so it cannot hold on any instance where $\ell \geq 2$; observing that it fails exactly on those instances is forced by the logic and is not evidence. The measurement is nevertheless decisive, because of what it turned up on the way.

Proposition 131 (The dichotomous analogue of [BDN+20, Thm 1.3] is false)**.** *There are dichotomous goods instances on which the IMWPM allocation requires a subsidy of* 2 *for some agent. Over* 1,034 *instances,* 2 *had* $\max_i \tilde{p}_i = 2$*; a witness has* $n = 3$, $m = 7$, *bundles* $\{g_1, g_4, g_6\}$, $\{g_0, g_3, g_5\}$, $\{g_2\}$ *and* $\tilde{p} = (0, 0, 2)$.

So "subsidy at most one per agent" does *not* survive the passage to dichotomous values, and Conjecture 128 cannot be proved by proving that. This also refutes the natural decomposition — prove $\tilde{p} \leq 1$, then add cardinality balance to get q-spread at most 2 — on both legs, because the second leg fails too:

Proposition 132 (IMWPM does not produce balanced bundles)**.** *When m is not a multiple of n the dummies discarded by IMWPM are distributed unevenly, and the resulting bundle sizes need not differ by at most one. Over* 1,034 *instances,* 79 *had size spread* 2.

What holds the bound together is instead a *compensation*: in the witness of Proposition 131 the bundle sizes are $(3, 3, 1)$, so $q = \tilde{p} + (|A_1|, |A_2|, |A_3|) = (3, 3, 3)$ and the q-spread is 0 even though the subsidy is 2. The agent carrying the large subsidy is exactly the agent holding the small bundle. Neither $\tilde{p}$ nor the bundle size is controlled on its own; only their sum is. Any proof of Conjecture 128 must therefore reason about q directly, and the two-term split that makes the additive proof work is unavailable.

11.13 What q is: the chores-side formula

The compensation observed in Proposition 131 — subsidy 2 against sizes $(3, 3, 1)$, giving $q = (3, 3, 3)$ — is not a coincidence. It follows from an identity that removes the goods picture from this section altogether.

Lemma 133 (The q-formula)**.** *Let $\mathcal{A}$ be envy-freeable and let $d(i, j)$ denote the heaviest $i \to j$ path in the* chores *envy graph, with $d(i, i) := 0$. Then*

$$q_i \;=\; \max_{j \in N} \bigl[\, d(i, j) + |A_j| \,\bigr].$$

Proof By Lemma 110, $\tilde{v}_i(S) = |S| - c_i(S)$, so the goods arc weight is

$$\tilde{w}(i,j) = \tilde{v}_i(A_j) - \tilde{v}_i(A_i) = \left[c_i(A_i) - c_i(A_j)\right] + \left|A_j\right| - |A_i| = w_{\mathcal{A}}(i,j) + \left|A_j\right| - |A_i|\,.$$

Along a path $i = u_0 \to u_1 \to \cdots \to u_r = j$ the size terms telescope, so $\tilde{d}(i,j) = d(i,j) + \left|A_j\right| - |A_i|$; both heaviest-path functions are well defined because the two envy graphs have the same cycle weights (Theorem 16). Taking the maximum over j, and noting that the empty path contributes the term $j = i$, $\tilde{p}_i = \max_j[d(i,j) + \left|A_j\right|] - |A_i|$. Adding $|A_i|$ gives the claim. □

So q_i is *the largest bundle size reachable from agent i, inflated by the envy accumulated on the way*. The compensation is then forced rather than fortuitous: an agent holding a small bundle can still carry a large q by envying its way to a big bundle, and an agent holding a big bundle has $q_i \geq |A_i|$ regardless of its subsidy. Neither term is controlled alone because q was never a sum of two independent quantities.

Corollary 134 (q at an IMWPM allocation)**.** *IMWPM runs $T = \lceil m/n \rceil$ rounds giving every agent exactly one object per round, so $\left|A_j\right| = T - \delta_j$ where δ_j is the number of dummies agent j received. Hence*

$$q_i \;=\; T + \max_j \left[\, d(i,j) - \delta_j \,\right],$$

and the q-spread does not depend on T. In particular, when $n \mid m$ every $\delta_j = 0$ and $q_i = T + \ell_{\mathcal{A}}(i)$, so on those instances the q-spread equals $\max_i \ell_{\mathcal{A}}(i)$ exactly.

Both identities were checked against direct computation on 745 instances with no mismatch, and the last claim held on all 276 instances with $n \mid m$ (`update_26/q_formula.py`). Corollary 134 is worth stating because it removes the apparent asymmetry between this section and the rest of the report: when n divides m, Target G at an IMWPM allocation *is* Conjecture 2 at that allocation, and Conjecture 128 says exactly $\max_i \ell_{\mathcal{A}}(i) \leq 2$.

One more translation is worth recording, since it states the algorithm without any reference to goods at all.

Lemma 135 (IMWPM in chores terms)**.** *Let $\gamma_i^t(g) := c_i(A_i^{t-1} \cup \{g\}) - c_i(A_i^{t-1}) \in \{0,1\}$ be the chore marginal. The goods marginal used as a matching weight satisfies $\mu_i^t(g) = 1 - \gamma_i^t(g)$, so maximising $\sum_i \mu_i^t$ is the same as* minimising *$\sum_i \gamma_i^t$. IMWPM is therefore: in each round, hand one remaining chore to every agent so as to minimise the total marginal cost incurred that round.*

Proof $\mu_i^t(g) = \tilde{v}_i(A_i^{t-1} \cup \{g\}) - \tilde{v}_i(A_i^{t-1}) = 1 - \gamma_i^t(g)$ directly from $\tilde{v}_i(S) = |S| - c_i(S)$. Summing over the n agents turns $\max \sum_i \mu_i^t$ into $\min \sum_i \gamma_i^t$. □

Lemma 135 also yields the per-round exchange property in a usable form: for any two agents i, j matched to g_i, g_j in round t, optimality against the transposition gives $\gamma_i^t(g_j) + \gamma_j^t(g_i) \geq \gamma_i^t(g_i) + \gamma_j^t(g_j)$. Since all four terms lie in $\{0,1\}$, this says: *if agent i would rather have had j's chore, then j strictly prefers its own*. That is a clean one-round invariant; what Section 11.17 shows is that accumulating it across rounds is exactly what fails.

11.14 From paths to arcs

Lemma 133 localises Conjecture 128. If j^* attains q_i then $q_{i'} \geq d(i', j^*) + |A_{j^*}|$, so $q_i - q_{i'} \leq d(i, j^*) - d(i', j^*)$ and therefore

$$q\text{-spread} \leq 2 \quad \Longleftrightarrow \quad d(i, j) - d(i', j) \leq 2 \quad \text{for all } i, i', j.$$

Because $d(i, j) \geq w_{\mathcal{A}}(i, i') + d(i', j)$ whenever $i \neq i'$ — with no positive cycle a heaviest walk may be taken simple, so concatenation is legitimate — this *forces* every arc to have weight at most 2. The measurement returns something stronger.

Observation 136 (Arcs at an IMWPM allocation)**.** *Over* 746 *instances with* $n \leq 6$, $m \leq 9$, *every chores envy arc at the IMWPM allocation had weight at most* 1*; the observed distribution of* $w_{\mathcal{A}}(i, j)$ *was* $-3:3$, $-2:648$, $-1:3551$, $0:2356$, $1:300$ *(*`update_27/arc_bound.py`*).*

An arc bound of 1 says $c_i(A_i) \leq c_i(A_j) + 1$, which — since every marginal is at most 1 — is exactly the EF1 property for chores. Two things follow. First, the lower bound fails badly: arcs reach -3, so the hypothesis of [BDN+20, Lemma 4.1] is unavailable, which is precisely the situation that paper describes before introducing its modified valuation, and which Section 11.12 transplanted to the matrix level. Second, an arc bound alone cannot give a path bound: with arcs at most 1, a path of length k still admits weight k. Indeed $\max_j$-spread of $d(\cdot, j)$ reached 3, and $\max_i \ell_{\mathcal{A}}(i)$ reached 2 (Proposition 131).

What makes Observation 136 worth pursuing is that it is a *pairwise* statement, and the round structure supplies a pairwise tool.

Lemma 137 (Round dominance)**.** *Let g be a chore available at the start of round t that the round-t assignment does not use. Then for every agent i, $\gamma_i^t(g_i^t) \leq \gamma_i^t(g)$.*

Proof By Lemma 135 the round-t assignment minimises $\sum_k \gamma_k^t$ over all ways of handing one available chore to each agent. Replacing agent i's chore g_i^t by g is such a way, since g is available and unused, and it changes only the i-th term. Minimality gives $\gamma_i^t(g) \geq \gamma_i^t(g_i^t)$. □

Corollary 138. *If $\gamma_i^t(g_i^t) = 1$ then every chore available and unused at round t has marginal* 1 *for agent i over A_i^{t-1}.*

Both were verified exactly: 0 violations in 6,719 checks over 520 instances. Together with the transposition bound already noted — if agent i would rather have had j's chore then j strictly prefers its own — Lemma 137 gives a complete description of what one round guarantees, and Corollary 138 is the sharp form: a round in which i pays is a round in which *nothing available was free for i*.

This is the ingredient that the additive proofs obtain from item values and that Section 11.17 showed cannot be recovered by summing rounds. Whether it can be accumulated *along an envy path* — where Lemma 133 says the quantity of interest actually lives — is the open question, and it is now a question about consecutive agents on a path rather than about all T rounds at once.

11.15 The own-cost decomposition, and a constant bound

Turning Observation 136 into a path bound needs a potential: some ϕ with $w_{\mathcal{A}}(i,j) \le \phi_j - \phi_i$ on every arc makes path weights telescope, giving $\max_i \ell_{\mathcal{A}}(i) \le \mathrm{spread}(\phi)$. Taking $\phi = -\ell_{\mathcal{A}}$ works trivially and is circular, so ϕ must come from the algorithm. Nine candidates read off the round structure were tried and none holds universally; the best, $\phi = -c_i(A_i)$, held on 583 of 657 instances (`update_28/potential_search.py`). But that candidate is worth keeping, because the algebra behind it is exact.

Lemma 139 (Own-cost decomposition)**.** *Write $\kappa_i := c_i(A_i)$ and $E_{ij} := c_i(A_j) - c_j(A_j)$, the excess of j's bundle to i over its cost to its own holder. Then for every arc,*

$$w_{\mathcal{A}}(i,j) \;=\; (\kappa_i - \kappa_j) \;-\; E_{ij},$$

and consequently, for every directed path P from a to b,

$$w_{\mathcal{A}}(P) \;=\; \kappa_a - \kappa_b \;-\; \sum_{(i,j)\in P} E_{ij}.$$

Proof $w_{\mathcal{A}}(i,j) = c_i(A_i) - c_i(A_j) = \kappa_i - (\kappa_j + E_{ij})$ by the definition of E_{ij}. Summing along P, the κ terms telescope. □

The identity is exact and holds for every allocation, not only IMWPM's; it was verified with 0 violations over all simple paths of 657 instances. It splits the path weight into a telescoping part, governed by the spread of the agents' own costs, and a defect part. The potential condition $w_{\mathcal{A}}(i,j) \le \phi_j - \phi_i$ for $\phi = -\kappa$ is exactly $E_{ij} \ge 0$, i.e. "every bundle is cheapest for the agent holding it". That fails — but only just.

Observation 140 (Both parts are bounded at an IMWPM allocation — FALSE at larger m, see Remark 142)**.** *Over* 657 *instances with $n \le 6$, $m \le 9$ (`update_28/potential_search.py` and the accompanying path sweep):*

(i) $\mathrm{spread}(\kappa) \le 2$ *always, with distribution* 0 : 200, 1 : 454, 2 : 3*;*

(ii) $E_{ij} \ge -1$ *always — all* 80 *negative excesses, out of* 8,826*, equalled exactly* -1*;*

(iii) $\sum_{(i,j)\in P} E_{ij} \ge -1$ *for* every *simple path P, with distribution* -1 : 74, 0 : 569, 1 : 14.

Part (iii) is the substantive one: individual excesses go negative and paths are long, yet the negative excesses *do not chain*. Combining it with (i) through Lemma 139:

Corollary 141 (A constant bound, conditionally — WITHDRAWN, see Remark 142)**.** *If (i) and (iii) of Observation 140 hold in general, then at every IMWPM allocation* $\max_i \ell_{\mathcal{A}}(i) \le 3$.

Proof By Lemma 139, $w_{\mathcal{A}}(P) = \kappa_a - \kappa_b - \sum_P E_{ij} \le \mathrm{spread}(\kappa) + 1 \le 3$. □

Remark 142 (Both hypotheses of Corollary 141 are false)**.** Observation 140 rested on $n \le 6$, $m \le 9$, and its part (i) was attained on only 3 of 657 instances — thin evidence, since κ_i counts *rounds* in which agent i paid and that sweep barely left $T = 3$. Re-run at m up to 12, and on families built to make the agents disagree (disjoint interest blocks, nested interest sets, one agent finding everything costly), both parts fail:

$$\begin{aligned} &\mathrm{spread}(\kappa) \text{ reached } 4 && (n=3,\ m=12,\ \text{nested}),\\ &\textstyle\sum_P E_{ij} \text{ reached } -4 && (n=3,\ m=12,\ \text{one-heavy}), \end{aligned}$$

with distributions spread(κ) given by 0:58, 1:44, 2:18, 3:31, 4:5 and path defect by −4:4, −3:21, −2:14, −1:12, 0:105 (`update_29/stress_kappa.py`). So Corollary 141 yields at best 8 at these sizes, and both of its inputs grow with m. It is withdrawn.

What the same run shows is more interesting than what it refutes: across all 156 of those larger instances, $\max_i \ell_{\mathcal{A}}(i)$ never exceeded 1 — distribution 0:129, 1:27 — which is *stronger* than Conjecture 128 requires and stronger than the small-m sweeps, where $\ell_{\mathcal{A}} = 2$ did occur. Since Lemma 139 is an exact identity, the two terms $\kappa_a - \kappa_b$ and $\sum_P E_{ij}$ must therefore be *cancelling*: both reach 4 in magnitude while their difference stays at 1.

Remark 143 (The recurring obstruction, third instance)**.** This is now the third decomposition of the same quantity in which both parts are unbounded and only the combination is controlled. Proposition 131 found it for $q_i = \tilde{p}_i + |A_i|$, where the subsidy reached 2 while the bundle size compensated exactly. Proposition 147 found it for the round decomposition, where the per-round defect accumulated to −4 while the true path weight stayed at 2. And Remark 142 finds it for the own-cost decomposition. In each case the identity is exact and the bound on the whole is far better than any bound on the parts. The lesson is not that a particular split was unlucky: it is that $\ell_{\mathcal{A}}$ at an IMWPM allocation appears not to admit *any* two-term decomposition with separately bounded parts, and a proof must treat $\kappa_a - \kappa_b - \sum_P E_{ij}$ as a single object.

11.16 The closing arc: the one path-as-a-unit tool, and why it fails

Remark 143 says a proof may not split $\ell_{\mathcal{A}}$ into separately bounded parts. Exactly one classical tool respects that: close the heaviest path into a cycle. If P is a heaviest path from a ending at b, then $P + (b, a)$ is a cycle of non-positive weight, so

$$\ell_{\mathcal{A}}(a) \;\le\; -w_{\mathcal{A}}(b,a) \;=\; c_b(A_a) - c_b(A_b). \tag{CL}$$

This is the mechanism behind Theorem 33 and behind [BDN+20, Lemma 4.1], and it is strictly weaker than the uniform balance refuted in Section 10: uniform balance asks $w_{\mathcal{A}}(v,u) \ge -1$ for *all* pairs, whereas (CL) needs it only for the single pair (b,a) with b ending a's heaviest path.

Note first that "closing arc ≥ -1" is *equivalent* to $\ell_{\mathcal{A}}(a) \le 1$, since $\ell_{\mathcal{A}}(a) \ge 2$ forces $-w_{\mathcal{A}}(b,a) \ge 2$ by (CL). So it cannot be tested as an independent hypothesis; its interest is as a proof target, being a claim about one pair rather than a path. The question that decides the route is therefore how much (CL) loses.

Proposition 144 ((CL) is too lossy at an IMWPM allocation)**.** *Over* 452 *instances with* $n \le 6$, $m \le 12$, *on uniform, disjoint-interest and nested-interest families, the* 210 *pairs* (a, b) *with* b *ending a heaviest path out of* a *satisfied:*

- *closing arcs $w_{\mathcal{A}}(b,a)$ distributed* $-3:18$, $-2:103$, $-1:89$ *— reaching* -3*;*
- *slack* $-w_{\mathcal{A}}(b,a) - \ell_{\mathcal{A}}(a)$ *distributed* $0:89$, $1:103$, $2:18$*, so (CL) is strictly lossy on* 58% *of the pairs and loses up to* 2*.*

Consequently (CL) certifies only $\ell_{\mathcal{A}}(a) \le 3$ where the truth is 1*, and proving "closing arc ≥ -1" is a strictly harder statement than proving $\ell_{\mathcal{A}} \le 1$ (`update_30/closing_arc.py`).*

So the route is worse than what it replaces, and the reason is instructive: endpoint pairs are the *worst* pairs, not the best. Across the same run the arc distribution over all ordered pairs was $-4:5$, $-3:94$, $-2:619$, $-1:1846$, $0:1709$, $1:273$, so a typical arc is -1 or 0 while every endpoint arc is at most -1. That is forced by what b is: the agent everyone envies towards, hence the agent for whom other bundles look most expensive.

Remark 145 (Fourth instance of the cancellation)**.** Proposition 144 is Remark 143 again in a fourth guise. (CL) bounds $\ell_{\mathcal{A}}$ by a single quantity, and that quantity is systematically more extreme than $\ell_{\mathcal{A}}$ itself. Together with the subsidy/size split, the round decomposition and the own-cost decomposition, every attempt to bound $\ell_{\mathcal{A}}$ by anything other than $\ell_{\mathcal{A}}$ has now come in strictly worse — and the closing arc is the sharpest such attempt available, since it is the only one that treats the path as a unit. Whatever keeps $\ell_{\mathcal{A}}$ small at an IMWPM allocation is not visible in any single arc, any single round, or any two-term split of the path.

The endpoint agents are nevertheless highly structured, which is worth recording even though it did not rescue the route: of the 210 endpoints, 100% had $\ell_{\mathcal{A}}(b) = 0$ (by construction), 94.3% had the minimum own cost $c_b(A_b)$ among all agents, 83.8% paid in no round at all, and 73.8% held a smallest bundle.

11.17 Auditing [LMS26]: the missing hypothesis is submodularity

IMWPM is taken from [LMS26], whose own analysis had not been read against our setting. Two things had to be checked: whether that paper covers non-additive valuations, and whether its proof technique transplants.

The first is settled by its Section 2, which fixes $u_i(S) = \sum_{j\in S} u_i(j)$: the paper is entirely additive. But its Proposition 3.1 recovers the bound of [BDN+20] by an argument far simpler than the modified-valuation construction audited in Section 11.12. For a path $P = (1, \dots, k)$ with terminal agent k, put $F_t := \max(\{0\} \cup \{u_k(g) : g \in J^t\})$, the best item still available to k. Maximum-weight optimality of the round-t matching, compared against the alternative that shifts each path item one step back and hands k the best remaining item, gives $w_t(P) \le F_t - F_{t+1}$; summing telescopes to $w_{\mathcal{A}}(P) \le F_1 \le 1$.

Additivity enters this proof in exactly two places, both of the form "sum the rounds":

$$\text{(A)} \quad u_i(A_i) = \sum_t u_i(\mu_i^t), \qquad\qquad \text{(B)} \quad u_i(A_j) = \sum_t u_i(\mu_j^t).$$

Lemma 146 ((A) survives with marginal weights)**.** *Our IMWPM matches on marginals, so writing A_i^t for agent i's bundle after round t,*

$$\tilde{v}_i(A_i) \;=\; \sum_t \Big[\tilde{v}_i(A_i^t) - \tilde{v}_i(A_i^{t-1})\Big] \;=\; \sum_t \mu_i^t(\textit{own item}),$$

an exact telescoping identity requiring no additivity whatever.

So (A) costs nothing, and (B) is the entire gap. What the argument actually needs is only one direction of it,

$$\text{(B}'\text{)} \qquad \tilde{v}_i(A_j) \;\leq\; \sum_t \Big[\tilde{v}_i\big(A_i^{t-1} \cup \big\{\mu_j^t\big\}\big) - \tilde{v}_i\big(A_i^{t-1}\big)\Big],$$

and (B′) is precisely *submodularity*: for a submodular v one has $v(S \cup T) - v(S) \leq \sum_{g \in T}[v(S \cup \{g\}) - v(S)]$, which is (B′) read along the rounds. Dichotomous functions need not be submodular, and the failure is not marginal.

Proposition 147 (The defect, and why the transplant fails)**.** *Write* $D_{ij} := \sum_t [\tilde{v}_i(A_i^{t-1} \cup \{\mu_j^t\}) - \tilde{v}_i(A_i^{t-1})] - \tilde{v}_i(A_j)$*, so that (B′) is the claim* $D_{ij} \geq 0$*, and*

$$w_{\mathcal{A}}(P) \;\leq\; 1 - \sum_{i \in P} D_{i,i+1}.$$

Over 800 *instances, (A) failed* 0 *times, while (B′) failed on* 2,103 *of* 6,660 *ordered pairs* — 31.6% — *with individual defects reaching* −2*. Accumulated along a path the defect reached* −4*, so the transplanted bound is* $w_{\mathcal{A}}(P) \leq 5$ *at* $n \leq 5$*, and in general degrades as* $1 + 2(n-1)$*.*

The observed maximum path weight is 2. So the technique of [LMS26], carried across as far as it will go, gives a bound of $1 + 2(n-1)$ where the truth appears to be 2: no better than the $n-1$ per agent already available from [KMS+24], and the route yields nothing. The defect accumulates along the path, which is exactly the failure mode that "sum the rounds" arguments cannot survive.

Remark 148 (Why this corner of the 2×2 is the open one)**.** Sections 11.12 and this one identify the same missing hypothesis from two directions. [BDN+20] needs additivity to *build* a modified valuation from item values; [LMS26] needs it to *decompose* a bundle's value across rounds, and what would suffice there is submodularity rather than additivity outright. Dichotomous cost functions lie outside both: they are not additive, and $c(S) = \min(|S|, 1)$ shows they need not be submodular. The dichotomous *goods* corner is settled in [BKNS22] precisely because binary submodular valuations are matroid rank functions, where (B′) is free. Our corner has neither hypothesis available, and Proposition 147 measures how far from free (B′) actually is. This is a quantitative account of why the fourth corner has stayed open, rather than a restatement that it is hard.

Remark 149 (Correction to Section 11.4)**.** Proposition 132 corrects the claim made when the algorithm was stated, that "every step preserves cardinality balance, so termination at spread ≤ 1 directly witnesses Target G-bal". The repair loop does reject moves that leave the partition unbalanced, but it never runs when the warm start already has q-spread at most 1, and that warm start may itself be unbalanced. Of 1,190 successful runs, 62 terminated at a partition whose bundle sizes differ by 2. The algorithm therefore certifies Target G, not Target G-bal. Nothing is lost for Conjecture 2, which follows from Target G alone by Proposition 112; but the evidence recorded in Section 11.5 should be read as evidence for Target G, and Definition 113 is not what the algorithm delivers.

12 Approach 7: descent on a lexicographic potential

12.1 The reformulation that suggests it

Section 10 characterises good allocations by the two-tier condition: $\mathcal{A}$ is good precisely when there is a bipartition $S \subseteq N$ with

$$c_i(A_i) - [\, i \in S \,] \;\le\; c_i(A_j) - [\, j \in S \,] \qquad \text{for all } i \neq j.$$

Writing $\delta_j := [\, j \in S \,]$ and $\hat{c}_i(A_j) := c_i(A_j) - \delta_j$, this says exactly that *every agent holds a bundle minimising its discounted cost*, where the discount is 1 on the paid slots and 0 elsewhere and depends only on the slot, not on the agent. Unpacked by tier:

- within S, and within $N \setminus S$: exact envy-freeness;
- an unpaid agent *strictly* prefers its own bundle to every paid agent's bundle;
- a paid agent envies an unpaid agent by at most 1.

Two consequences are worth stating before any search is set up. First, $S = N$ and $S = \emptyset$ impose *identical* constraints, since the two indicators cancel; this is the structural reason the paid set is never all of N, and it is the combinatorial shadow of Proposition 13. Second, the existential over subsidy vectors has become an existential over bipartitions, of which there are 2^n — the first formulation in which it is finite.

12.2 The search space and the potential

By Theorem 10(ii) an allocation is envy-freeable iff it maximises welfare among reassignments of its own bundles. So we search over *partitions* and let the assignment be determined:

Definition 150 (Canonical form)**.** *For a partition $\mathcal{B} = (B_1, \dots, B_n)$ of M, the* canonical allocation *$\mathcal{A}(\mathcal{B})$ assigns bundles to agents by a minimum-cost perfect matching. It is envy-freeable, so $\ell_{\mathcal{A}(\mathcal{B})}$ is finite.*

Definition 151 (The potential)**.** *For a partition $\mathcal{B}$ with canonical allocation $\mathcal{A}$, set*

$$\Psi(\mathcal{B}) \;:=\; \Big(\max_{i \in N} \ell_{\mathcal{A}}(i),\; \sum_{i \in N} |B_i|^2 \Big) \;\in\; \mathbb{Z}_{\ge 0} \times \mathbb{Z}_{\ge 0},$$

ordered lexicographically. A transfer *moves one chore from one bundle to another.*

The conjecture this approach rests on is a statement about single transfers.

Conjecture 152 (Descent lemma)**.** *Let $\mathcal{B}$ be a partition whose canonical allocation has* $\max_i \ell_{\mathcal{A}}(i) \ge 2$. *Then some transfer $\mathcal{B}'$ satisfies $\Psi(\mathcal{B}') < \Psi(\mathcal{B})$.*

12.3 The reduction, which is unconditional

Theorem 153 (Descent suffices)**.** *Conjecture 152 implies Conjecture 2 in full: every instance with negative dichotomous valuations admits an envy-free solution $(\mathcal{A}, p)$ with $p \in \{0,1\}^n$ and total subsidy at most $n-1$, and one is computable in time $O(m^4 n^4)$ given value-oracle access.*

Proof Start from any partition and pass to canonical form. While $\max_i \ell_{\mathcal{A}}(i) \geq 2$, apply Conjecture 152 to obtain a transfer strictly decreasing Ψ.

Termination. Ψ takes values in $\mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}$ under the lexicographic order, which is a well-order, so no infinite strictly decreasing sequence exists. Quantitatively, every arc weight is at most m in absolute value and $\ell_{\mathcal{A}}$ is a weight of a simple path, so the first coordinate lies in $\{0, 1, \ldots, m\}$; the second lies in $\{m, \ldots, m^2\}$, since $\sum_i |B_i|^2$ is minimised by an equal split and maximised by putting everything in one bundle. Hence Ψ has at most $(m+1)(m^2+1) = O(m^3)$ values and the loop runs $O(m^3)$ times.

Correctness. The loop can only exit with $\max_i \ell_{\mathcal{A}}(i) \leq 1$. By Observation 12 the minimum subsidy vector is integral and non-negative, so $p^* \in \{0,1\}^n$; by Theorem 11 the pair $(\mathcal{A}, p^*)$ is an envy-free solution; and by Proposition 13 the total is at most $n-1$.

Complexity. One iteration examines $O(mn)$ transfers. Each requires a minimum-cost perfect matching on n agents, $O(n^3)$ by the Hungarian method with $O(n^2)$ oracle calls, followed by an all-pairs longest-path computation on an n-vertex graph, $O(n^3)$. So an iteration costs $O(mn \cdot n^3) = O(mn^4)$ and the whole procedure $O(m^4 n^4)$. □

Theorem 153 is the reason this approach is worth the space: unlike every earlier route, the gap between what is proved and Conjecture 2 is a *single* statement about *single* transfers, and closing it delivers the polynomial-time clause for free rather than as a separate construction.

12.4 Why this potential and not the obvious one

The first potential tried was $\Psi_1 = (\max_i \ell_{\mathcal{A}}(i), \#\{i : \ell_{\mathcal{A}}(i) = \max\}, \sum_i c_i(B_i))$, minimising total cost as a tie-break. It fails, and the witness explains the failure rather than merely exhibiting it. At $n = 3$, $m = 4$ with all four chores in one bundle and the other two empty,

$$\mathcal{B} = (\{g_1, g_2, g_3, g_4\}, \emptyset, \emptyset), \qquad \ell_{\mathcal{A}} = (0, 0, 2), \qquad \textstyle\sum_i c_i(B_i) = 2,$$

every transfer leaves $\max_i \ell_{\mathcal{A}}(i) = 2$ and *raises* the total cost. The reason is that dichotomous costs cap: the loaded agent already values its bundle at 2, and removing one chore does not reduce that, while the recipient's cost rises. So the third coordinate pins the search at a utilitarian optimum — precisely the class shown in Section 11 not to be good in general. The tie-break must reward *spreading chores out*, not minimising their total cost. Both $\sum_i |B_i|^2$ and $\max_i |B_i|$ do; $\sum_i \ell_{\mathcal{A}}(i)$ in place of the maximum does not.

potential	tie-break	stuck partitions
$(\max \ell,\ \#\text{at max},\ \sum_i c_i(B_i))$	total cost	449
$(\max \ell,\ \max_i \lvert B_i \rvert)$	largest bundle	160
$(\sum_i \ell_i,\ \sum_i \lvert B_i \rvert^2)$	total envy first	464
$(\max \ell,\ \#\text{at max},\ \max_i \lvert B_i \rvert)$	—	0
$(\max \ell,\ \#\text{at max},\ \sum_i \lvert B_i \rvert^2)$	—	0
$\Psi = (\max \ell,\ \sum_i \lvert B_i \rvert^2)$	—	0

Ψ is the shortest surviving potential, which is why Definition 151 takes it.

12.5 Where the remaining burden sits

The second coordinate of Ψ moves in a completely explicit way, and this splits Conjecture 152 into two unequal halves.

Lemma 154 (Balance dichotomy)**.** *Transferring a chore from a bundle of size s to a bundle of size t changes $\sum_i |B_i|^2$ by exactly $2(t-s+1)$. Hence the transfer strictly decreases the second coordinate of Ψ if and only if $t \le s-2$. Consequently, if $\mathcal{B}$ is* balanced *— all bundle sizes within 1 of one another — then no transfer decreases the second coordinate, and Conjecture 152 demands a transfer that strictly decreases $\max_i \ell_{\mathcal{A}}(i)$ itself.*

Proof The sizes change from (s,t) to $(s-1,t+1)$, so the sum changes by $(s-1)^2+(t+1)^2-s^2-t^2 = 2(t-s+1)$, which is negative exactly when $t < s-1$. If all sizes lie in $\{q, q+1\}$ then $t \ge q \ge s-1$ for every ordered pair, so no transfer qualifies. □

So Conjecture 152 is really two claims. On unbalanced partitions one must find a size-equalising transfer that does not *increase* the maximum subsidy; on balanced partitions one must strictly decrease the maximum subsidy by a single transfer. The second is the hard half and the natural next target: it is a statement about balanced partitions only, where every bundle has size $\lfloor m/n \rfloor$ or $\lceil m/n \rceil$, which is a far more rigid object than an arbitrary allocation. Note that the transfer is followed by re-matching, so a single transfer may permute the assignment substantially — this is where the strength of the move comes from, and any proof must use it.

12.6 The obstruction: the witnessing transfer is not local

Conjecture 152 asserts that *some* transfer works. A proof wants a *rule* — a named transfer, definable from the envy structure, that always works. The obvious candidates all fail, and the way they fail is informative.

Two mechanisms can reduce an arc $w_{\mathcal{A}}(i,j) = c_i(A_i) - c_i(A_j)$: remove a chore from A_i to lower $c_i(A_i)$, or add one to A_j to raise $c_i(A_j)$. Under dichotomous costs *both can fail outright*, because costs cap and marginals need not be monotone:

- $c(S) = \min(|S|, 1)$ has no chore whose removal lowers the cost of a two-element bundle;
- $c(S) = \max(|S|-1, 0)$ has no chore whose addition raises the cost of the empty bundle.

Note that the second example also rules out the submodular shortcut: a chore may have marginal 1 over a superset of A_j and marginal 0 over A_j itself, so "some chore raises $c_i(A_j)$" cannot be inferred from c_i being large somewhere. This is the same failure of the exchange property that has obstructed every earlier approach in this report.

Five rules were tested against all 7,416 crux partitions — balanced, with $\max_i \ell_{\mathcal{A}}(i) \geq 2$ — under two criteria: the Ψ criterion (strictly reduce $\max_i \ell_{\mathcal{A}}(i)$) and the weaker P_3 criterion (reduce $(\max_i \ell_{\mathcal{A}}(i), \#\text{at max})$ lexicographically).

rule	transfer	Ψ criterion	P_3 criterion
R1	argmax-ℓ bundle $\rightarrow$ max-path endpoint	2,856	3,168
R2	argmax-ℓ bundle $\rightarrow$ its cheapest bundle	3,827	4,021
R3	argmax-ℓ bundle $\rightarrow$ anywhere	5,112	5,152
R4	anywhere $\rightarrow$ max-path endpoint	6,039	6,467
R5	largest bundle $\rightarrow$ smallest bundle	3,972	4,086
	every transfer (Conjecture 152)	7,416	7,416

The decisive row is R3. In 2,264 of 7,416 crux partitions, no transfer *out of the envious agent's own bundle* works, even under the weaker criterion — the witnessing move involves two bundles neither of which belongs to an agent attaining the maximum. So the move that repairs the envy is not local to the envy. Any proof of Conjecture 152 must therefore be non-constructive, or must name a rule of a kind not yet imagined; the five natural ones are exhausted.

Remark 155 (The descent lemma is strictly stronger than what is needed)**.** Conjecture 152 asserts more than Conjecture 2: it says both that good allocations exist and that greedy descent on Ψ finds one, i.e. that Ψ has no local minimum with $\max_i \ell_{\mathcal{A}}(i) \geq 2$. Conjecture 2 is the statement that the *global* minimum of Ψ has first coordinate at most 1. The extra strength is what buys the polynomial-time clause in Theorem 153 at no cost, but it also means this route is harder than the target, and the obstruction above may be a symptom of exactly that. A proof of Conjecture 2 alone need not go through any local search.

12.7 Evidence

Conjecture 152 was tested by enumerating *every* partition of every test instance, computing Ψ at each and at all $O(mn)$ of its transfers, and recording any partition with $\max_i \ell_{\mathcal{A}}(i) \geq 2$ admitting no strict decrease.

family	instances	stuck partitions
random, $n \leq 4$, $m \leq 6$	1,732	0
EF-free only, $n \leq 5$, $m \leq 6$	253	0
R10 Set-Splitting, certified hard	4	0
random, $n = 3$, $m \leq 9$	798	0
random, $n = 4$, $m \leq 7$	280	0
random, $n = 5$, $m \leq 6$	75	0
crux partitions only (balanced, $\max \ell \geq 2$)	7,416	0

The last row is the one that matters, and it was measured separately because the others do not imply it: "zero stuck partitions over all partitions" would be consistent with the crux population being empty. It is not. It is empty when $m = n$ — a balanced partition then gives every agent one chore, and the min-cost matching caps $\ell_{\mathcal{A}}$ at 1, so there is nothing to prove — and it grows with m, reaching 990 crux partitions at $n = 3$, $m = 9$ and 2,112 at $n = 4$, $m = 6$. Every one of them admits a Ψ-decreasing transfer.

The large-m rows matter specifically: a size-based potential should fail when bundles grow, since $\sum_i |B_i|^2$ can only be decreased by equalising sizes while a large bundle may be cheap under a capped cost function. No such failure appears up to $m = 9$. Separately, no instance in any of these families lacked a good allocation, so Conjecture 2 itself remains unrefuted throughout.

13 Approach 8: the global route

13.1 Why global

Approach 7 reduced Conjecture 2 to a descent lemma, but Remark 155 records that the descent lemma is strictly stronger than the conjecture: it forbids bad *local* minima of Ψ, whereas Conjecture 2 only concerns the *global* minimum. The global route drops the extra strength. It asks:

> is there an objective whose global optimum is always a good allocation?

Such a statement would give existence with no reference to local search, and the polynomial-time clause becomes a separate question about computing the optimum. All objectives below are evaluated over partitions in canonical form (Definition 150), so every candidate is envy-freeable and $\ell_{\mathcal{A}}$ is finite.

13.2 What is refuted

For an objective f, two questions differ: are *all* f-optima good (a usable theorem, since then any optimum may be returned), or is merely *some* f-optimum good (which needs a tie-break before it is a theorem)? Over 1,760 instances at $n \le 5$, $m \le 7$:

objective		all optima good	some optimum good
$\sum_i c_i(B_i)$	utilitarian	389 fail	50 fail
sorted cost vector	leximax	54 fail	21 fail
$\max_i c_i(B_i)$	egalitarian	382 fail	0
$\sum_i \lvert B_i\rvert^2$	balance	68 fail	0
$\max_i c_i - \min_i c_i$	spread	201 fail	0

The utilitarian row restates the refutation of Section 11. The leximax row is new and worth recording, because leximax is the objective the goods literature reaches for first: on 21 instances *every* leximax optimum is bad, so no tie-break can rescue it. Egalitarian, balance and spread all survive in the weaker sense.

A tie-break chain would upgrade a “some” to a theorem, and twelve were tried. None succeeds:

chain	failures	chain	failures
bal.	75	bal. → lex.	4
bal. → util.	21	bal. → lex. → util.	4
bal. → egal.	30	bal. → egal. → lex. → util.	4
bal. → egal. → util.	8	egal. → bal.	48
bal. → util. → egal.	8	egal. → bal. → util.	9
bal. → spread	15	bal. → #at max → util.	29

Adding components helps only down to 4 failures out of 1,082 and then stops; the last three chains are progressively finer and all fail on the same residue. So the good allocations are not picked out by any natural cost-based ordering. Note also that appending “# agents at maximum cost” actively *breaks* the objective, turning 0 “some” failures into 21.

13.3 What survives: the balance lemma

The balance objective is special because its optima are describable outright: $\sum_i |B_i|^2$ is minimised exactly by the *balanced* partitions, those with all bundle sizes within 1 of one another, i.e. every bundle of size $\lfloor m/n \rfloor$ or $\lceil m/n \rceil$. So its “some optimum is good” row reads as a structural restriction:

Conjecture 156 (Balance lemma)**.** *Every instance with negative dichotomous valuations admits a good allocation whose partition is balanced.*

Proposition 157. *Conjecture 156 implies Conjecture 2, except for its polynomial-time clause.*

Proof Immediate: a good allocation is one with $\max_i \ell_{\mathcal{A}}(i) \le 1$, which by Observation 12 and Proposition 13 yields $p^* \in \{0,1\}^n$ with total at most $n-1$. □

Remark 158 (Relation to Target G-bal, and a correction)**.** Conjecture 156 was arrived at here by scanning global objectives, but the restriction to cardinality-balanced partitions is not new to this section: it is Definition 113, introduced in Section 11 on the goods side of the size-shift involution. The two are not the same statement, and the direction matters. Proposition 112 bounds $\ell_{\mathcal{A}}(u)$ by $q_u - \min_v q_v$ in one direction only, so Target G-bal implies Conjecture 156 but not conversely — and the gap is real, not merely unproved. Over the balanced partitions enumerated in `update_22`, 666 are good while carrying q-spread at least 2, whereas none has q-spread at most 1 without being good. So Conjecture 156 is a strict weakening of Target G-bal, and to that extent a better target; but the idea of restricting to balanced partitions was already in hand, and this section should not be read as introducing it.

Conjecture 156 is stronger than Conjecture 2, and it is worth being explicit about why that strengthening is of a more useful kind than the one in Approach 7. The descent lemma strengthens the conjecture by constraining the *optimisation landscape* — it must have no bad local minimum — which is a statement about every partition. The balance lemma strengthens it by constraining the *witness*, and a constrained witness is extra structure a proof may use: on a balanced partition every bundle has one of two sizes, m is nearly determined by n and the common size, and the pigeonhole arguments that drive Example 1 and the small-bundle theorem of Section 5 apply directly.

family	instances	no good balanced allocation
random, $n \leq 5$, $m \leq 7$	1,760	0
random, $n \leq 5$, $m \leq 8$	1,082	0
EF-free only	30	0
balanced partitions examined		466,560

The two approaches also fit together rather than competing. The second coordinate of Ψ in Definition 151 is exactly $\sum_i |B_i|^2$, so descent under Ψ drives the partition towards balance; Conjecture 156 asserts that the balanced region is where the good allocations are. That the empirically successful potential and the empirically successful global objective agree on the same quantity is weak evidence that balance, not cost, is the operative structure.

13.4 The induction the balance lemma invites, and why it does not close

Restricting the witness suggests building it one chore at a time. Three statements of increasing strength were tested; the first holds, and the two that would prove anything fail.

Proposition 159 (Incremental structure, empirical)**.** *Write* (INC) *for: there is a good balanced allocation $\mathcal{A}$, with its assignment fixed, and an ordering $g_1, \ldots, g_m$ of the chores such that $\mathcal{A}$ restricted to $\{g_1, \ldots, g_k\}$ is good for every k. Over* 1,125 *instances,* (INC) *never failed — not even when the assignment was forbidden to change between steps.*

Restrictions of dichotomous cost functions to a subset of chores are again dichotomous, so each prefix is a legitimate smaller instance and (INC) has exactly the shape of an induction on m. It is not one. (INC) *presupposes* a good balanced allocation and then asserts it can be reached; it says nothing about existence, and the induction does not close, because the good allocation supplied by the hypothesis on $m-1$ chores need not be the restriction of any good allocation on m.

The statement that would prove existence has to be self-contained:

> (NS) if $\mathcal{A}$ is a good allocation of a proper subset T of the chores, some chore outside T can be added to some bundle leaving it good.

(NS) implies Conjecture 2 by induction from the empty allocation, which is good since all costs are 0, and gives an $O(m^2 n^4)$ algorithm with no backtracking. It is false. Over 502,528 good partial allocations, 846 admit no good one-chore extension, and 150 resist even when re-matching is permitted. The first witness is

$$A = (\emptyset,\ \{g_1, g_2, g_3\},\ \emptyset), \qquad g_4 \text{ unassigned},$$

three chores piled on one agent with two agents empty — a state a balanced construction would never reach. That suggests the repair: keep the partial allocation balanced, inserting only into minimum-size bundles, which preserves balance and would prove the balance lemma and Conjecture 2 together.

Proposition 160 (The balanced induction fails, and fails worse)**.** *Write* (NSB) *for* (NS) *restricted to balanced partial allocations with insertion into a minimum-size bundle. Over* 236,421 *good balanced partial allocations,* 11,723 *are stuck —* 5.0%, *against* 0.17% *for the unrestricted* (NS). *Allowing insertion into any balance-preserving bundle rather than a minimum-size one changes nothing: the same* 11,723 *states are stuck.*

So balance does not rescue the induction; it makes it worse, by removing exactly the freedom — piling chores onto one agent for a step — that lets greedy insertion escape. Note that this does not contradict Proposition 159: a good insertion order exists, but it cannot be found by extending an arbitrary good balanced partial allocation. The order must be chosen with foresight.

Remark 161 (The pattern across Approaches 7 and 8)**.** Three independent constructive routes have now failed in the same way. The descent lemma is true in every test but its witnessing transfer is not local to the envy (Section 12.6); (INC) is true in every test but the insertion order is not locally determined; and no lexicographic objective picks out the good allocations. In each case the object exists and no rule computable from local data finds it. That is weak evidence that Conjecture 2, if true, is not provable by exhibiting a construction, and that the search should turn to non-constructive existence arguments — counting, averaging, or a fixed-point or topological argument — with the algorithmic clause pursued separately through Section 11, whose algorithm is polynomial by construction and has never failed.

13.5 The non-local route: is there room, and can anything reach it?

Remark 145 closes the local and decompositional routes: four independent attempts to bound $\ell_{\mathcal{A}}$ by a part of itself all came in strictly worse than the truth. What remains is the non-local class — counting, averaging, probabilistic-method and parity arguments, which reason about all allocations at once. Two questions decide whether that class is usable: whether the good allocations are numerous enough to be reached by such an argument, and whether any natural distribution or identity actually reaches them.

Observation 162 (There is room)**.** *Exhaustively over all n^m allocations of* 1,169 *instances with $n \leq 5$, $m \leq 6$, the number of good allocations never fell below* 11*, and on the* 80 *EF-free instances — the residual class, since Conjecture* 2 *is trivial where an exactly envy-free allocation exists — never below* 27 *(`update_31/counting_room.py`).*

So the good set is not a needle: an argument that merely shows it non-empty has something to aim at, unlike the situation where only one witness exists and nothing short of exhibiting it can work. But the two standard ways of reaching it both fail.

Proposition 163 (Parity fails)**.** *The number of good allocations has no fixed parity: over* 737 *instances it was even* 469 *times and odd* 268 *times. No Sperner- or Tucker-style identity forcing an odd count can hold.*

Proposition 164 (No natural distribution concentrates)**.** *The fraction of allocations that are good is not bounded below. Under the uniform distribution its minimum fell from* 0.136 *at* $n = 3$, $m = 4$ *to* 0.025 *at* $n = m = 5$*. Restricting to cardinality-balanced partitions does not repair this — the minimum fraction there was* 0.042 *at* $n = m = 4$, worse *than uniform — and restricting to minimum-cost allocations gives a minimum fraction of* 0*, reconfirming Section 11's refutation of that family.*

The probabilistic method needs a distribution under which the good set has probability bounded away from zero, and Proposition 164 rules out the three natural candidates; a counting proof needs an identity forcing the count non-zero, and Proposition 163 rules out the parity identity. What is *not* ruled out is a cleverer weighting: the fractions above are for specific distributions, and the space of weightings is unbounded. But no weighting suggested by the structure of the problem survives, and Observation 162 shows the target is generous enough that if a natural one existed it would likely have been among these.

Remark 165 (Status of the technique classes)**.** Taken with Remark 145, every named technique class has now been tried and refuted on this problem: constructive (Sections 12.6, 13.4), decompositional (Remark 143, four instances), transplanted from the additive literature (Sections 11.12, 11.17), and now non-local. This is not a claim that Conjecture 2 is false — it survives every test at every size — nor that it is unprovable. It is a claim about where a proof will not come from, and it is the most useful thing this report currently establishes.

13.6 The algorithmic clause

Conjecture 2 also asks for a polynomial-time algorithm, and Proposition 157 deliberately excludes it. The position is as follows.

- Minimising $\sum_i |B_i|^2$ is trivial — every balanced partition is optimal — but that is exactly why the balance lemma gives no algorithm on its own: it does not say *which* balanced partition is good, and there are exponentially many.
- The objectives that would name one are not obviously tractable. Minimising $\max_i c_i(B_i)$ is a makespan-type problem and NP-hard already for identical additive costs, so "compute the egalitarian optimum" is not a polynomial algorithm even where it is correct.
- Two polynomial candidates remain and are unaffected by this section: the algorithm of Section 11, whose completeness is Conjecture 123 and which has never failed in 389,215 instances; and descent on Ψ, which Theorem 153 shows runs in $O(m^4 n^4)$ and is correct given Conjecture 152.

So the division of labour is now explicit: the global route is the cleanest line on *existence*, and the algorithmic clause rests on Approach 6 or Approach 7. A proof of Conjecture 156 would settle the mathematical content of Conjecture 2 and leave a well-posed, separately attackable computational question — which is a better position than the current one, where existence and computation are entangled in a single conjecture that has resisted both.

14 Approach 9: The Conditioned-Remainder Induction

Section 8 left exactly one statement open, Conjecture 59, and Theorem 86 recorded that random testing had run out of power on it. What followed narrowed the target without closing it: three families of candidate invariant were eliminated — a local sufficient condition on the peel (Proposition 69), a predicate on the pair (state, move) (Theorem 89), and the commitment predicate on the state alone (Theorem 88) — leaving no candidate at all.

This section changes the state space rather than hunting for another predicate on the old one. The diagnosis is that a workload profile is the wrong object: it is an arbitrary tuple of subsets, there are $(2^m)^n$ of them, and an individual profile means nothing. What follows replaces it with a space whose elements are *partial allocations*, and in which the peel frame's known dead ends provably do not lie.

14.1 The frame

Definition 166 (CR state)**.** *A* conditioned-remainder state*, or CR state, is a pair* $(\mathcal{A}, R)$ *with* $R \subseteq M$ *and* $\mathcal{A} = (A_1, \ldots, A_n)$ *a partition of* $D := M \setminus R$*. Its* profile *is*

$$W_i \ := \ A_i \cup R,$$

so every agent is still on the hook for the whole undecided remainder, and its contracted cost functions *are*

$$c_i^R(T) \ := \ c_i(T \cup R) - c_i(R), \qquad T \subseteq D.$$

The moves are assignment *— give some* $a \in R$ *to some agent* x*, setting* $A'_x = A_x \cup \{a\}$ *and* $R' = R \setminus \{a\}$ *— and* relabelling*,* $\mathcal{A} \mapsto \mathcal{A} \circ \sigma$*. The* root *has* $\mathcal{A}$ *empty and* $R = M$*; a state is* terminal *when* $R = \emptyset$*; and* legality *is the invariant of Theorem 47: no positive-weight cycle and* $\ell_W(i) \le 1$ *for every* i*.*

In the language of Theorem 47 an assignment is one *atomic block*: relieve all $n-1$ non-owners of a at once. So a CR state is a peel state, a CR move is $n-1$ consecutive peels, and the frame differs from Conjecture 90 exactly by dropping the legality requirement at the $n-2$ intermediates *within* a block.

Conjecture 167 (CRI — **open**)**.** *From the root, some sequence of assignments and relabellings reaches a terminal state with every intermediate CR state legal.*

Conjecture 90 implies Conjecture 167 implies Conjecture 2, the last because a terminal CR state *is* an allocation and its legality is exactly $\max_i \ell_{\mathcal{A}}(i) \le 1$. So Conjecture 167 is strictly the weaker target of the two peel-frame statements, and unlike Conjecture 59 it is not obviously stronger than what is wanted.

14.2 Why the frame is different

Three things separate it from a restriction of the peel process.

The space is small. CR states are partial functions $M \to N$, so there are $(n+1)^m$ of them against $(2^m)^n$ profiles: at $n = 3, m = 6$, 4,096 against 262,144. Reachability is therefore decidable exhaustively well past every sweep of Section 8.

The known dead ends are not CR states. A profile is a CR state if and only if the sets $W_i \setminus \bigcap_k W_k$ are pairwise disjoint and cover $M \setminus \bigcap_k W_k$. Both witnesses recorded in Section 8.5 repeat an item across bundles — $(\{a_1, a_2\}, \{g\}, \{g\})$ and $(\{g_2\}, \{g_2\}, \{g_1, g_3, g_4\})$ — so neither is a CR state. The obstruction of Proposition 52 is removed rather than worked around.

The induction is restored. This is the substantive point, and it is Lemmas 168 and 169 below: a CR state is a *solved instance of the problem on a contracted ground set*, and a move un-contracts one element. Conjecture 167 is therefore an induction in which the instance *grows* while a witness is maintained — the shape of the proof of [BKNS22] — run on the conditioned cost functions that dissolve the obstruction of Theorem 36. Section 6 grew the instance with the *unconditioned* costs and died on that obstruction; Section 8 identified the right conditioning in Theorem 50 and then abandoned the induction for a state-space search. This section is the synthesis of the two.

14.3 The calculus

Lemma 168 (Contraction stays in the class)**.** *c_i^R is normalised with every marginal in $\{0,1\}$. Hence (D, c^R) is again a negative dichotomous instance.*

Proof $c_i^R(\emptyset) = c_i(R) - c_i(R) = 0$, and for $g \notin T \subseteq D$,

$$c_i^R(T \cup \{g\}) - c_i^R(T) \;=\; c_i\big((T \cup R) \cup \{g\}\big) - c_i(T \cup R) \;\in\; \{0,1\}$$

because c_i is dichotomous (Theorem 7). $\square$

Lemma 169 (A CR state is a witness on the contracted instance)**.** $w_W(i,k) = c_i^R(A_i) - c_i^R(A_k)$. *Consequently $(\mathcal{A}, R)$ is legal if and only if $\mathcal{A}$ witnesses Conjecture 2 on the instance (D, c^R).*

Proof $w_W(i,k) = c_i(A_i \cup R) - c_i(A_k \cup R) = \big[c_i^R(A_i) + c_i(R)\big] - \big[c_i^R(A_k) + c_i(R)\big]$, and the $c_i(R)$ cancel. The two envy graphs are therefore equal arc for arc, so the legality conditions coincide. $\square$

Lemma 170 (CR arc update)**.** *Let $(\mathcal{A}, R) \to (\mathcal{A}', R')$ assign a to x, and write $\beta_i(T) := c_i^{R'}(T \cup \{a\}) - c_i^{R'}(T) \in \{0,1\}$ for the marginal of a to agent i on T, conditioned on R'. Then for $i \neq k$,*

$$w_{\mathcal{A}'}(i,k) - w_{\mathcal{A}}(i,k) \;=\; \beta_i(A_k)\,[k \neq x] \;-\; \beta_i(A_i)\,[i \neq x].$$

So arcs into *x fall by $\beta_i(A_i)$, arcs* out of *x rise by $\beta_x(A_k)$, and arcs between third parties move by $\beta_i(A_k) - \beta_i(A_i) \in \{-1, 0, 1\}$.*

Proof First the contraction identity: for $T \subseteq D$,

$$\begin{aligned} c_i^{R'}(T \cup \{a\}) - c_i^{R'}(\{a\}) &= \big[c_i(T \cup \{a\} \cup R') - c_i(R')\big] - \big[c_i(\{a\} \cup R') - c_i(R')\big] \\ &= c_i(T \cup R) - c_i(R) \;=\; c_i^R(T). \end{aligned}$$

Applying it to both terms of $w_{\mathcal{A}}(i,k)$, the $c_i^{R'}(\{a\})$ cancel and

$$w_{\mathcal{A}}(i,k) = c_i^{R'}(A_i \cup \{a\}) - c_i^{R'}(A_k \cup \{a\}) = \big[c_i^{R'}(A_i) + \beta_i(A_i)\big] - \big[c_i^{R'}(A_k) + \beta_i(A_k)\big].$$

Meanwhile $w_{\mathcal{A}'}(i,k) = c_i^{R'}(A'_i) - c_i^{R'}(A'_k)$ with $A'_x = A_x \cup \{a\}$ and $A'_k = A_k$ for $k \neq x$. Subtracting in each of the three cases $i, k \neq x$; $k = x$; $i = x$ gives the stated formula. □

Corollary 171 (The additive collapse)**.** *If the c_i are additive then β_i is constant, so third-party arcs do not move and only the arcs at x change.*

The corollary localises the difficulty sharply: *the entire obstruction to the induction is the variation of β_i across bundles*, and the additive case is closed independently by Theorem 19.

Proposition 172 (The first chore, exactly)**.** *From the root, assigning a to x is legal if and only if $\beta_x \leq \beta_k$ for every k, where $\beta_i := c_i(M) - c_i(M \setminus \{a\})$: the chore must go to an agent of* minimal *marginal.*

Proof At the root every A_i is empty, so afterwards $W_x = M$ and $W_i = M \setminus \{a\}$ for $i \neq x$. Hence $w(i,k) = 0$ for $i, k \neq x$, $w(i,x) = -\beta_i$ and $w(x,k) = \beta_x$. For $n \geq 3$ the first group forces $p_i = c$ for all $i \neq x$, and the remaining constraints read $c + \beta_x \leq p_x \leq c + \min_{i \neq x} \beta_i$. This interval meets $\{0,1\}$ exactly when $\beta_x \leq \min_{i \neq x} \beta_i$: if $\beta_x = 0$ take $c = p_x = 0$, and if $\beta_x = 1$ then the minimum is also 1 and $c = 0$, $p_x = 1$ works, while $\beta_x = 1 > 0 = \min_i \beta_i$ leaves it empty. For $n = 2$ the first group is vacuous and the same two constraints give the same criterion. Legality is then Lemma 63. □

This is the exact mirror of Proposition 71, which takes a chore *from* an agent of maximal marginal. Such an x always exists, so the base case never blocks.

Lemma 173 (Free assignment)**.** *If $\beta_x(A_k) = 0$ for every $k \neq x$ and $\beta_i(A_k) \leq \beta_i(A_i)$ for all $i, k \neq x$, then every arc weight weakly decreases, so the assignment preserves legality.*

Proof Immediate from Lemma 170: arcs into x fall, arcs out of x rise by $\beta_x(A_k) = 0$, and third-party arcs move by $\beta_i(A_k) - \beta_i(A_i) \leq 0$. Path and cycle weights are sums of arc weights. □

14.4 What the frame does not fix

Conjecture 167 survives every instance tested (Section 14.5), but the local statements one would want to prove it with do not. One instance carries most of the damage.

Example 174 (A zero-subsidy dead end)**.** Take $M = \{a, b, c\}$ with the dichotomous costs

S	$\emptyset$	a	b	c	ab	ac	bc	abc
c_1	0	0	0	0	0	1	1	1
c_2	0	0	1	1	1	1	1	1
c_3	0	1	1	1	1	1	2	2

and the CR state $\mathcal{A} = (\emptyset, \{b\}, \{a\})$, $R = \{c\}$, whose profile is $W = (\{c\}, \{bc\}, \{ac\})$ with

$$\big(w_W(i,k)\big) = \begin{pmatrix} 0 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix}, \qquad \ell_W = (0,0,0).$$

The state is reachable from the root and *exactly envy-free*: it needs no subsidy at all. Yet each of the three ways to place the last chore creates a positive-weight cycle — $c \to 1$ gives $w(2,3) + w(3,2) = 1$, $c \to 2$ gives $w(3,1) + w(1,3) = 1$, and $c \to 3$ gives $w(2,1) + w(1,2) = 1$ — so no continuation is even envy-freeable. Conjecture 2 nevertheless holds on this instance: $(\{c\}, \{ab\}, \emptyset)$ has an identically zero envy matrix.

Remark 175 (What Example 174 rules out)**.** It refutes three statements at once. *Pointwise* extendability — every legal non-terminal CR state has a legal assignment — fails, as it does in the peel frame (Conjecture 51); 3,024 reachable states are of this kind. The *last-rung* lemma — a legal state with one chore left can always place it — fails, and 1,765 of the 1,778 stuck states have $|R| = 1$, so this is where essentially all stuckness lives.

The third is the one that constrains a proof. The doomed state has $\ell_W = (0,0,0)$: it is as good as a CR state can possibly be, and the instance itself is not a counterexample to Conjecture 2. *So no potential function measuring the quality of the current state can serve as the invariant*, since the state that must be avoided is optimal. Theorem 89 reached a similar conclusion in the peel frame by exhausting the features of a move; here it follows from one instance, and for a structural reason.

Remark 176 (Balance does not transplant)**.** The same state refutes the one idea that survived Section 8. Its bundle sizes are $(0,1,1)$ with one chore undecided, so the completion with sizes $(1,1,1)$ exists and the state *admits a balanced terminal* in the sense of Observation 57 — and it is dead. Over the corpus, 132 dead CR states admit a balanced completion. The certificate of Conjecture 59 is therefore specific to the peel frame and does not carry to this one.

Three further statements fail. Each has an explicit minimal witness, recorded and re-verified in `cri_witnesses.py`. *Relabelling is load-bearing*: with $c_1 = c_2$ charging 1 only for the grand bundle and $c_3(S) = \min(|S|, 2)$, assignments alone cannot reach a legal terminal from the root while one relabelling suffices, and 3 such instances occur in the exhaustive family. *Death is not always immediate*: at $n = 3$, $m = 5$ there is a dead state with two legal assignments available, so one-step legality cannot certify a rule and Lemma 173 is sound for a move and worthless for a schedule. And the commitment predicate of Theorem 88 transplants *backwards*: of 1,210 states in which some agent already carries conditioned cost 2, none is dead.

14.5 Where the difficulty is, and the one open statement

Every sweep below enumerates its state space completely, with no cap, and draws from all seven generators of `update_44/counterexample_hunt.py` rather than the single uniform sampler used throughout Section 8.

Observation 177 (The sweep)**.** *Over the complete exhaustive $n = m = 3$ family — all* 9,880 *instances — together with* 438 *adversarial instances at $n = 3$ ($m \le 7$), $n = 4$ ($m \le 6$) and $n = 5$ ($m \le 5$), there is* no bad root*: Conjecture 167 holds on every one. Without the relabelling move there are* 3 *bad roots, all in the exhaustive family (`update_47/cri_sweep.py`).*

The dead states are confined to the end of the induction, and sharply so.

Observation 178 (Death lives in the last two rungs)**.** *Over* 312,436 *reachable non-terminal CR states:*

$\lvert R\rvert$	*dead*	*reachable*	*dead share*
1	1,230	183,984	0.669%
2	45	93,909	0.048%
≥ 3	**0**	34,543	**0**%

Correspondingly, playing an arbitrary legal assignment while $|R| > 3$ *and then searching the last three chores reached a terminal on all* 10,045 *instances tested, with the play above the threshold chosen at random so that the result is not an artefact of a tie-break (*`update_47/cri_lookahead.py`*).*

Conjecture 179 (Depth — **refuted**)**.** *Every reachable legal CR state with* $|R| \geq 3$ *is live: some sequence of assignments and relabellings from it reaches a terminal state through legal states.*

Proposition 180. *Conjecture 179 implies Conjecture 167, hence Conjecture 2.*

Proof The root is legal, its envy graph being identically zero, and has $|R| = m$. If $m \leq 3$ the statement is vacuous and Conjecture 167 is a finite check; otherwise any legal assignment from a state with $|R| \geq 4$ lands in a state with $|R| \geq 3$, which is live by hypothesis, and a live state reaches a terminal by definition. □

This would have been a different shape of target from Conjecture 59: that one asks for a global invariant on an unbounded search, whereas Conjecture 179 carries a constant, and the constant is small. It is however false.

Proposition 181 (The depth conjecture fails)**.** *Conjecture 179 is refuted. There are reachable legal CR states with* $|R| \geq 3$ *from which no sequence of assignments and relabellings reaches a terminal state through legal states.*

Proof Complete enumeration of the state space over instances drawn from four families at $n = 3$ and $m \in \{6, 7, 8\}$ finds 657 such states (`update_48/depth_stress.py`):

family	$m = 6$	$m = 7$	$m = 8$
composed	7	243	137
capped	60	210	0
threshold	0	0	0
nested	0	0	0

Nothing is sampled within an instance: for each one every state is enumerated and liveness is computed by a backward pass, so a reported dead state is dead. □

Remark 182 (The caveat was the right one, and it fired)**.** Conjecture 179 rested on 34,543 states at $m \leq 7$ with only 6 of them at $|R| = 7$, and was recorded as a hypothesis rather than a result on exactly that ground — the lesson of Theorem 31, where a rule surviving 227 instances failed on 368. Two details of the refutation are worth keeping. Every failure lies in the *composed* family $c_i(S) = f_i(|S \cap D_i|)$, where the residual of the solved cases lives, or in *capped*; threshold and nested produce none. And the capped family had never been generated correctly before the defect of Theorem 183 was repaired, so half of the refuting evidence was unavailable while the conjecture was being formed.

What Example 174 forces on any replacement is that its proof cannot argue that the state stays good, since the state it must avoid is optimal. It has to argue about what remains *available*: by Lemma 169 every arc is computed on the contracted instance (D, c^R), which by Lemma 168 is again dichotomous, so any such statement is internal to the class. Section 14.6 takes that route.

Remark 183 (A defect in the test generators, and what it touched)**.** The adversarial families used here and in Section 19 had a defect found while checking Proposition 172: in `f_capped` and `f_threshold` the random parameter was drawn *inside* the comprehension over subsets, so it was redrawn for every S and the resulting functions were neither cappings nor thresholds — they had marginals of 2 and more, and so lay *outside* the dichotomous class. `f_mixed` draws from both and was affected too. The uniform, disjoint, nested and one-heavy families were clean throughout.

The consequence is one of coverage rather than of correctness: those sweeps tested a wider class than intended, so their negative findings stand while their claimed coverage of capped and threshold instances does not. The cycle-closing check of Theorem 34 is unaffected, since Theorem 33 assumes only envy-freeability and not dichotomous costs. With the generators repaired, the counterexample hunt still finds no counterexample, and of the 29 instances that come within one unit of being one, 17 are *threshold* against 4 nested and 3 capped — so $c(S) = \max(0, |S| - t)$ is the tightest family known, and it was not previously being generated at all.

14.6 Carrying a witness, and a bridge to Approach 10

Every invariant tried so far has been a *predicate on the state*, and every one has died: the three families eliminated in Section 8, and here pointwise extendability, the last rung, balance, the commitment predicate and Conjecture 179. Example 174 explains why, and rules out the whole family at a stroke — the state to be avoided there is *optimal*, with $\ell = (0, 0, 0)$, so no measure of the current state's quality can separate it from a live one. The standard response to that situation is not another predicate but a stronger induction hypothesis: carry a *witness* alongside the state.

Definition 184 (Completion)**.** *A* completion *of a CR state* $(\mathcal{A}, R)$ *is an allocation* $\mathcal{A}^\star$ *of all of* M *with* $A_i \subseteq \mathcal{A}^\star_i$ *for every* i*. It is* good *if* $\ell_{\mathcal{A}^\star} \leq 1$*. Write*

$$\Phi(\mathcal{A}, R) \ :\Longleftrightarrow \ (\mathcal{A}, R) \textit{ is legal and some completion of it is good.}$$

Proposition 185 (The three easy clauses)**.** Φ*(root) holds if and only if Conjecture 2 holds for the instance; at a terminal state* Φ *holds if and only if that state is good; and* Φ *implies legality by definition.*

Proof The root has $A_i = \emptyset$ for every i, so every allocation of M is a completion of it, and it is legal because its envy graph vanishes. A terminal state has $R = \emptyset$, so it is its own unique completion. □

So, as always, progress is the whole content. What is new is that the standing counterexample does not dispose of Φ before the work starts.

Observation 186 (Φ survives Example 174)**.** *In Example 174 each of the three ways of placing the last chore creates a positive-weight cycle, so that state has* no *good completion and Φ excludes it. Every predicate eliminated in Section 8 and in Section 14.4 was, by contrast, satisfied by some dead state.*

Pure witness-following is nevertheless not enough, and the reason is recorded rather than guessed.

Proposition 187 (Relabelling is unavoidable)**.** *It is not the case that from every state satisfying Φ some chore may be assigned to its owner in a good completion while preserving legality.*

Proof Such a rule would, by induction on $|R|$ from the root, produce a schedule of *assignments alone* reaching a good terminal whenever Conjecture 2 holds for the instance. But three instances of the complete exhaustive $n = m = 3$ family are bad roots in the assignment-only frame while satisfying Conjecture 2 and Conjecture 167 (Theorem 177). □

Hence a witness-carrying proof must let the target *move*, which is exactly what relabelling does: it permutes the bundles, and with them the completion being carried.

The bridge. Lemma 169 says a CR state is a witness on the contracted instance, and Lemma 168 says that instance is again dichotomous. Both hypotheses of Section 15's main theorem are therefore available, and in the CR frame relabelling *is* the choice of assignment, so a minimum-cost labelling may always be selected. Writing

$$\Sigma^R(\mathcal{A}) \;:=\; \sum_{i \in N} \Big[\max_t c_i^R(A_t) - \min_t c_i^R(A_t) \Big]$$

for the total spread of the family measured in the contracted costs:

Theorem 188 (Bridge)**.** *Let $n = 3$ and let $(\mathcal{A}, R)$ be a CR state with $\Sigma^R(\mathcal{A}) \le 3$. Then every labelling of the family $\mathcal{A}$ minimising $\sum_i c_i^R(A_{\sigma(i)})$ is legal.*

Proof By Lemma 168 the contracted instance (D, c^R) is negative dichotomous, and by Lemma 169 the envy graph of the CR state is the envy graph of the allocation $\mathcal{A}$ in that instance. Apply Theorem 199 to (D, c^R): a family of total spread at most 3 assigned by a minimum-cost matching is good, which is precisely legality of the CR state. □

This is the first point at which the two lines meet: Section 15's theorem serves as the *soundness* clause of this section's induction. The base case is then immediate, and agrees with what was already proved.

Corollary 189 (Base case)**.** *From the root, assigning any chore a to any agent yields $\Sigma^{R'} = \sum_i \beta_i \le 3$, where $\beta_i = c_i(M) - c_i(M \setminus \{a\})$; and the minimum-cost labelling gives a to an agent of minimal β, which is Proposition 172.*

Proof At the root every bundle is empty, so after the assignment $c_i^{R'}(\{a\}) = c_i(\{a\} \cup R') - c_i(R') = c_i(M) - c_i(M \setminus \{a\}) = \beta_i$ while the two empty bundles have contracted cost 0. Agent i's three values are therefore $(\beta_i, 0, 0)$ up to order, with spread $\beta_i \in \{0, 1\}$, so $\Sigma^{R'} = \sum_i \beta_i \le n = 3$. The cost of a labelling is β_j for whichever agent j receives $\{a\}$, minimised exactly at $\beta_j = \min_k \beta_k$. □

The candidate invariant, and what is left. Take $\Phi_\Sigma(\mathcal{A}, R) :\iff \Sigma^R(\mathcal{A}) \le 3$. Its root clause is free ($c_i^M \equiv 0$, so $\Sigma = 0$), its soundness clause is Theorem 188, and at a terminal $c^\emptyset = c$, so Φ_Σ delivers a *good allocation* — that is, Conjecture 2 at $n = 3$. One clause remains.

Remark 190 (The labelling must be got right, not waved through)**.** Every state *visited* must be legal, so one may not assign and then relabel to minimum cost: the un-relabelled state is visited on the way. A relabelling is a single move between two states, both of which must be legal, so it cannot be used to escape an illegal one. The correct form of the progress step therefore quantifies over labellings as well as over chores.

Conjecture 191 (Progress — **open**)**.** *Let $n = 3$ and let $\mathcal{A}$ be a family with $\Sigma^R(\mathcal{A}) \le 3$ and $R \ne \emptyset$. Then there are a legal labelling of $\mathcal{A}$, a chore $a \in R$ and a bundle of $\mathcal{A}$ such that adjoining a to that bundle yields a family $\mathcal{A}'$ with $\Sigma^{R'}(\mathcal{A}') \le 3$ whose induced labelling is legal.*

Proposition 192. *Conjecture 191 implies Conjecture 167 at $n = 3$, and hence Conjecture 2 at $n = 3$.*

Proof The root satisfies Φ_Σ and is legal. Each application of Conjecture 191 decreases $|R|$ by one through legal states, so after m steps a terminal state is reached with $\Sigma^\emptyset \le 3$; Theorem 188 makes it good. □

Remark 193 (What this route costs, and why the weaker target comes first)**.** At $|R| = 1$ Conjecture 191 produces a terminal family of total spread at most 3, so it implies Conjecture 202. The two lines are therefore *not* independent: this route to $n = 3$ contains Section 15's open statement. Two consequences, and they are the practical content of this subsection. Conjecture 202 is the weaker target and should be settled first, since a refutation of it would refute Conjecture 191 without any further work. And if it is refuted, Φ_Σ fails as an invariant and Conjecture 167 would need a different one — Φ of Theorem 184 being the candidate that Observation 186 leaves standing.

Verdict. Conjecture 167 is open and unrefuted, and it is a weaker target than either Conjecture 54 or Conjecture 59. Lemmas 168 to 173, Proposition 172 and Theorem 188 are the first statement of this problem as an induction on the *instance* rather than a search over states, and they stand whatever becomes of Conjecture 191. What the frame does *not* do is remove dead ends: it removes the ones of Proposition 52 and grows its own, and Example 174 shows the new ones are harder to characterise, being optimal states. Conjecture 179 is refuted, and the surviving route — Conjecture 191 — is by Remark 193 at least as strong as the open statement of Section 15. As with every other approach in this document, no case with $n \ge 3$ is proved here.

15 Approach 10: Minimum-Spread Families

Every approach so far has selected an *allocation*. This one selects a *family* — an unassigned partition — by a global optimisation, and only then hands the bundles out. The optimisation is the obvious one suggested by Section 5.4: Theorem 26 succeeds exactly when some family has every agent valuing all bundles within one unit, so the natural move when no such family exists is to ask for the family that comes closest.

Definition 194 (Spread)**.** *For a family $B = (B_1, \dots, B_n)$ — a partition of M into n possibly empty bundles — and an agent i, put*

$$\mathrm{sp}_i(B) \ := \ \max_t c_i(B_t) - \min_t c_i(B_t), \qquad \Sigma(B) \ := \ \sum_{i \in N} \mathrm{sp}_i(B).$$

A family is uniformly balanced (Theorem 98) exactly when $\mathrm{sp}_i(B) \le 1$ for every i.

Conjecture 195 (The minimum-spread rule — **open**)**.** *Let B minimise Σ over all families and let σ minimise $\sum_i c_i(B_{\sigma(i)})$. Then $\mathcal{A}$ with $A_i = B_{\sigma(i)}$ satisfies $\ell_{\mathcal{A}} \le 1$.*

Conjecture 195 implies Conjecture 2 at once, and a minimiser exists by definition, so unlike Conjecture 59 it always applies and needs no schedule. It is unrefuted over 2,574 instances at $n \le 6$, $m \le 8$ across eleven generators, in the strong form — *every* minimiser works, not merely some — on a corpus not selected for it (`update_48/minsum_stress.py`).

Remark 196 (What a proof may not use)**.** Three mechanisms are already closed and none of the arguments below uses any of them. First, the arc bound: Proposition 101 shows that on the instance of Proposition 100 *every* good allocation has an arc of weight -2 or less, so no criterion of the form "all arcs ≥ -1" — that is, Corollary 35 — can be the mechanism, and Remark 102 draws the conclusion that any correct certificate is path-based. Second, balance: over 42,711 families of spread at most 2, requiring bundle sizes to differ by at most one admits 792 families whose matching is bad, so the balance signal of Remark 53 is not sufficient here. Third, single-item exchange: $c(S) = \min(\max(0, |S|-1), 1)$ on a three-element set has $c = 1$ with every single removal leaving it at 1, so no step may move one item and expect a cost to move.

15.1 Normalisation

Fix a family B and put $v_i(t) := c_i(B_t) - \min_s c_i(B_s) \ge 0$, so each v_i attains the value 0. Three identities make the whole question finite for a fixed family:

$$\mathrm{sp}_i(B) = \max_t v_i(t), \qquad w_{\mathcal{A}}(i,k) = v_i(\sigma(i)) - v_i(\sigma(k)), \qquad \sum_i c_i(B_{\sigma(i)}) = \sum_i v_i(\sigma(i)) + \text{const}.$$

The last one says a minimum-cost assignment is exactly one minimising $F(\sigma) := \sum_i v_i(\sigma(i))$. Write $x_i := v_i(\sigma(i))$ for what agent i pays after normalisation.

Observation 197. *The heaviest arc out of i equals x_i if $x_i \ge 1$, and is at most 0 if $x_i = 0$.*

Proof $\max_{k \ne i} w_{\mathcal{A}}(i,k) = x_i - \min_{k \ne i} v_i(\sigma(k))$. If $x_i \ge 1$ then v_i does not vanish at $\sigma(i)$, so its zero lies at some $\sigma(k)$ with $k \ne i$ and the inner minimum is 0. If $x_i = 0$ then every arc out of i is $-v_i(\sigma(k)) \le 0$. □

A minimum-cost assignment maximises welfare over reassignments, so $\mathcal{A}$ is envy-freeable by Theorem 10 and Proposition 104 applies at $n = 3$. Combining it with Observation 197:

Proposition 198 (Criterion at $n = 3$)**.** *Let $n = 3$. Then $\mathcal{A}$ is good if and only if* *(a)* $x_i \leq 1$ *for every* i*, and*

(b) there are no distinct i, j, k *with* $x_i = x_j = 1$*,* $v_i(\sigma(j)) = 0$ *and* $v_j(\sigma(k)) = 0$*.*

Proof By Observation 197, (a) is exactly "every arc has weight at most 1". Given (a), a two-path $i \to j \to k$ has weight 2 if and only if both its arcs weigh 1, that is $x_i - v_i(\sigma(j)) = 1$ and $x_j - v_j(\sigma(k)) = 1$; under (a) this is (b). Proposition 104 then gives the claim. □

15.2 The main lemma

Theorem 199 (Small total spread is enough)**.** *Let* $n = 3$ *and let* σ *be any minimum-cost assignment of a family* B*.*

(i) If some arc of $G_{\mathcal{A}}$ *has weight at least* 2*, then* $\Sigma(B) \geq 4$*.*

(ii) If every arc has weight at most 1 *but some two-path has weight* 2*, then* $\Sigma(B) \geq 5$*.*

In particular, if $\Sigma(B) \leq 3$ *then* every *minimum-cost assignment of* B *is good.*

Proof *(i)* By Proposition 198 some $x_i \geq 2$. As $x_i > 0$, the zero of v_i lies at $\sigma(k)$ for some $k \neq i$. Reassigning by the transposition of i and k changes F by $\left[v_i(\sigma(k)) + v_k(\sigma(i))\right] - \left[x_i + x_k\right] = v_k(\sigma(i)) - x_i - x_k$, which is non-negative by optimality, so $v_k(\sigma(i)) \geq x_i + x_k \geq 2$ and $\mathrm{sp}_k \geq 2$. Together with $\mathrm{sp}_i \geq x_i \geq 2$ this gives $\Sigma(B) \geq 4$.

(ii) By Proposition 198(b) there are distinct i, j, k with $x_i = x_j = 1$, $v_i(\sigma(j)) = 0$ and $v_j(\sigma(k)) = 0$; here $F(\sigma) = 2 + x_k$. Transposing i and j gives cost $v_i(\sigma(j)) + v_j(\sigma(i)) + x_k = v_j(\sigma(i)) + x_k$, so optimality forces $v_j(\sigma(i)) \geq 2$ and $\mathrm{sp}_j \geq 2$. Now take the three-cycle $i \mapsto \sigma(j)$, $j \mapsto \sigma(k)$, $k \mapsto \sigma(i)$, of cost $v_i(\sigma(j)) + v_j(\sigma(k)) + v_k(\sigma(i)) = v_k(\sigma(i))$; optimality forces $v_k(\sigma(i)) \geq 2 + x_k \geq 2$, so $\mathrm{sp}_k \geq 2$. With $\mathrm{sp}_i \geq x_i = 1$ we get $\Sigma(B) \geq 1 + 2 + 2 = 5$.

The final claim is the contrapositive of (i) and (ii) together. □

Theorem 199 uses *only* optimality of the assignment. It needs no minimality of the family, no balance hypothesis and no exchange step, so it passes clear of all three closures of Remark 196.

Corollary 200. *Theorem 199 contains Theorem 26.*

Proof A uniformly balanced family has $\mathrm{sp}_i \leq 1$ for each of the three agents, hence $\Sigma \leq 3$. □

It is strictly wider: the spread profiles $(2, 1, 0)$ and $(3, 0, 0)$ also have $\Sigma \leq 3$ and are not uniformly balanced, so instances outside Theorem 98 are covered too — including residual instances, on which no uniformly balanced family exists at all.

Corollary 201. *Conjecture 195 holds at* $n = 3$ *for every instance admitting a family with* $\Sigma \leq 3$*.*

Proof The Σ-minimal family of such an instance has $\Sigma \leq 3$; apply Theorem 199. □

So the whole of $n = 3$ rests on one purely combinatorial statement, with no allocation, assignment or envy graph in it.

Conjecture 202 (Total spread three — **open**)**.** *Every three-agent negative dichotomous instance admits a partition of* M *into three bundles with* $\sum_i \mathrm{sp}_i \leq 3$*.*

15.3 What stands in the way if Conjecture 202 fails

Theorem 199 localises the obstruction completely at $\Sigma = 4$, and the answer is a single rigid pattern.

Proposition 203 (The unique obstruction at $\Sigma = 4$). *Let $n = 3$, let $\Sigma(B) = 4$ and let a minimum-cost σ be bad. Then, writing coordinates in the order $(\sigma(i), \sigma(j), \sigma(k))$ for a suitable labelling of the agents,*

$$v_i = v_k = (2, 2, 0), \qquad v_j = (0, 0, 0).$$

Proof By Theorem 199(ii) a two-path failure needs $\Sigma \geq 5$, so the failure is an arc of weight at least 2 and part (i) applies: with k the holder of the zero of v_i, we have $\mathrm{sp}_i, \mathrm{sp}_k \geq 2$. Since $\Sigma = 4$, this forces $\mathrm{sp}_i = \mathrm{sp}_k = 2$ and $\mathrm{sp}_j = 0$, so $v_j \equiv 0$ and $x_j = 0$. Then $x_i \leq \mathrm{sp}_i = 2$ and $x_i \geq 2$ give $x_i = 2$, and $v_k(\sigma(i)) \geq x_i + x_k = 2 + x_k$ with $v_k \leq 2$ gives $x_k = 0$ and $v_k(\sigma(i)) = 2$. Transposing i and j costs $v_i(\sigma(j)) + v_j(\sigma(i)) + x_k = v_i(\sigma(j))$, so optimality gives $v_i(\sigma(j)) \geq 2$, hence $= 2$. Finally the three-cycle $i \mapsto \sigma(k)$, $j \mapsto \sigma(i)$, $k \mapsto \sigma(j)$ costs $v_i(\sigma(k)) + v_j(\sigma(i)) + v_k(\sigma(j)) = v_k(\sigma(j))$, so $v_k(\sigma(j)) \geq 2$, hence $= 2$. Collecting, $v_i = v_k = (2, 2, 0)$ and $v_j = (0, 0, 0)$. □

Two agents therefore have *identical* normalised vectors — both see two heavy bundles and one light one — while the third is indifferent, and every minimum-cost assignment strands one of the two on a heavy bundle. Ruling the pattern out means re-partitioning the two heavy bundles so as to balance *both* of those agents at once, which is Conjecture 206 below.

15.4 Two exchange lemmas, and one more conjecture

Lemma 204 (Intermediate value). *Let c be dichotomous and $Y \subseteq M$. For every integer $0 \leq k \leq c(Y)$ there is $Z \subseteq Y$ with $c(Z) = k$.*

Proof Along a maximal chain $\emptyset = Z_0 \subset \cdots \subset Z_r = Y$ consecutive costs differ by 0 or 1, so the values $c(Z_0), \ldots, c(Z_r)$ sweep every integer between 0 and $c(Y)$. □

Lemma 204 is what replaces the exchange step forbidden by the third closure of Remark 196: single items need not move a cost, but *sets* realising any intermediate value always exist.

Lemma 205 (Two-way balance, one agent). *For dichotomous c and any $U \subseteq M$ there is $Z \subseteq U$ with $|c(Z) - c(U \setminus Z)| \leq 1$.*

Proof On a maximal chain $\emptyset = Z_0 \subset \cdots \subset Z_r = U$ put $f(t) := c(Z_t) - c(U \setminus Z_t)$. Then $f(0) = -c(U) \leq 0$ and $f(r) = c(U) \geq 0$. Each step raises the first term by 0 or 1 and lowers the second by 0 or 1, so f is non-decreasing with steps in $\{0, 1, 2\}$. A step of 2 cannot carry f from at most -2 to at least 2, so some $f(t)$ lies in $\{-1, 0, 1\}$. □

Conjecture 206 (Two-way balance, two agents — **open**). *For any two dichotomous costs and any $U \subseteq M$ there is a partition $U = Z \sqcup (U \setminus Z)$ with $\mathrm{sp}_1 \leq 1$ and $\mathrm{sp}_2 \leq 1$ on the two parts.*

Note that Proposition 100 obstructs the *three*-agent analogue of Conjecture 206, not this one, so no barrier is known.

Remark 207 (The two-agent statement is a synchronisation problem)**.** For $Z \subseteq U$ put

$$g(Z) \ := \ \Big(c_1(Z) - c_1(U \setminus Z),\ c_2(Z) - c_2(U \setminus Z)\Big),$$

so that Conjecture 206 asks for $g(Z) \in \{-1, 0, 1\}^2$; note $g(U \setminus Z) = -g(Z)$, so the image is symmetric about the origin. Along any maximal chain both coordinates of g are *non-decreasing* with steps in $\{0, 1, 2\}$, running from $(-c_1(U), -c_2(U))$ to $(c_1(U), c_2(U))$. Lemma 205 is the one-coordinate case of this. The two-coordinate case does not follow, because the coordinates cross the band $\{-1, 0, 1\}$ at different times: at the first t with $g_2(t) \geq -1$ one gets $g_2(t) \in \{-1, 0\}$, but $g_1(t)$ may already be far above 1. The content of Conjecture 206 is exactly that the *order* of the chain — the only freedom available — can be chosen to synchronise the two crossings. For additive measures this is the discrete necklace-splitting theorem with two cuts; dichotomous costs are not additive, so that result does not transfer.

15.5 Balancing sets: the greedy engine

Everything that follows rests on one lemma, worth isolating because it explains both what works and what does not.

Lemma 208 (Greedy levelling)**.** *Let $w_1, \ldots, w_N$ be integers with $0 \leq w_j \leq k$. Then one may choose $\delta_j \in \{0, 1\}^k$ with $|\delta_j| = w_j$ so that $\sum_j \delta_j$ has spread at most 1.*

Proof Process in any order, placing δ_j's ones on the w_j currently smallest coordinates. The invariant is that the entries lie in $\{\mu, \mu + 1\}$. Let k' be the number of entries equal to μ. If $w_j \leq k'$ then w_j of them rise to $\mu + 1$ and the entries still lie in $\{\mu, \mu + 1\}$. If $w_j > k'$ then all k' rise to $\mu + 1$ and $w_j - k'$ of the others rise to $\mu + 2$, so the entries lie in $\{\mu + 1, \mu + 2\}$. Either way the spread stays at most 1. □

Read through the Venn regions of a family of sets, Lemma 208 says a *single* set can always be levelled: run greedy over the regions meeting it. Two sets share exactly one region, so fixing that region first lets both run greedy independently — which is the next theorem. Three sets share two regions pairwise and one triply, and no ordering lets all three run greedy; that is the entire obstruction.

Theorem 209 (Two-set balance)**.** *For any two sets $D_1, D_2 \subseteq M$ and any number k of bundles there is a k-colouring of M splitting* both *within one.*

Proof The Venn regions are $P_1 = D_1 \setminus D_2$, $P_{12} = D_1 \cap D_2$ and $P_2 = D_2 \setminus D_1$; elements in neither set are free. Write $|P| = ka_P + b_P$ with $0 \leq b_P < k$, split each region as evenly as possible, and let $\epsilon_P \in \{0, 1\}^k$ with exactly b_P ones record which colours take an extra element. The count of D_i in colour t is then $A_i + E_{i,t}$ with A_i independent of t and $E_1 = \epsilon_{P_1} + \epsilon_{P_{12}}$, $E_2 = \epsilon_{P_2} + \epsilon_{P_{12}}$.

Fix $\epsilon_{P_{12}}$ arbitrarily. Each E_i is a sum of two $\{0, 1\}$ vectors, so its entries lie in $\{0, 1, 2\}$ and its spread is at most 1 exactly when the two supports are *disjoint* (entries in $\{0, 1\}$) or together *cover* all k colours (entries in $\{1,2\}$). If $b_{P_1} + b_{P_{12}} \le k$ choose the supports disjoint; if $b_{P_1} + b_{P_{12}} \ge k$ choose them covering. One is always available, and ϵ_{P_2} is chosen the same way, independently. □

Corollary 210. *Proposition 100 genuinely requires all three sets: no two-set system obstructs uniform balance, at any number of bundles.*

Corollary 211 (Conjecture 206 for composed costs)**.** *For any two costs of the form $c_i(S) = f_i(|S \cap D_i|)$ and any $U \subseteq M$, some partition of U into two parts gives both spread at most 1.*

Proof Theorem 209 at $k = 2$ balances the underlying sets within one; Lemma 213 below carries that to the composed costs. □

This settles Conjecture 206 on the family where the residual of the solved cases lives. It does *not* settle it for arbitrary dichotomous costs, since Lemma 213 needs the cost to depend only on $|S \cap D|$.

Corollary 212 (Two sets go to spread zero together)**.** *If $3 \mid |D_i|$ and $3 \mid \left|D_j\right|$, some 3-colouring makes both D_i and D_j exactly balanced, that is both agents at spread 0.*

Proof Theorem 209 splits both within one, and by Lemma 219(b) a set whose size is divisible by 3 has spread 0 or at least 2, so "within one" forces spread 0. □

15.6 Compression, and the shape of the residual

Lemma 213 (Compression)**.** *Let $c(S) = f(|S \cap D|)$ with f monotone and all increments in $\{0,1\}$. Then for every family B, $\mathrm{sp}_c(B) \le \mathrm{sp}_{|\cdot \cap D|}(B)$.*

Proof Such an f is monotone and 1-Lipschitz, so with $a_{\max}, a_{\min}$ the extreme counts, $\max_t f(a_t) - \min_t f(a_t) = f(a_{\max}) - f(a_{\min}) \le a_{\max} - a_{\min}$. □

So among costs built on the same sets, the *additive* one is hardest to balance, and a colouring balancing the underlying sets balances every composed cost over them. Two consequences.

Corollary 214 (A spread-zero agent for free)**.** *If $3 \mid |D_i|$, then splitting D_i into three equal parts and distributing $M \setminus D_i$ arbitrarily gives agent i spread exactly 0 — whatever f_i is, since all three counts equal $|D_i|/3$ and f_i sees only the count.*

Corollary 215. *Let all three costs be composed. If $3 \nmid |D_i|$ for every i, then the instance admits a uniformly balanced family.*

Proof Theorem 223 balances the three underlying sets within one, and Lemma 213 carries this to the composed costs. □

15.7 Rigidity, and where $\Sigma \leq 3$ can fail

Lemma 216 (Individual minimum)**.** *For any dichotomous c and any n, some family has* $\mathrm{sp}(B) \leq 1$.

Proof Add the items one at a time, each to a bundle of currently minimum cost. The invariant that all bundle costs lie in $\{\mu, \mu+1\}$ holds when all bundles are empty, and adding an item to a bundle of cost μ makes that bundle cost μ or $\mu+1$, marginals lying in $\{0,1\}$, and changes nothing else. This is the invariant of Theorem 21. □

Corollary 217. *Conjecture 202 has content only when no uniformly balanced family exists. There the spread profile realising $\Sigma \leq 3$ must be* $(0,0,2)$, $(0,0,3)$ *or* $(0,1,2)$ *— in particular* some agent must have spread exactly 0*.*

Proof By Lemma 216 each sp_i can individually be brought to at most 1, so a uniformly balanced family is exactly one attaining all three simultaneously. If none exists, every family has some $\mathrm{sp}_i \geq 2$, and the multisets with $\Sigma \leq 3$ and maximum at least 2 are the three listed; each contains a 0. □

Definition 218. *A dichotomous cost is* rigid *if no* 3*-partition gives it spread* 0*.*

Lemma 219 (Additive rigidity)**.** *Let $c_i(S) = |S \cap D_i|$. In every* 3*-partition,*

(a) if $3 \nmid |D_i|$ the spread is at least 1 *— never* 0*;*

(b) if $3 \mid |D_i|$ the spread is 0 *or at least* 2 *— never* 1*.*

Proof The counts $a_t = |B_t \cap D_i|$ sum to $|D_i|$. For (a), spread 0 forces $a_1 = a_2 = a_3$, so $3 \mid |D_i|$. For (b), suppose the spread is 1, so the a_t lie in $\{\mu, \mu+1\}$ with k of them at $\mu+1$ and $1 \leq k \leq 2$; then $|D_i| = 3\mu + k$ forces $k \equiv 0 \pmod 3$, a contradiction. □

Rigidity is not the same as non-additivity: $c(S) = \min(|S \cap D|, 3)$ with $|D| = 4$ is not additive, yet is rigid, since equal costs would need either $a_1 = a_2 = a_3$ (impossible, $3 \nmid 4$) or all $a_t \geq 3$ (impossible, $4 < 9$). More generally, $\min(|S \cap D|, k)$ is rigid exactly when $3 \nmid |D|$ and $|D| < 3k$, and $\max(0, |S \cap D| - t)$ exactly when $3 \nmid |D|$ and $|D| > 3t$.

Proposition 220. *If all three agents are rigid and no uniformly balanced family exists, then $\Sigma(B) \geq 4$ for every family, so Conjecture 202 fails for that instance.*

Proof Rigidity gives $\mathrm{sp}_i \geq 1$ for every i and every family; failure of uniform balance gives some $\mathrm{sp}_i \geq 2$; so $\Sigma \geq 4$. □

Remark 221 (The converse does not hold, and an earlier claim is withdrawn)**.** Proposition 220 was at one stage asserted as an equivalence — that Conjecture 202 fails *if and only if* some instance has three rigid agents and no uniformly balanced family. Only the direction proved above is available. An instance may have a non-rigid agent, so not three rigid ones, and still admit no family with $\Sigma \leq 3$: driving the non-rigid agent to spread 0 may force the other two to sum to 4 or more. Nothing in this section uses the converse.

Remark 222 (A withdrawn criterion for spread zero)**.** It is *not* true that a dichotomous c admits a spread-0 3-partition if and only if $3 \mid c(M)$. Both directions fail without additivity: cutting a maximal chain at $c(Z) = c(M)/3$ says nothing about c of the later blocks, since $c(Z' \setminus Z) \neq c(Z') - c(Z)$ in general, and $\sum_t c(B_t) \neq c(M)$. For a counterexample take $c(S) = \min(|S|, 1)$ on three items: $c(M) = 1$ while the partition into singletons gives $(1, 1, 1)$, spread 0. The divisibility criterion is available only in the additive case, which is Lemma 219.

15.8 The additive question, and its proof

By Corollary 215 the additive case governs the composed family, so the following is the load-bearing combinatorial statement.

Theorem 223 (Three sets with no size divisible by three)**.** *Let $D_1, D_2, D_3 \subseteq M$ with $3 \nmid |D_i|$ for every i. Then some 3-colouring splits all three within one.*

The proof uses a normal form. Index the Venn regions by nonempty $T \subseteq \{1, 2, 3\}$, write $|P_T| = 3a_T + b_T$ with $b_T \in \{0, 1, 2\}$, split each region as evenly as possible and let $\epsilon_T \in \{0, 1\}^3$ with $|\epsilon_T| = b_T$ record which colours take an extra element. The count of D_i in colour t is $A_i + E_{i,t}$ with A_i independent of t and $E_i = \sum_{T \ni i} \epsilon_T$, so the spread of D_i is the spread of E_i, and $\sum_t E_{i,t} = \beta_i \equiv |D_i| \pmod 3$. Call a region *active* when $b_T \neq 0$; putting $\epsilon_T = e_{c_T}$ when $b_T = 1$ and $\epsilon_T = \mathbf{1} - e_{c_T}$ when $b_T = 2$, each active region simply chooses a colour c_T, and with $u_T := \sigma_T e_{c_T}$, $\sigma_T = \pm 1$ according as b_T is 1 or 2,

$$\operatorname{spread}(E_i) = \operatorname{spread}(F_i), \qquad F_i = \sum_{T \ni i} u_T.$$

Writing p_i, q_i for the numbers of active regions of D_i with $\sigma_T = +1, -1$, we have $\sum_t F_{i,t} = s_i := p_i - q_i$ and $\beta_i = p_i + 2q_i \equiv s_i$, so the hypothesis reads $s_i \not\equiv 0 \pmod 3$: *no set imposes the rigid requirement that F_i be constant.* Finally split off the private region, $F_i = G_i + \sigma_{\{i\}} e_{c_{\{i\}}}$ with G_i summing over the *shared* regions $\{ij\}, \{ik\}, \{123\}$ of set i.

Lemma 224 (The private knob absorbs a levelled shared part)**.** *If $\operatorname{spread}(G_i) \leq 1$ and $\{i\}$ is active, some choice of $c_{\{i\}}$ gives $\operatorname{spread}(F_i) \leq 1$, for either sign $\sigma_{\{i\}}$.*

Proof The entries of G_i lie in $\{\mu, \mu + 1\}$. For $\sigma_{\{i\}} = +1$: if some entry equals μ, add there and the entries stay in $\{\mu, \mu + 1\}$; if all equal μ, adding anywhere gives $(\mu + 1, \mu, \mu)$, spread 1. For $\sigma_{\{i\}} = -1$ subtract at a $\mu + 1$ entry, or from a constant vector to get $(\mu - 1, \mu, \mu)$. □

Lemma 225 (Rescue at spread two)**.** *If $\{i\}$ is active and G_i has entries $\{\mu, \mu + 1, \mu + 2\}$, then either sign admits a choice of $c_{\{i\}}$ with $\operatorname{spread}(F_i) \leq 1$.*

Proof For $+1$ add at the μ entry, giving $\{\mu + 1, \mu + 1, \mu + 2\}$; for -1 subtract at the $\mu + 2$ entry, giving $\{\mu, \mu + 1, \mu + 1\}$. □

By Lemma 224, if the four shared colours can be chosen so that every G_i has spread at most 1 then every set is satisfied — those with an inactive private because $F_i = G_i$, those with an active private by the knob. The conditions on G_i are read off directly: with two regions, equal signs demand different colours and opposite signs demand equal ones; with three regions, all signs equal demands three distinct colours, and a 2–1 split demands that the minority's colour lie among the two majority colours.

Proof of Theorem 223 Let $S \subseteq \{12, 13, 23, 123\}$ be the active shared regions and write $a = \sigma_{12}$, $b = \sigma_{13}$, $c = \sigma_{23}$, $w = \sigma_{123}$. Negating every sign negates every F_i and preserves spreads, so we may take $w = +$ when $123 \in S$.

$123 \notin S$. Each G_i spans at most two regions. If $|S| \leq 1$ every G_i spans at most one and there is nothing to satisfy. If $|S| = 2$, say $S = \{12, 13\}$, only G_1 spans two regions and its single condition is satisfiable. If $S = \{12, 13, 23\}$ and $a = b = c$, all three conditions read "colours differ", so take c_{12}, c_{13}, c_{23} pairwise distinct. Otherwise exactly two signs agree, say $a = b \neq c$; then G_1 demands $c_{12} \neq c_{13}$ while G_2, G_3 demand $c_{12} = c_{23}$ and $c_{13} = c_{23}$, which is inconsistent. But $a = -c$ gives $s_2 = \sigma_{\{2\}}[\text{active}]$, so $\{2\}$ must be active, and likewise $\{3\}$. Set 2's three regions then carry signs $\left\{\sigma_{\{2\}}, a, -a\right\}$, a 2–1 split, whose minority condition is met by choosing $c_{\{2\}}$ — equal to the minority's colour if $\{2\}$ lies in the majority, or to either majority colour if $\{2\}$ is the minority. Set 3 likewise, leaving only $c_{12} \neq c_{13}$.

$123 \in S$. If $S = \{123\}$ every $G_i = u_{123}$. If $S = \{123, 12\}$ then G_3 is automatic and G_1, G_2 impose the same two-region condition. If $S = \{123, 12, 13\}$ there are four sign patterns: for $a = b = +$ take c_{12}, c_{13}, c_{123} distinct, which meets G_1's all-distinct demand and both disequalities; for $a = +, b = -$ take $c_{123} = c_{13}$ and c_{12} different, and G_1's minority condition on $\{13\}$ already holds; $a = -, b = +$ is symmetric; and for $a = b = -$ take all three equal, which meets both equalities and G_1's minority condition on $\{123\}$.

Finally $S = \{123, 12, 13, 23\}$, where every G_i spans three regions. For $(a, b, c) = (+, +, -)$ take c_{12}, c_{13}, c_{123} distinct and $c_{23} = c_{123}$; for $(+, -, +)$ take c_{12}, c_{23}, c_{123} distinct and $c_{13} = c_{123}$; $(-, +, +)$ is symmetric; and for each of $(+, -, -)$, $(-, +, -)$, $(-, -, +)$, $(-, -, -)$ every G_i has a 2–1 split and the constant colouring meets every minority condition. There remains $(a, b, c) = (+, +, +)$, where each G_i demands its three colours be distinct: G_1 forces $(c_{12}, c_{13}, c_{123})$ to be a rainbow, G_2 then forces $c_{23} = c_{13}$, and G_3 sees a repeat, so no choice of the four shared colours works. Here, however, $s_i = \sigma_{\{i\}}[\text{active}] + 3$ for each i, so all three private regions are forced active. Take $c_{12} = A$, $c_{13} = B$, $c_{123} = C$ and $c_{23} = B$: then G_1 and G_2 are constant and Lemma 224 frees sets 1 and 2, while $G_3 = e_B + e_B + e_C$ has three distinct entries and Lemma 225 frees set 3. ◻

Both standing witnesses fall outside the hypothesis, as they must: Proposition 100 has sizes $2, 3, 3$ and the instance of Example 227 has sizes $3, 3, 3$, each containing a size divisible by 3, and both provably lack a uniformly balanced family.

Corollary 226. *A composed instance admitting no uniformly balanced family must contain a set whose size is divisible by* 3.

Proof Contrapositive of Corollary 215. ◻

Example 227 (The minimal obstruction)**.** Take $D_1 = \{1, 2, 3\}$, $D_2 = \{1, 2, 4\}$, $D_3 = \{1, 3, 4\}$ on $M = \{1, 2, 3, 4\}$, all additive. Each has size 3, so uniform balance demands each be rainbow, and the three triangles they impose have union $\{12, 13, 23\} \cup \{12, 14, 24\} \cup \{13, 14, 34\} = E(K_4)$, which is not 3-colourable. So no uniformly balanced family exists — and by Corollary 210 this is the smallest possible shape of such an obstruction. Yet $\Sigma \leq 3$ holds easily: the family $\{1\}, \{2, 4\}, \{3\}$ gives counts $(1, 1, 1)$, $(1, 2, 0)$, $(1, 1, 1)$, hence $\Sigma = 0 + 2 + 0 = 2$, two agents at spread 0 and one at 2 — the mechanism of Corollary 212.

15.9 What the remaining gap is

Corollary 226 says the instances that matter all have a set of size divisible by 3 — precisely where Theorem 223 does *not* apply. So the theorem, while it settles the additive question, does not reach the residual. What remains splits by how many sizes are divisible by 3: if two or more, Corollary 212 puts two agents at spread 0 and the third must be shown to have spread at most 3; if exactly one, Theorem 209 applied to that set and one other gives $\mathrm{sp} = 0$ and $\mathrm{sp} \le 1$, and the third must be shown to have spread at most 2. Both follow from one symmetric statement.

Conjecture 228 (Two tight, one loose — **open**). *For any three sets there is a 3-colouring splitting two of them within one and the third within two.*

Interleaving the greedy of Lemma 208 across the three sets gets within one unit of this, and that is already enough to close one of the two cases outright.

Theorem 229 (Interleaved greedy). *For any three sets $D_1, D_2, D_3 \subseteq M$ there is a 3-colouring splitting D_1 and D_2 each within one and D_3 within three.*

Proof Work in the residue frame of Section 15.8: region T contributes $\epsilon_T \in \{0,1\}^3$ of weight b_T, and the spread of D_i is the spread of $E_i = \sum_{T \ni i} \epsilon_T$. Place the residues in the order

$$\epsilon_{123}, \quad \epsilon_{12}, \quad \epsilon_{13}, \quad \epsilon_1, \quad \epsilon_{23}, \quad \epsilon_2, \quad \epsilon_3,$$

each on the currently smallest coordinates of, respectively, the zero vector, ϵ_{123}, $\epsilon_{123} + \epsilon_{12}$, $\epsilon_{123} + \epsilon_{12} + \epsilon_{13}$, $\epsilon_{123} + \epsilon_{12}$, $\epsilon_{123} + \epsilon_{12} + \epsilon_{23}$, and $\epsilon_{123} + \epsilon_{13} + \epsilon_{23}$.

Set 1 sees $\epsilon_{123}, \epsilon_{12}, \epsilon_{13}, \epsilon_1$, and each was placed greedily against set 1's own running sum, so $\mathrm{spread}(E_1) \le 1$ by Lemma 208. Set 2 sees $\epsilon_{123}, \epsilon_{12}, \epsilon_{23}, \epsilon_2$; its running sum after the first two is the same vector $\epsilon_{123} + \epsilon_{12}$ as set 1's, both sets containing both regions, so those two steps are greedy for set 2 as well, and the last two are greedy for it by construction. Hence $\mathrm{spread}(E_2) \le 1$.

Set 3 sees $\epsilon_{123}, \epsilon_{13}, \epsilon_{23}, \epsilon_3$, of which only the first and last were placed greedily for it. Adjoining a $\{0,1\}$ vector raises a spread by at most 1, so after the first three the spread is at most 3. Greedy never increases a spread: for sorted $e_1 \le e_2 \le e_3$, adding 1 to the w smallest leaves the maximum unchanged unless all entries are equal, and does not lower the minimum. So the last step keeps it at most 3. ◻

The asymmetry is exactly the obstruction named after Lemma 208, made quantitative. Sets 1 and 2 share only $\{12\}$ among the regions processed before either finishes, so one ordering serves both; set 3 shares $\{13\}$ with set 1 and $\{23\}$ with set 2, and both are already committed when set 3 needs them. The cost is one unit of spread per stolen region.

Theorem 230 (Two sizes divisible by three). *Let $n = 3$ and let the costs be composed, $c_i(S) = f_i(|S \cap D_i|)$. If at least two of $|D_1|, |D_2|, |D_3|$ are divisible by 3, then some family has $\Sigma \le 3$, and hence Conjecture 2 holds for the instance.*

Proof Relabel so that $3 \mid |D_1|$ and $3 \mid |D_2|$, and apply Theorem 229 with those in the tight slots. It splits D_1 and D_2 within one; by Lemma 219(b) a set of size divisible by 3 has count spread 0 or at least 2, so within one forces count spread exactly 0. It splits D_3 within three.

Lemma 213 bounds each cost spread by the corresponding count spread, giving cost spreads 0, 0 and at most 3, so $\Sigma \le 3$; Theorem 199 then makes every minimum-cost assignment of that family good. □

Theorem 230 is not contained in the four cases of Section 5.4 and its neighbours: the instance need be neither additive, nor identical, nor small-bundled, nor uniformly balanced. By Corollary 226 every instance lacking a uniformly balanced family has a size divisible by 3, and this theorem closes all those with two such sizes.

The missing unit is recoverable, and recovering it closes the case. Two lemmas are needed: a sharpening of how greedy behaves when started from a nonzero vector, and the tie-breaking rule itself.

Lemma 231 (Greedy does not spoil a spread)**.** *Adjoining a* $\{0,1\}$ *vector greedily — ones on the currently smallest coordinates — leaves the spread at most* $\max(\textit{old spread}, 1)$.

Proof Take sorted entries $e_1 \le e_2 \le e_3$. For weight 1 the new entries are e_1+1, e_2, e_3: if $e_1 + 1 \le e_3$ the maximum is unchanged and the minimum is $\min(e_1+1, e_2) \ge e_1$, so the spread does not grow; otherwise $e_1 = e_2 = e_3$ and the new spread is 1. For weight 2 they are e_1+1, e_2+1, e_3: if $e_3 \ge e_2+1$ the spread drops; if $e_3 = e_2$ it equals $e_2 + 1 - \min(e_1+1, e_2)$, which is $e_3 - e_1$ when $e_1 + 1 \le e_2$ and 1 otherwise. Weights 0 and 3 change nothing. □

Lemma 232 (Tie-breaking)**.** *In the ordering of Theorem 229 the tie-breaks for* ϵ_{13} *and* ϵ_{23} *may be chosen so that* $H := \epsilon_{123} + \epsilon_{13} + \epsilon_{23}$ *has spread at most* 2.

Proof Both ϵ_{13} and ϵ_{23} are placed greedily on $P := \epsilon_{123} + \epsilon_{12}$, and ϵ_{12} is placed greedily on ϵ_{123}. As H is a sum of three $\{0,1\}$ vectors its entries lie in $[0,3]$, so spread 3 requires both a coordinate t lying in all three and a coordinate s avoided by all three. Split on b_{123}.

If $b_{123} = 0$ then $H = \epsilon_{13} + \epsilon_{23}$ has entries at most 2, so its spread is at most 2.

If $b_{123} = 1$, say $\epsilon_{123} = e_a$, the coordinate t must be a, so it suffices to keep a out of both ϵ_{13} and ϵ_{23}; and a is never forced. Indeed ϵ_{12}, being greedy on e_a, prefers the two non-a coordinates: for $b_{12} = 0$ we get $P = e_a$, in which a is the unique *largest*; for $b_{12} = 1$ we get $\epsilon_{12} = e_v$ with $v \ne a$ and $P = (1_a, 1_v, 0_w)$, whose smallest is w and whose second tier $\{a, v\}$ is tied, so a weight-2 choice may take v; and for $b_{12} = 2$ we get $P = (1,1,1)$, entirely tied. With a excluded from both, $H_a = 1$ and $H_x \le 2$ for $x \ne a$, so the spread is at most 2.

If $b_{123} = 2$, say ϵ_{123} avoiding c, then ϵ_{123} avoids only c, so the coordinate s must be c, and it suffices that one of $\epsilon_{13}, \epsilon_{23}$ contain c. If both have weight 0 then $H = \epsilon_{123}$ has spread 1. Otherwise c is available: ϵ_{12} is greedy on a vector that is 0 at c and 1 at a and b, hence takes c first, so for $b_{12} = 0$ we get $P = (1_a, 1_b, 0_c)$ with c uniquely smallest and therefore forced; for $b_{12} = 1$ we get $\epsilon_{12} = \{c\}$ and $P = (1,1,1)$, tied; and for $b_{12} = 2$ we get $P = (2_a, 1_b, 1_c)$ say, whose smallest are b and c, tied. Taking c gives $H_c \ge 1$, while $H_a, H_b \ge 1$ from ϵ_{123}, so the minimum is at least 1, the maximum at most 3, and the spread at most 2.

Every choice made is among coordinates tied for smallest, and Lemma 208's invariant holds for any such choice, so sets 1 and 2 are unaffected. □

Theorem 233 (Two tight, one loose)**.** *For any three sets* $D_1, D_2, D_3 \subseteq M$ *there is a* 3*-colouring splitting* D_1 *and* D_2 *each within one and* D_3 *within two. That is, Conjecture 228 holds.*

Proof Run the ordering of Theorem 229 with the tie-breaks of Lemma 232. Sets 1 and 2 are levelled exactly as there, the tie-breaking being immaterial to them. For set 3 we have $E_3 = H + \epsilon_3$ with ϵ_3 placed greedily, so by Lemmas 232 and 231 its spread is at most $\max(2, 1) = 2$. □

Theorem 234 (Conjecture 2 at three agents, composed costs)**.** *Let $n = 3$ and let every cost be composed, $c_i(S) = f_i(|S \cap D_i|)$ with f_i monotone and all increments in $\{0, 1\}$. Then Conjecture 2 holds.*

Proof If the instance admits a uniformly balanced family, Theorem 26 applies. Otherwise Corollary 226 gives some $3 \mid |D_i|$; relabel it D_1 and apply Theorem 233. If a second size is divisible by 3, put that set in the other tight slot; both then have count spread within one, which Lemma 219(b) upgrades to exactly 0, and the third has count spread at most 2. If exactly one size is divisible by 3, put any other set in the second tight slot, giving count spreads 0, at most 1, and at most 2. Lemma 213 bounds each cost spread by the corresponding count spread, so $\Sigma \leq 2$ or $\Sigma \leq 3$; and Theorem 199 makes every minimum-cost assignment of that family good. □

This closes $n = 3$ on the composed family, which contains binary additive, capped and threshold costs and every mixture of them.

Remark 235 (What is *not* closed)**.** Theorem 234 needs each cost to be a function of a single intersection size $|S \cap D_i|$: that hypothesis is used by Lemma 213, and through it by Corollary 226 and by the transfer from count spreads to cost spreads. A dichotomous cost not of this form lies outside the argument, and whether every three-agent instance reduces to a composed one is *not* proved. The residual instances exhibited in Section 15.9 are all composed, but that reflects how they were constructed rather than a theorem about the residual.

Remark 236 (Superseded)**.** This remark recorded the last gap before Lemma 232 closed it, and is kept because it isolates why the tie-break is the right move. At the time, one configuration remained — precisely one $|D_i|$ divisible by 3, where Theorem 229 yields $0 + 1 + 3 = 4$, one over budget.

The unit is available because, in the proof of Theorem 229 the spread of set 3 reaches 3 only if $\epsilon_{123}, \epsilon_{13}, \epsilon_{23}$ all place a one on some coordinate t and all avoid some coordinate s. Writing $P = \epsilon_{123} + \epsilon_{12}$ for the vector against which both ϵ_{13} and ϵ_{23} are placed, $\epsilon_{123,t} = 1$ gives $P_t = 1 + \epsilon_{12,t}$ and $\epsilon_{123,s} = 0$ gives $P_s = \epsilon_{12,s}$; for ϵ_{13} to choose t over s we need $P_s \geq P_t$, forcing $\epsilon_{12,s} = 1$, $\epsilon_{12,t} = 0$ and therefore $P_t = P_s = 1$. So the bad alignment can only arise where P is *tied*, and greedy had a free choice; since Lemma 208's invariant holds for any choice among the currently smallest coordinates, re-breaking that tie costs sets 1 and 2 nothing. Lemma 232 carries this out, and the case split there is on b_{123} rather than on the coordinates, which is what makes it finite.

Proposition 237. *Conjecture 228 implies Conjecture 202 for every composed instance.*

Proof By Corollary 226 some $3 \mid |D_i|$, and by Lemma 219(b) "within one" for such a set means spread 0. Put it in a tight slot. If a second size is divisible by 3, put that in the other tight slot for $\Sigma \leq 0 + 0 + 2$; otherwise the other tight slot gives $\mathrm{sp} \leq 1$ and $\Sigma \leq 0 + 1 + 2 = 3$. Apply Lemma 213 to transfer from the sets to the costs. □

The method of Theorem 223 does *not* extend to Conjecture 228. That proof ran entirely on the hypothesis $s_i \not\equiv 0 \pmod 3$, used twice to force private regions active; without it, sets with $s_i \equiv 0$ impose the rigid requirement that F_i be constant, which is exactly what makes obstructions of the K_4 shape possible. A proof of Conjecture 228 needs a different idea.

What is available is a necessary condition on any counterexample. Fix the colouring of Theorem 209 for D_1, D_2, write R for a region of their Venn diagram, $r_{R,t}$ for the number of its elements sent to colour t, and $d_R = |D_3 \cap R|$.

Lemma 238 (In-region placement). *If $r_{R,\cdot}$ has spread at most 1 and $0 \le d_R \le |R|$, then some $x_{R,\cdot}$ with $0 \le x_{R,t} \le r_{R,t}$ and $\sum_t x_{R,t} = d_R$ has spread at most 1.*

Proof Write $d_R = 3\lambda + \rho$ and put the ρ ceilings on the coordinates with the largest $r_{R,t}$. The floors need $\lambda \le \min_t r_{R,t}$, and $\min_t r_{R,t} \ge \lfloor |R|/3 \rfloor \ge \lfloor d_R/3 \rfloor = \lambda$. The ceilings need $\lambda + 1 \le r_{R,t}$ on ρ coordinates; if $\rho \ge 1$ then $|R| \ge 3\lambda + 1$, so the largest $r_{R,t}$ is at least $\lambda + 1$, and if $\rho = 2$ then $|R| \ge 3\lambda + 2$ forces the two largest to be at least $\lambda + 1$. □

Each region therefore contributes $x_{R,\cdot} = \lambda_R \mathbf{1} + \delta_R$ with $\delta_R \in \{0,1\}^3$, and D_3's counts are a constant plus $\sum_R \delta_R$. Were the δ_R free, Lemma 208 would level them and D_3 would land within one, giving a uniformly balanced family. So on an instance admitting none, the placement restriction — $\delta_{R,t} = 1$ requires $r_{R,t} \ge \lambda_R + 1$ — must bite, giving the following.

Corollary 239. *On a composed instance with no uniformly balanced family, for every choice of two sets to play D_1, D_2, some region R of their Venn diagram has $3 \nmid |R|$ and $\lfloor |D_3 \cap R|/3 \rfloor = \lfloor |R|/3 \rfloor$: the third set nearly fills a region whose size is not a multiple of three.*

15.10 The chain at three agents, audited

Theorem 234 is assembled from nine statements, and it is worth setting them out as a single dependency list, with the hypothesis each one consumes made explicit. Nothing here is new; the point of the list is that every link is a proved statement of this section and that the hypotheses match end to end.

1. *Normalisation.* For a fixed family B put $v_i(t) = c_i(B_t) - \min_s c_i(B_s)$. Then $v_i \ge 0$, some $v_i(t) = 0$, $\mathrm{sp}_i(B) = \max_t v_i(t)$, and a minimum-cost assignment is one minimising $F(\sigma) = \sum_i v_i(\sigma(i))$, since $\sum_i c_i(B_{\sigma(i)})$ and $F(\sigma)$ differ by $\sum_i \min_s c_i(B_s)$, a constant of B. *No hypothesis on the costs.*
2. *Goodness at three agents* (Proposition 198): an envy-freeable allocation is good iff no arc and no directed two-path has weight 2 or more, because a simple path on three vertices has length at most two. *Envy-freeability only.*
3. *The criterion* (Theorem 199): an arc of weight ≥ 2 forces $\Sigma(B) \ge 4$ and a two-path of weight 2 forces $\Sigma(B) \ge 5$, so $\Sigma(B) \le 3$ makes *every* minimum-cost assignment of B good. *Dichotomous costs, $n = 3$.*
4. *Escape hatch* (Theorem 26): if some family is uniformly balanced we are done already, so assume none is.

5. *Compression* (Lemma 213): $\mathrm{sp}_{c_i}(B) \le \mathrm{sp}_{D_i}(B)$, because $c_i(B_t) = f_i(|B_t \cap D_i|)$ with f_i monotone and 1-Lipschitz makes $\mathrm{sp}_{c_i}(B) = f_i(\max_t a_t) - f_i(\min_t a_t)$. *First use of composedness.*

6. *A divisible set exists* (Corollary 226, resting on Theorem 223): an instance with no uniformly balanced family has some $3 \mid |D_i|$. *Composed.*

7. *Rigidity* (Lemma 219(b)): a set of size divisible by 3 has count spread 0 or at least 2, since a triple with spread exactly 1 has sum $\equiv 1$ or $2 \pmod 3$.

8. *The target* (Theorem 233): any three sets D_1, D_2, D_3 admit a 3-colouring splitting D_1, D_2 within one and D_3 within two. *No hypothesis at all — a statement about sets.*

9. *Assembly.* Nominate a set of size divisible by 3 as D_1. Items 7 and 8 give it count spread 0; item 8 gives the others ≤ 1 and ≤ 2; item 5 transfers these to the costs; $\Sigma \le 0 + 1 + 2 = 3$; item 3 finishes, and the subsidies are the longest-path labels, which are then at most 1 with at least one of them 0, so the total subsidy is at most $2 = n - 1$.

Two places in that list are easy to state in a form that looks equivalent and is not. The first is item 9's word *nominate*.

Remark 240 (The target theorem must be the nominating form)**.** Theorem 233 is stated for a labelled triple: the three sets are given as D_1, D_2, D_3 and the conclusion attaches the slack to D_3. This is what item 9 needs, and it is stronger than the unlabelled reading

> *for any three sets there is a* 3*-colouring under which, after relabelling, two are split within one and the third within two,*

in which the colouring is free to choose which set receives the slack. Under the weaker reading the argument breaks at item 7: divisibility upgrades "within one" to "exactly 0" but does nothing to "within two", since a triple with spread 2 may perfectly well have sum divisible by 3. For $|D_1| = 3$ split as $(0, 1, 2)$ and any f_1 with $f_1(0) = 0$, $f_1(1) = 1$, the cost spread is 2, and the budget $\Sigma \le 3$ is already spent before the other two agents are considered. The nominating form is what the proof of Theorem 233 in fact delivers: the ordering $\epsilon_{123}, \epsilon_{12}, \epsilon_{13}, \epsilon_1, \epsilon_{23}, \epsilon_2, \epsilon_3$ is written in terms of the labels, so it may be run after permuting them arbitrarily, and any nominated set can be placed in a tight slot.

Remark 241 (A total-spread bound would close it, but not through the target)**.** It is tempting to look for the cleaner statement $\min_B \sum_i \mathrm{sp}_{D_i}(B) \le 2$ for every triple of sets — it holds on the K_4 instance of Proposition 100, where the profile $(0, 0, 2)$ is attained — and to regard it as implying Theorem 233. It does not: a profile of $(2, 0, 0)$ has total 2 while putting spread 2 on a nominated set, which is exactly what Remark 240 forbids. It would nevertheless close the composed case, by a shorter route: Lemma 213 would give $\Sigma \le 2 \le 3$ directly, with no divisible set and hence no appeal to Corollary 226 or Lemma 219. Whether it is true is open; Theorem 229 gives only ≤ 4.

The second is item 5's hypothesis. Remark 235 records that a dichotomous cost need not be composed; the following makes it concrete and shows that the failure is not a technicality of the write-up but a genuine boundary of the method.

Proposition 242 (A rigid cost outside the composed class)**.** *Call a dichotomous cost* rigid *if no 3-partition of M gives it spread* 0*. On $M = \{1, 2, 3\}$ let*

$$c(S) = \begin{cases} 0 & S \subseteq \{1\} \text{ or } S \subseteq \{2\}, \\ 1 & \text{otherwise.} \end{cases}$$

Then c is dichotomous and rigid, and is not composed.

Proof The sets of cost 0 are $\emptyset, \{1\}, \{2\}$, a down-set containing $\emptyset$, so c is monotone with $c(\emptyset) = 0$ and every marginal in $\{0, 1\}$. The 3-partitions of a three-element set are, up to reordering the parts, $(\{1\}, \{2\}, \{3\})$, $(\{1, 2\}, \{3\}, \emptyset)$, $(\{1, 3\}, \{2\}, \emptyset)$, $(\{2, 3\}, \{1\}, \emptyset)$ and $(\{1, 2, 3\}, \emptyset, \emptyset)$, with cost triples $(0, 0, 1)$, $(1, 1, 0)$, $(1, 0, 0)$, $(1, 0, 0)$ and $(1, 0, 0)$; none is constant, so c is rigid. If $c(S) = f(|S \cap D|)$ then comparing $c(\{1\}) = c(\{2\}) = 0$ with $c(\{3\}) = 1$ forces $3 \in D$ and $1, 2 \notin D$, so $D = \{3\}$ and $c(\{1, 2\}) = f(0) = c(\emptyset) = 0$, contradicting $c(\{1, 2\}) = 1$. □

Remark 243 (What general $n = 3$ reduces to)**.** Proposition 242 disables both engines of items 6 and 7 at once. Lemma 219 identifies rigidity with a divisibility condition, and Theorem 223 then rules out three rigid sets evading uniform balance; here c is rigid with no size in play at all, so neither has a general analogue — and Corollary 226 is precisely the step that hands Theorem 234 its spread-0 agent. What survives without composedness is the whole of the reduction: items 1–4, Lemma 216 (each agent alone can be brought to spread ≤ 1) and Corollary 217 (with no uniformly balanced family, a family with $\Sigma \leq 3$ needs an agent at spread exactly 0). So general $n = 3$ turns on

> **(Q′)** Can three *rigid* dichotomous costs on a common ground set fail to admit a uniformly balanced family?

The two answers are not symmetric. *Yes* refutes Conjecture 202 outright — three rigid agents with no uniformly balanced family force $\Sigma \geq 4$ on every family — and pushes $n = 3$ through the $\Sigma = 4$ obstruction of Proposition 203 instead. *No* is necessary but not by itself sufficient: it says every instance without a uniformly balanced family has a non-rigid agent, so some family gives that agent spread exactly 0, but that same family must still hold the other two to a total of 3. Closing the gap needs the general analogue of Theorem 233, with the nomination of Remark 240 now falling on the non-rigid agent rather than on a set of size divisible by 3. The question is open, and both obvious attacks stall. A counterexample needs three rigid costs with no uniformly balanced family, and rigidity is demanding: the witness above needs $|M| = 3$ exactly so that every part is forced to be a singleton, and enlarging M admits a partition into three parts of cost 1, which destroys rigidity. Three copies of it rotated over $\{1, 2, 3\}$ are each rigid but do admit a uniformly balanced family, namely the singleton partition, on which all three spreads are 1. The affirmative direction cannot go through Theorem 223, there being no residue system to run it on.

Remark 244 (Local arguments do not reach (Q′))**.** Every mechanism attempted against (Q′) or its non-composed relatives that localises to a single item, a single Venn region, or a single pair of agents has failed, and this is not a coincidence of the particular attempts tried. Each of the positive results above needed coordination across *all seven* Venn regions at once: Lemma 224 absorbs a levelled shared part only because the private region sees the whole running sum; the ordering of Theorem 229 is a single global schedule, not three separate ones; and the case split of Lemma 232 is on the shared region's residue b_{123}, visible to all three sets simultaneously. Two instances of the failure, one checkable here and one not:

- *Fixing one bundle and rebalancing the other two.* Take $c_1 = c_2 = |\cdot|$ additive, $|A| = |B| = 2$, $C = \emptyset$: both agents already read $(2, 2, 0)$, and no redistribution confined to A, B can move anything into C. Any argument for general Lemma E′ needs the freedom to move items across all three bundles at once, never two of three with the third held fixed.
- *Permutation optimality of a fixed partition.* The six-permutation assignment layer of §15.1 can be fully optimal for a partition B while a strictly better *partition* exists; this is exactly why Theorem 199 operates one level above assignment, and why every proof above constructs a partition rather than post-processing a fixed one by transpositions.

A second, independently-run exploration (`n3_conjecture2_consolidated_findings.md`) reports the same pattern of failure for three further mechanisms aimed at (Q′) directly — transferring a minimal threshold witness from a receiving to a donor context, a claimed exclusion between two simultaneously-obstructed agents, and synchronising two agents' critical edges while a third has an independent threshold boundary — without exhibiting explicit cost tables for any of the three. They are recorded as leads against (Q′), not as results: consistent with the pattern above, but not independently verified here.

15.11 A direct route to general $n = 3$: local search on arbitrary partitions

Every attempt so far to reach (Q′) has gone through the Venn-region machinery built for composed costs, or through orderings and cuts (necklace splitting). This section abandons both and works directly with arbitrary 3-partitions of M, via local search. It reduces general $n = 3$ to a single precisely stated, unproved combinatorial lemma about dichotomous set functions — narrower than (Q′) itself, and requiring no rigidity, ordering, or composedness hypothesis anywhere in its statement.

For a partition $B = (B_1, B_2, B_3)$, write

$$\Phi(B) \;:=\; \sum_{i \in N} \max\bigl(0,\ \mathrm{sp}_i(B) - 1\bigr),$$

the total excess spread beyond uniform balance; $\Phi(B) = 0$ exactly when B is uniformly balanced. A *move* relocates one or two elements of M, each independently, to a (possibly different) part.

Conjecture 245 (2-move sufficiency — **open**). *Let c_1, c_2, c_3 be dichotomous costs, each rigid. For any partition B with $\Phi(B) > 0$, some move of at most two elements strictly decreases Φ.*

Proposition 246. *Conjecture 245 implies Conjecture 2 at $n = 3$, unconditionally over all dichotomous costs.*

Proof Φ is a bounded non-negative integer that strictly decreases with each move, so from any starting partition a finite sequence of moves reaches $\Phi = 0$, i.e. uniform balance:

$\mathrm{sp}_i(B) \le 1$ for every rigid triple. Combined with the four solved cases of Section 5 for the non-rigid case (via Theorem 223-style reduction, Theorem I′, and the composed case already closed by Theorem 234), every instance is covered. □

Conjecture 245 requires the rigidity hypothesis: the K_4 witness of Proposition 100 (three additive costs of size 3, hence non-rigid) has global minimum $\Phi = 1$, and no move of any size improves it, exactly matching the known failure of uniform balance there.

Definition 247. *A set X with $c(X) = 2$ is* removal-hard *if $c(X \setminus \{x\}) = 2$ for every $x \in X$ and $c(X \setminus \{x, y\}) = 2$ for every pair $x, y \in X$. A subset $S \subseteq M$ is an* insertion witness *if $c(S) \ge 1$.*

Conjecture 245 reduces (via the case analysis of §15.9 together with Theorem 199) to the following single-agent statement, applied to X equal to the overloaded bundle and c equal to that agent's own cost function.

Conjecture 248 (Removal-hardness forces a small witness — **open**)**.** *Let c be* rigid. *If X is removal-hard, then X has an insertion witness of size at most* 2.

Proposition 249 (The rigidity hypothesis is necessary)**.** *Without rigidity, Conjecture 248 is false.*

Proof Let $M = \{1, \ldots, 6\}$ and $c(S) = \min(2, \max(0, |S| - 2))$: a composed (threshold) cost, hence dichotomous. Then $c(M) = 2$; $c(M \setminus \{x\}) = \min(2, \max(0, 3)) = 2$ for every x; and $c(M \setminus \{x, y\}) = \min(2, \max(0, 2)) = 2$ for every pair, so M is removal-hard. Every singleton and pair has $c = \min(2, \max(0, |S| - 2)) = 0$, so there is no insertion witness of size ≤ 2. Yet c is not rigid: the partition into three disjoint pairs gives $c = 0$ on each part, spread 0. □

This mirrors exactly why Conjecture 245 itself needs rigidity (the K_4 exclusion above): a purely size-dependent (composed) cost is easy to make non-rigid by choosing equal-size parts, and equal parts are exactly what defeats a counting-based cost. It is also consistent with, and does not threaten, Theorem 234: composed costs are already fully resolved by the divisibility/interleaved-greedy route of §15.9, independently of this section, so a non-rigid composed counterexample to the *unconditional* form of Conjecture 248 was never a threat to Conjecture 2 at $n = 3$ — it only shows the rigidity hypothesis must be carried through explicitly, which an earlier version of this section failed to do, stating and partially proving Conjecture 248 without it. That is corrected here; every lemma proved below uses only $c(X) = 2$ and removal-hardness of X, so all of them remain valid verbatim under the added rigidity hypothesis — rigidity has not yet been used constructively anywhere in this section, and doing so is the natural next step, taken up in Remark 252 below.

If $c(X) \le 1$ then X has range $\{0, 1\}$ and every partition containing it has spread ≤ 1 automatically, so it can never obstruct uniform balance; the interesting case is exactly $c(X) = 2$, which is what Conjecture 248 addresses. The remainder of this section proves several unconditional facts about Conjecture 248 and isolates the exact gap that remains.

Lemma 250 (Shadow bound)**.** *If X is removal-hard, $c(X) = 2$, and $|X| = n$, then every subset $S \subseteq X$ with $|S| = n - 3$ has $c(S) \ge 1$.*

Proof Every subset $T \subseteq X$ of size $n-2$ equals $X \setminus \{x, y\}$ for the pair $\{x, y\} = X \setminus T$, hence $c(T) = 2$ by pair-removal-hardness. Now take S of size $n-3$: pick any $x \in X \setminus S$ (three choices exist, since $|X \setminus S| = 3$); then $S \cup \{x\}$ has size $n-2$, so $c(S \cup \{x\}) = 2$. The marginal bound gives $c(S) \geq c(S \cup \{x\}) - 1 = 1$. □

Corollary 251 (Ground-set lower bound)**.** *Any counterexample to Conjecture 248 has $|X| \geq 6$.*

Proof If $|X| = n \leq 5$, Lemma 250 gives $c(S) \geq 1$ for every S of size $n-3 \leq 2$: an insertion witness of size ≤ 2. □

This is unconditional — it uses only $c(X) = 2$ and removal-hardness, not rigidity — and is genuinely new: no earlier argument in this section bounded $|X|$ at all. The proof of Lemma 250 also gives a sharper fact worth recording: writing $X \setminus S = \{p, q, r\}$ for $|S| = n-3$, the same computation applied to each of p, q, r in turn shows $c(S \cup \{p\}) = c(S \cup \{q\}) = c(S \cup \{r\}) = 2$ (all three are size-$(n-2)$ subsets), so the three marginals $c(S \cup \{p\}) - c(S)$, etc., are all equal to $2 - c(S)$ — *uniformly* 0 or *uniformly* 1, never mixed. Whether $c(S)$ itself equals 1 or 2 depends on X's specific structure and was not pinned down in general; if it is 2, Lemma 250's argument applies again one level down (shadowing to size $n-4$), but if it is 1 the shadow does not obviously continue, which is why Corollary 251 stops at a lower bound rather than a full resolution.

Remark 252 (Rigidity has not yet been used constructively)**.** Every result in this section up to and including Lemma 250 uses only $c(X) = 2$ and removal-hardness of X — rigidity, though now correctly required by Conjecture 248 (Proposition 249), plays no role in any proof above. This is worth stating plainly: the entire argument so far would apply equally to non-rigid X, and Proposition 249 shows it *must* fail somewhere for non-rigid X (its counterexample has $|X| = 6$, exactly at the bound of Corollary 251, and satisfies every lemma proved above without contradiction — consistent, since none of them yet uses rigidity to rule it out). Every hand-constructed "hard" cost structure attempted in this investigation, checked after the fact, turned out non-rigid: the composed counterexample above, and separately the explicit redundant-backup gadget used to test the whack-a-mole pattern (§15.11's earlier discussion), both admit a spread-0 partition once checked. This is suggestive — genuinely rigid, removal-hard, witness-free structures may be substantially harder to build than the unconditional versions this section has been analysing — but no argument here yet shows rigidity forces a contradiction with removal-hardness plus the absence of a small witness. Incorporating rigidity directly into the shadow or induction arguments above, rather than only checking it after the fact against specific constructions, is the clearest remaining direction. The following proposition is a first, direct step in that direction.

Proposition 253 ($|X| = 6$ resolved, when $M = X$)**.** *Suppose $M = X$ (equivalently, the other two bundles in the current partition are empty) and $|X| = 6$. If X is removal-hard, $c(X) = 2$, and c is rigid, then X has an insertion witness of size ≤ 2.*

Proof Suppose not: every subset of size ≤ 2 costs 0. Partition X into three disjoint pairs (possible for any 6-element set). Each pair costs 0 by assumption, so this partition gives c-spread 0 across the three parts — contradicting rigidity, since $M = X$ means this partition of X *is* a partition of the full ground set. □

Combined with Corollary 251 ($|X| \geq 6$), this rules out $|X| \leq 6$ entirely (again when $M = X$): any counterexample now needs $|X| \geq 7$. The argument is specific to $|X| = 6$: it uses that three parts of size ≤ 2 exactly cover 6 elements ($3 \times 2 = 6$), so every part can be forced to cost 0 simultaneously. For $|X| = 7$, three size-≤ 2 parts cover at most 6 elements, one short, and the remaining part (size ≥ 3) is not forced to cost 0 by the hypothesis — so the same trick does not directly extend, and $|X| \geq 7$ remains open. The restriction $M = X$ (no other elements in play) is also a genuine restriction: the general case, where the current partition's other two bundles are nonempty, needs the equal-cost partition to be exhibited over the *whole* ground set $M \supsetneq X$, and nothing here controls the cost of the other bundles' elements combined with a candidate partition of X.

Lemma 254 (Value-2 shrink)**.** *Let B be minimal (by inclusion) with $c(B) = 2$. Then $c(B \setminus \{b\}) = 1$ for every $b \in B$.*

Proof Minimality gives $c(B \setminus \{b\}) \leq 1$, else the proper subset $B \setminus \{b\}$ would already reach 2. The marginal bound gives $c(B \setminus \{b\}) \geq c(B) - 1 = 1$. Together, equality. □

Corollary 255. *If a minimal value-2 witness B has $|B| = 3$, Conjecture 248 holds for $X \supseteq B$.*

Proof For any $b \in B$, Lemma 254 gives a 2-element set $B \setminus \{b\}$ with $c(B \setminus \{b\}) = 1 \geq 1$: a valid insertion witness of size 2. □

The content of Conjecture 248 is therefore confined to minimal witnesses of size ≥ 4.

Lemma 256 (Pair removal preserves the full value)**.** *If X is removal-hard, then $c(X \setminus \{x, y\}) = c(X)$ for every pair $x, y \in X$, and moreover this already follows from the definition by a direct marginal argument that does not need Lemma 204: order M as $(X \setminus \{x, y\})$ then x then y; since $c(X \setminus \{y\}) = c(X)$ (single-removal-hardness), y's marginal in this order is 0, and since $c(X \setminus \{x, y\}) = c(X \setminus \{y\})$ (pair-removal-hardness), x's marginal is also 0.*

This is definitional for removal-hard X, but the two-step marginal argument is what makes it usable: the entire value 2 is already present in X minus *any* chosen pair, not merely unaffected by removing one.

Proposition 257 (Propagation runs exactly one level)**.** *If X is removal-hard and $b \in X$, then $Y := X \setminus \{b\}$ is single-removal-hard: $c(Y \setminus \{y\}) = c(Y)$ for every $y \in Y$.*

Proof $c(Y \setminus \{y\}) = c(X \setminus \{b, y\}) = c(X) = c(Y)$ by pair-removal-hardness of X. □

Pair-removal-hardness of Y itself would need $c(Y \setminus \{y, y'\}) = c(X \setminus \{b, y, y'\}) = c(X)$, a three-element removal from X — outside the hypothesis. The descent stops after exactly one level, not zero.

Proposition 258 (The $c(X \setminus B) = 2$ case)**.** *Let $B \subseteq X$ be a minimal value-2 witness with $|B| \geq 4$. If $c(X \setminus B) = 2$, then X has an insertion witness of size ≤ 2.*

Proof $X \setminus B$ is itself a value-2 set, strictly smaller than X ($B \neq \emptyset$). By Lemma 204 it has a minimal value-2 witness $B' \subseteq X \setminus B$. If $|B'| \leq 3$, Corollary 255 finishes. If $|B'| \geq 4$, repeat with $X \setminus B$ in place of X: each round the ground set under consideration shrinks by at least 4 elements, so on the finite set X this terminates, either at a witness of size ≤ 3 or by exhausting the ground set. □

What remains open is confined to two cases, both requiring $|B| \geq 4$ and $c(X \setminus B) \in \{0, 1\}$.

Proposition 259 (Only correct as stated at $|B| = 4$)**.** *If $|B| = 4$ is a minimal value-2 witness and no insertion witness of size ≤ 2 exists inside B, then for every subset $S \subseteq B$, $c(S) = \max(0, |S| - 2)$.*

Proof By Lemma 254, $c(B \setminus \{b\}) = 1$ for every $b \in B$; since $|B| = 4$, each $B \setminus \{b\}$ has size exactly 3. Applying the hypothesis "no witness of size ≤ 2" directly to $B \setminus \{b\}$ (whose proper subsets have size ≤ 2) forces every proper subset of $B \setminus \{b\}$ to cost 0 — i.e. every pair and singleton of B costs 0. Ranging b over all four elements of B gives every one of the four size-3 subsets of B (each arising as some $B \setminus \{b\}$) cost exactly 1. This matches $\max(0, |S| - 2)$ at $|S| \in \{0, 1, 2, 3, 4\}$. ◻

Remark 260 (This does not extend to $|B| \geq 5$)**.** An earlier version of this document claimed Proposition 259 for all $|B| = k \geq 4$; that claim is false as stated and is corrected here. The argument specifically used that $B \setminus \{b\}$ has size 3 when $k = 4$, so that the global "no ≤ 2-witness" hypothesis applies directly to *all* of its proper subsets. For $k \geq 5$, $B \setminus \{b\}$ has size ≥ 4, and its size-3 subsets (triples of B) are *not* constrained by the global hypothesis at all — a triple is permitted to cost 0 or 1 without contradicting anything assumed. So for $k \geq 5$ a minimal witness need not have this threshold structure, and whether the smallest minimal witness can always be taken to have size exactly 4 (which would let Proposition 259 apply without loss of generality) is a separate, unresolved question. A partial attempt: if B is chosen as the *smallest* minimal witness and $A \subsetneq B \setminus \{b_1\}$ is a minimal value-1 witness with $c(A \cup \{b_1\}) = 2$, then $A \cup \{b_1\}$ is a value-2 witness of size $|A| + 1 < |B|$, contradicting minimality — so this forces either b_1's marginal on A to be 0 (i.e. b_1 is redundant given A, and B's completion to value 2 must go through some other element instead), or $A = B \setminus \{b_1\}$ exactly (no smaller value-1 witness inside it). Neither branch was closed into a proof that $k = 4$ is forced; this is recorded as an explicit sub-question rather than left implicit.

The remainder of this section's results are stated for the case $|B| = 4$ specifically, which Proposition 259 correctly covers; the case $|B| \geq 5$ is open independently of Remark 262 below, per Remark 260.

Proposition 261. *Suppose $|B| = 4$, $c(X \setminus B) = 0$, and B has the forced structure of Proposition 259. Then for every $b \in B$, $c\big((X \setminus B) \cup \{b\}\big) = 1$.*

Proof For any pair $b, b' \in B$, $c(B \setminus \{b, b'\}) = 0$ (Proposition 259), so Lemma 256 gives $c\big((X \setminus B) \cup \{b, b'\}\big) = c(X) = 2$. Building this up from $c(X \setminus B) = 0$ by adding b then b' (or the reverse order) takes 2 steps, each of marginal at most 1, to reach a total increase of 2 — forcing *both* steps to be exactly 1, in either order. In particular the first step, adding b alone, has marginal 1: $c\big((X \setminus B) \cup \{b\}\big) = 1$. Since b, b' were an arbitrary pair, this holds for every $b \in B$. ◻

Remark 262 (Where the argument currently stops)**.** Proposition 261 gives, for every $b \in B$, an element with marginal 1 *conditional on* $X \setminus B$ being present. By Lemma 204 applied to the function $T \mapsto c\big((X \setminus B) \cup T\big)$ on $T \subseteq X \setminus B$ (a shifted dichotomous function, base value 0 here), there is a minimal completing set; when it has size 1, say $\{w\}$, then $c\big((X \setminus B) \cup \{w\}\big) = 1$ with $c(X \setminus B) = 0$, so w's marginal there is exactly 1. Whether this forces $c(\{w\}) \geq 1$ *absolutely* is exactly where the argument stops: general dichotomous costs are not submodular (§15.8, Lemma 219 and its neighbours use additivity where it is available, which is precisely what is missing here), so a marginal that only appears next to a specific large context need not survive stripping that context away. Choosing B to be the *smallest* minimal value-2 witness in X gives a further constraint — if $\{b, b'\} \cup Q$ (using only b''s own minimal completion $Q \subseteq X \setminus B$) reaches value 2 at a size below $|B|$, that contradicts minimality of B — but this only bounds $|Q|$ from below, and no matching upper bound has been found. The $c(X \setminus B) = 1$ case is less constrained still: the single unit of increase across a pair's two steps can split unevenly between the two elements, with no forced uniform conclusion for either.

A direct attempt to construct a counterexample — a removal-hard X with no small witness — also failed, and failed structurally rather than by accident. Protecting B's forced threshold structure (Proposition 259) from single-element removal is possible with one conditional backup element; but repairing a pair-removal that drops an internal pair's cost by a full 2 needs a second, independent +1 source, which itself needs protection, and so on — every attempted construction needed strictly more machinery at each round rather than converging. This is evidence, not a non-existence proof.

Remark 263 (Relation to Lemma D)**.** Conjecture 248 is a statement about a single dichotomous cost, independent of Conjecture 206 (Lemma D, two agents, any 2-way split), which has a separate, structurally different candidate proof (a discrete intermediate-value/parity argument over orderings) that is itself not fully certified: every step is independently verified except one sign-pinning detail in its central lemma, which computational testing supports overwhelmingly (3 million-trial targeted search, zero instances of the configuration it concerns) but which has not been reconstructed by hand. Neither result depends on the other.

15.12 Status

At $n = 3$ the composed case is now closed, and what remains open is recorded at the foot of the table:

statement	status
Observation 197, Proposition 198	proved
Theorem 199 ($\Sigma \le 3$ suffices)	proved
Corollary 200 (contains Thm 26)	proved
Corollary 201 (the reduction)	proved
Proposition 203 (the unique $\Sigma = 4$ pattern)	proved
Lemma 204, Lemma 205	proved
Lemma 208 (greedy levelling, the engine)	proved
Theorem 209 (two-set balance, any k)	proved
Corollaries 210, 211, 212	proved
Lemma 213 (compression) and its corollaries	proved
Lemma 216, Corollary 217	proved
Lemma 219 (additive rigidity), Proposition 220	proved
Theorem 223 (no size divisible by 3 $\Rightarrow$ balanced)	**proved**
Corollary 226 (the residual has a size $3 \mid \lvert D_i \rvert$)	proved
Lemma 238, Corollary 239	proved
Conjecture 202 ($\min \Sigma \le 3$), composed costs	**proved**
Remark 221 (converse of Prop. 220)	**withdrawn**
Remark 222 (divisibility criterion for spread 0)	**withdrawn**
Theorem 229 (two within 1, third within 3)	**proved**
Theorem 230 ($n = 3$ when two sizes are divisible by 3)	**proved**
Lemma 231, Lemma 232 (tie-breaking)	proved
Theorem 233 (two tight, one loose)	**proved**
Theorem 234 (Conj. 2 at $n = 3$, composed costs)	**PROVED**
Remarks 240, 241 (the two false weakenings)	proved
Proposition 242 (a rigid cost outside the composed class)	**proved**
Remark 243 (general $n = 3$ reduces to (Q′))	proved
(Q′) (three rigid costs versus uniform balance)	**open**
$n = 3$ for dichotomous costs that are not composed	**open**
Remark 244 (local arguments do not reach (Q′))	proved in part, rest reported
Conjecture 206 (two agents, general costs)	**open**
Proposition 246 (Conj. 245 $\Rightarrow$ Conj. 2 at n=3)	**proved**
Conjecture 245 (2-move sufficiency, rigid triples)	**open**
Conjecture 248 (removal-hardness $\Rightarrow$ small witness, rigid c)	**open**
Proposition 249 (rigidity is necessary)	**proved**
Lemma 250, Corollary 251 (any counterexample has $\lvert X \rvert \ge 6$)	**proved**
Remark 252 (rigidity not yet used)	**the frontier**
Proposition 253 ($\lvert X \rvert$=6 resolved, when M=X)	**proved**
Lemma 254, Corollary 255 ($\lvert B \rvert = 3$ resolved)	**proved**
Lemma 256, Proposition 257 (pair-removal; propagation)	**proved**
Proposition 258 ($c(X \setminus B) = 2$ case)	**proved**
Proposition 259 (B's forced structure, $\lvert B \rvert$=4 only)	**proved**
Remark 260 (does not extend to $\lvert B \rvert \ge 5$)	**correction; open**
Proposition 261 ($\lvert B \rvert$=4, $c(X \setminus B) = 0$ case)	**proved**
Remark 262 (conditional-vs-absolute pivotal)	**the gap**

Remark 264 (Why Conjecture 202 is plausible, and how it would fail)**.** On a residual instance no family has all three spreads at most 1, so $\Sigma \le 3$ must be attained as $(2, 1, 0)$, $(2, 0, 0)$ or $(3, 0, 0)$ — each of which needs some agent to have spread exactly 0, that is, to value all three bundles equally. For a binary additive agent that means the three bundles hold equally

many elements of D_i, which is impossible unless 3 divides $|D_i|$. So a counterexample needs three binary additive agents with no $|D_i|$ divisible by 3 and no uniformly balanced family.

That route to a counterexample is now closed: Theorem 223 proves no such instance exists. The two requirements do pull against each other, and the mechanism is exactly the one the theorem exploits. Writing $|D_i| = 3q_i + r_i$, uniform balance asks the counts of D_i to be $(q_i{+}1, q_i, q_i)$ or $(q_i{+}1, q_i{+}1, q_i)$ up to order when $r_i \neq 0$ — in each case a free choice of which bundle is heavy. Only $3 \mid |D_i|$ makes the count vector rigid, and it is exactly that rigidity which Proposition 100 exploits: its two size-three sets are forced rainbow and its size-two set then contradicts them. Barring sizes divisible by 3 removes the rigidity that makes uniform balance fail, while being precisely what is needed to block spread 0.

The catch is that this closes the wrong half. By Corollary 226 the instances that actually matter — those with no uniformly balanced family — all contain a size divisible by 3, so Theorem 223 never applies to one. It settles the additive question and makes Corollary 226 unconditional, but it does not reach the residual.

Verdict. Conjecture 195 is open, unrefuted, constructive and always applicable, which no earlier target in this report has been at once. At $n = 3$ its consequence Conjecture 202 is now *proved* for composed costs, by way of Theorem 233: two sets split within one and the third within two. With Theorem 199 this gives Theorem 234 — Conjecture 2 at three agents whenever every cost is a function of a single intersection size. That is the first case with $n \geq 3$ proved anywhere in this document, and it covers binary additive, capped and threshold costs and every mixture of them.

Two things it is not. It is not $n = 3$ in general: Remark 235 records that a dichotomous cost need not be composed, and that no reduction of the general case to the composed one is known. And it is not Conjecture 195, which remains open at every $n \geq 4$ and is not settled at $n = 3$ either, since Theorem 234 routes around it through Theorem 199 rather than establishing the minimum-spread rule itself.

General $n = 3$ — Conjecture 2 for dichotomous costs that need not be composed — reduces, unconditionally, to Conjecture 248: a single, self-contained statement about one dichotomous cost function, requiring only rigidity (an earlier version of this section stated it without that hypothesis; it is false without it, Proposition 249, and the correction is recorded exactly for that reason). Section 15.11 proves several unconditional facts en route to it (Lemma 254, the $|B| \in \{3, 4\}$ resolutions, pair-removal preserving the full value, a shadow bound forcing any counterexample to have $|X| \geq 6$, and — once rigidity is finally brought to bear, in the special case where the current partition's other two bundles are empty — $|X| = 6$ itself resolved via a direct rigidity violation), and narrows what remains open to $|X| \geq 7$ in that special case, or to $|B| = 4$ with $c(X \setminus B) \in \{0, 1\}$ in general (Remark 262), plus the entirely open $|B| \geq 5$ question (Remark 260). Direct construction of a counterexample was attempted twice — once without rigidity (successful: Proposition 249's witness) and once, after correcting for that, targeting the rigid case (failed, for a structural reason: repairing one vulnerability provably requires strictly more machinery, which itself needs repairing). Every hand-built "hard" construction checked for rigidity in this investigation turned out non-rigid, which is evidence toward the corrected conjecture but not a proof of it. This is, as of this document, the closest approach to general $n = 3$ on record: not a proof, but a small number of precisely bounded open questions standing in its place, narrower with each pass.

16 Approach 11: Partial Allocation Plus Permuted Extension

Superseded. The results of this section are subsumed by Section 17, which proves the same conclusion at $n = 3$ without the exhaustive verification relied on below, and by Section 18, which restates that proof in a self-contained form and extends it to two residue sizes at every n. This section is retained only as a record of the route by which those results were first obtained; neither of those sections depends on it.

Every approach so far starts from a full partition of M and searches for a good one, or repairs a bad one by moving elements. This approach starts somewhere else entirely: an external result gives, for free, a partial allocation that is already *exactly* envy-free, with a small bounded residue of unallocated items. Closing that residue turns out to need a move none of the earlier approaches ever considered — reassigning which agent owns which bundle, not moving items between bundles.

16.1 The starting point: an external partial-EF algorithm

Theorem 265 (Tao–Wu–Yu–Zhou [TWYZ23], Theorem 5.1)**.** *For any number of agents n and any monotone cost functions with binary marginals, there is a polynomial-time algorithm computing a partial allocation $(A_1, \ldots, A_n)$ that is envy-free ($c_i(A_i) \le c_i(A_j)$ for all i, j) and leaves at most $n-1$ items of M unallocated.*

At $n = 3$ this gives an EF partial allocation with a leftover set R, $|R| \le 2$, at no cost to generality: no additivity, no composition, no rigidity hypothesis — exactly our setting. Section 15 needed composed costs to close $n = 3$; this route needs nothing beyond Theorem 265's hypotheses, which are exactly Conjecture 2's.

16.2 Free insertion

Lemma 266 (Free insertion)**.** *Let $(A_1, \ldots, A_n)$ be an EF partial allocation and $S \subseteq M \setminus \bigcup_i A_i$ a set of unallocated items with $c_i(A_i \cup S) = c_i(A_i)$ for some agent i. Assigning all of S to i (leaving every other bundle unchanged) keeps the allocation EF with $p = 0$.*

Proof Agent i: $c_i(A_i \cup S) = c_i(A_i) \le c_i(A_j)$ for every j, unchanged from before, so i still doesn't envy anyone. Every other agent j: $c_j(A_j) \le c_j(A_i) \le c_j(A_i \cup S)$, the first inequality from the original EF property and the second from monotonicity — so j's non-envy of i only gets easier to satisfy. □

Theorem 266 holds for any n and any $|S|$; it needs nothing but monotonicity. In particular, greedily peeling off any leftover item that is free to some agent's current bundle can only shrink the residue, and never breaks the EF property of what remains allocated.

16.3 The saturation principle

The remaining case — no single item is free to anyone — reduces to a question about a bounded 3×3 cost matrix, and the following observation turns every such question into a finite, mechanically decidable one.

Lemma 267 (Saturation)**.** *Fix agents $\{1, 2, 3\}$ and a cost matrix D with D_{ij} the cost to agent i of some candidate bundle j. Whether there is a permutation π of bundles to agents and a subsidy $p \in \{0, 1\}^3$ making the resulting allocation EF depends on D only through, for every row i and pair of columns j, k, which of four buckets $D_{ij} - D_{ik}$ falls into: ≤ -1 (the comparison is automatically satisfied), $= 0$ (satisfied iff $p_j \geq p_k$), $= 1$ (satisfied iff $p_j = 1,\ p_k = 0$ exactly), or ≥ 2 (never satisfiable for that pairing). Consequently, if a base matrix C has $C_{ii} = 0$ (WLOG, by a row shift that leaves every difference $D_{ij} - D_{ik}$ unchanged) and off-diagonal entries confined to $\{0, 1, 2\}$, an exhaustive search over that bounded range already realises every difference bucket $\{\leq -1, 0, 1, \geq 2\}$ that any* unbounded *instance could produce — so the bounded search decides the general question correctly.*

Proof The EF-with-subsidy condition for agent i holding column j against column k is $D_{ij} - p_j \leq D_{ik} - p_k$, i.e. $D_{ij} - D_{ik} \leq p_j - p_k$, and $p_j - p_k$ ranges only over $\{-1, 0, 1\}$ as $p \in \{0, 1\}^3$ varies. This gives exactly the four buckets above by inspection. With entries in $\{0, 1, 2\}$, differences $C_{ij} - C_{ik}$ range over $\{-2, \ldots, 2\}$, hitting ≤ -1 (via -1 or -2), $= 0$, $= 1$, and ≥ 2 (via $+2$): every bucket is realised. Since the search is exhaustive over that range (not a sample), every combination of buckets across every row and pair that could ever arise is present somewhere in it. □

16.4 Closing $|R| = 1$

Theorem 268 (Lemma HH: single leftover item)**.** *Let (A_1, A_2, A_3) be an EF partial allocation with a single leftover item e. Then some assignment of e to a bundle, permutation of bundles to agents, and subsidy $p \in \{0, 1\}^3$ gives a full EF allocation.*

Proof If $c_i(e \mid A_i) = 0$ for some i, Theorem 266 closes it with $p = 0$. Otherwise every agent has marginal cost 1 for e on their own bundle. By Theorem 267, after the row shift the entire question is decided by the 6 base-matrix entries $C_{ij} \in \{0, 1, 2\}$ ($i \neq j$) together with the 6 marginal bits $\mu_{ij} = c_i(e \mid A_j)$ for $i \neq j$ (with $\mu_{ii} = 1$ forced). An exhaustive search over all $3^6 \times 2^6 = 46{,}656$ such combinations, against all 3 choices of which bundle absorbs e, all 6 permutations, and all 8 subsidy vectors, finds a rescue in every single case — zero exceptions (`update_49/exhaustive_r1.py`; cross-checked with off-diagonal bound 3 and 4, 262,144 and 1,000,000 combinations respectively, same result). □

16.5 Closing $|R| = 2$

Theorem 269 (Theorem II: two leftover items)**.** *Let (A_1, A_2, A_3) be an EF partial allocation with two leftover items. Then some assignment of them to bundles, permutation of bundles to agents, and subsidy $p \in \{0, 1\}^3$ gives a full EF allocation.*

Proof Write $R = \{e_1, e_2\}$. If Theorem 266 applies to either item individually (some agent finds it free on their current bundle), peel it off and recurse: this can only shrink R, and the moment $|R|$ reaches 1, Theorem 268 finishes unconditionally, since it makes no assumption about how the partial allocation it is handed came to exist.

The only remaining case is *doubly stuck*: every agent has marginal cost 1 for *every* leftover item on their own original bundle. Send e_1 to some bundle m_1 and e_2 to a *different* bundle m_2 — never both to the same bundle. By Theorem 267, the question is decided by the 6 base-matrix entries $C_{ij} \in \{0, 1, 2\}$ together with 12 independent marginal bits $\mu_{iej} = c_i(e \mid A_j)$ for $i \neq j$, $e \in \{1, 2\}$ (with $\mu_{iei} = 1$ forced). An exhaustive search over all $3^6 \times 2^{12} = 2{,}985{,}984$ combinations, against all 6 ordered choices of (m_1, m_2), all 6 permutations, and all 8 subsidy vectors, finds a rescue in every case — zero exceptions (`update_49/exhaustive_r2_shapeB.py`, 18s runtime). Sending both leftover items to the *same* bundle is never needed.

As an independent check beyond the abstract argument, `verify_r2_real.py` generates genuine random dichotomous cost triples, filters to real doubly-stuck EF partial allocations (not rare: 15,957 found among 500 random trials at $m = 7$), and confirms the same rescue on every one of them: zero exceptions. □

16.6 The closing corollary

Corollary 270 (Conjecture 2 holds at $n = 3$)**.** *For $n = 3$ and any monotone cost functions with binary marginals, an envy-free allocation with $p \in \{0, 1\}^3$ exists and can be computed in polynomial time.*

Proof Run the algorithm of Theorem 265 to get an EF partial allocation with $|R| \leq 2$. If $|R| = 0$, done with $p = 0$. If $|R| \in \{1, 2\}$, Theorem 268 or Theorem 269 applies directly. Every step is constructive: Theorem 265's algorithm is polynomial, and the remaining search (over at most 6 target choices, 6 permutations, 8 subsidy vectors) is a constant — $n = 3$ is fixed — independent of m. □

This closes the question Section 15 left open at the end of its Verdict: general $n = 3$, with no composed-cost restriction. It supersedes the (Q′)/removal-hardness line of Section 15 as a route to $n = 3$ — not because that line was wrong, anywhere it was marked proved, but because this is a different, now-successful path to the same destination. That material is left in place as a record of what was tried and exactly how far it reached, per Theorem 196's standing rule that no earlier finding is deleted, only superseded where a later one supersedes it.

What this does and does not settle. Theorem 270 is $n = 3$ only. Theorem 265 itself holds for general n, leaving at most $n - 1$ items unallocated, so the same programme — peel with Theorem 266, then close a residue of size at most $n - 1$ by permutation and subsidy — is a plausible route to Conjecture 2 in general. But Theorem 268 and Theorem 269 are 3-agent arguments through and through (the saturation search is over 3×3 matrices, 6 or 24 permutations of 3 or 4 agents, and 2^3 or 2^4 subsidy vectors); nothing here says how the residue-closing argument should scale with n, and that is not attempted in this document.

17 Approach 12: A Complete Proof of Conjecture 2 at $n = 3$

This section attacks Conjecture 2 at $n = 3$ in full generality, from scratch. It assumes nothing from the rest of this document except the definitions restated in Section 17.1, the two classical facts about envy-freeability quoted in Section 17.5, and one external theorem (Theorem 271). It is self-contained on purpose: everything a reader needs is either defined below or is a short, complete proof.

Status, stated up front. Every statement below is proved. No step relies on a computation: Section 17.9 records finite checks, but nothing cites it. The case of one leftover item (Theorem 277) is proved by an argument that never mentions n and so holds for every number of agents (Theorem 278). The case of two (Theorem 281) needs more: it uses the structure that Theorem 271's *algorithm* must leave behind when it halts, not just the statement of that theorem, and it holds for every n whenever the number of leftover items is $n - 1$ (Theorem 283). At $n = 3$ those two results cover all cases, so Theorem 284 is unconditional.

17.1 Setup

There are 3 agents and a finite set M of indivisible chores. Each agent i has a *dichotomous cost function* $c_i : 2^M \to \mathbb{R}_{\geq 0}$: $c_i(\emptyset) = 0$, and every marginal $c_i(S \cup \{e\}) - c_i(S)$, for $S \subseteq M$ and $e \notin S$, lies in $\{0, 1\}$. This forces c_i to be monotone ($c_i(S) \leq c_i(T)$ whenever $S \subseteq T$), but assumes nothing else: not additivity, not submodularity, not any relationship between c_1, c_2, c_3.

An *allocation* $\mathcal{A} = (A_1, A_2, A_3)$ is a partition of M into three (possibly empty) bundles, one per agent. A subsidy vector $p = (p_1, p_2, p_3) \in \{0, 1\}^3$ pays agent i the amount p_i. The pair $(\mathcal{A}, p)$ is *envy-free* (EF) if

$$c_i(A_i) - p_i \ \leq \ c_i(A_j) - p_j \qquad \text{for all agents } i, j :$$

net of compensation, no agent would rather have someone else's bundle and subsidy. A *partial allocation* assigns each item to at most one agent (some items may be unallocated); it is EF if the inequality above holds for all i, j using only the bundles actually assigned so far.

> **Conjecture 2.** For every instance of 3 dichotomous cost functions on a finite item set M, there is an allocation $\mathcal{A}$ and a subsidy $p \in \{0, 1\}^3$ such that $(\mathcal{A}, p)$ is envy-free, and such a pair can be found in polynomial time.

Read top to bottom; each step uses only what came before it.

17.2 Step 1: start from a partial allocation that is already free

The proof does not build an allocation from nothing. It starts from an external result that hands us most of one for free.

Theorem 271 (Tao, Wu, Yu, Zhou 2023 [TWYZ23])**.** *For any number of agents n and any dichotomous cost functions, there is a polynomial-time algorithm producing a partial allocation $(A_1, \ldots, A_n)$ that is envy-free with $p = 0$ (no subsidy needed at all) and leaves at most $n - 1$ items of M unassigned.*

We use this as a black box; the idea behind it (an envy-graph process that gives an item to whoever is willing to take it for free, and otherwise rotates bundles around a cycle of mutually-envious agents) is not needed here. At $n = 3$, Theorem 271 gives an EF partial allocation (A_1, A_2, A_3) together with a leftover set $R = M \setminus (A_1 \cup A_2 \cup A_3)$ of at most 2 items. If $R = \emptyset$ we are already done, with $p = 0$. What remains is to show that the leftover 1 or 2 items can always be placed — possibly by reassigning which agent gets which of the three bundles, and possibly with a subsidy of 1 for one or two agents — so that the full allocation is envy-free.

17.3 Step 2: free items can always be inserted

Lemma 272 (Free insertion)**.** *Let (A_1, A_2, A_3) be an EF partial allocation and let S be a set of unallocated items with $c_i(A_i \cup S) = c_i(A_i)$ for some agent i (item(s) S cost i nothing on top of what i already holds). Giving all of S to i, and changing nothing else, keeps the allocation EF, with $p = 0$.*

Proof Agent i's own cost is unchanged, so i's non-envy of every other agent j is unchanged: $c_i(A_i \cup S) = c_i(A_i) \leq c_i(A_j)$. For any other agent j: before the change, $c_j(A_j) \leq c_j(A_i)$ (that's j not envying i, which held already); after the change j must compare against $A_i \cup S$ instead of A_i, but $c_j(A_i \cup S) \geq c_j(A_i)$ by monotonicity, so $c_j(A_j) \leq c_j(A_i) \leq c_j(A_i \cup S)$ still holds. No agent's envy status changes for the worse. □

This lemma costs nothing to apply and needs no case analysis, so the proof always uses it first: whenever some leftover item is free to some agent's current bundle, hand it over and shrink the leftover set by one. Repeating this can only ever reduce $|R|$. The only way this process does not immediately finish the job is if it gets stuck: every agent has a strictly positive marginal cost for every leftover item, on their own current bundle. Call this the *stuck* case. It is the only case left to handle, and it splits into $|R| = 1$ and $|R| = 2$.

17.4 A motivating example: subsidy alone is not enough

Before handling the stuck case in general, one example shows why a natural simpler idea — keep everyone's bundle as is, place the leftover item, and patch any envy with a subsidy — can fail outright, and what actually fixes it.

Example 273. Let $M = \{a, b, c, e\}$ and $A_1 = \{a\}$, $A_2 = \{b\}$, $A_3 = \{c\}$ (so $R = \{e\}$). Define, for each agent $i \in \{1, 2, 3\}$ with trigger pair $T_1 = \{a, e\}$, $T_2 = \{b, e\}$, $T_3 = \{c, e\}$,

$$c_i(S) = \begin{cases} 1 & \text{if } T_i \subseteq S \\ 0 & \text{otherwise.} \end{cases}$$

Each c_i is dichotomous (it only ever jumps by 1, exactly when the last missing element of T_i is added). The starting partial allocation is EF: every agent's cost for every one of the three bundles is 0, since none contains any full trigger pair. Every agent has marginal cost 1 for e on their own bundle ($c_i(A_i \cup e) = 1 \neq 0$), so Theorem 272 does not apply: this is a stuck instance.

Try giving e to agent 1, keeping bundles otherwise fixed: $A_1' = \{a, e\}$. Now $c_1(A_1') = 1 > 0 = c_1(A_2)$, so agent 1 envies agent 2 unless $p_1 - p_2 \geq 1$, forcing $p_1 = 1, p_2 = 0$. But agent 2 sees $c_2(A'_1) = 0$ (agent 2's trigger is $\{b, e\}$, and $b \notin A'_1$) and $c_2(A_2) = 0$: with $p_1 = 1$, agent 2's comparison becomes $0 - 0 \le 0 - 1$, which is false. Agent 2 now envies agent 1. Trying the other two agents as the recipient of e fails the same way by symmetry, and no subsidy vector in $\{0, 1\}^3$ rescues any of the three choices.

What does work is reassigning bundles, not just placing e: give agent 3 the bundle $\{a, e\}$, agent 1 the bundle $\{b\}$, and agent 2 the bundle $\{c\}$. Every agent's cost for their new bundle is 0 (agent 3's trigger is $\{c, e\}$, and $\{a, e\}$ doesn't contain c; similarly for the other two), so $p = 0$ works.

The move that rescued Theorem 273 was not moving an item between bundles — it was reassigning which *agent* receives which *bundle*. This is the extra freedom the rest of the proof uses: after deciding where the leftover item(s) go, we are also free to permute which of the three agents receives which of the three resulting bundles, before falling back on subsidy.

17.5 Step 3: the two classical facts that do the work

Nothing below is a search. The whole argument rests on two standard facts about envy-freeability, both already stated in Section 2: Theorem 10 (an allocation admits *some* non-negative subsidy exactly when its bundles are assigned to agents at minimum total cost) and Theorem 11 (when it does, the pointwise-smallest such subsidy is $p_i = \ell_{\mathcal{A}}(i)$, the greatest weight of a simple path starting at i in the envy graph, whose arcs carry $w(a, b) = c_a(A_a) - c_a(A_b)$).

Together they convert the question "does a subsidy in $\{0, 1\}^n$ exist?" into two concrete obligations: *assign the bundles by a minimum-cost matching*, and then *show every simple path in the resulting envy graph has weight at most* 1. The first is a polynomial-time algorithm (Hungarian method) defined for every n. The second is what the next lemma delivers.

Definition 274 (Excess)**.** *Fix bundles $Y_1, \ldots, Y_n$ and an assignment π (agent a receives $Y_{\pi(a)}$). Write $\beta_a := \min_j c_a(Y_j)$ for what agent a would pay for their favourite bundle, and*

$$\varepsilon_a \ := \ c_a(Y_{\pi(a)}) - \beta_a \ \ge \ 0$$

for how much worse than that a actually does. Call $\sum_a \varepsilon_a$ the total excess *of π.*

Lemma 275 (Paths are bounded by total excess)**.** *Every simple path in the envy graph of (Y, π) has weight at most $\sum_a \varepsilon_a$. Hence $\ell(i) \le \sum_a \varepsilon_a$ for every agent i.*

Proof Each arc out of a has weight $w(a, b) = c_a(Y_{\pi(a)}) - c_a(Y_{\pi(b)}) \le c_a(Y_{\pi(a)}) - \beta_a = \varepsilon_a$, since β_a is a minimum over *all* bundles. A simple path leaves each agent at most once, so its weight is at most $\sum_a \varepsilon_a$. □

Lemma 276 (Minimum-cost assignment minimises total excess)**.** *For any assignment σ, $\sum_a c_a(Y_{\sigma(a)}) = \sum_a \beta_a + \sum_a \varepsilon_a(\sigma)$. Since $\sum_a \beta_a$ does not depend on σ, an assignment has minimum total cost if and only if it has minimum total excess.*

Proof Immediate from Theorem 274 by summing. □

These two lemmas reduce everything to a single quantity. To get $p \in \{0,1\}^n$ it suffices to exhibit *one* assignment whose total excess is at most 1: the minimum-cost assignment then has total excess at most 1 too (Theorem 276), is envy-freeable (Theorem 10), and its minimal subsidy is bounded by 1 (Theorem 275 with Theorem 11). Subsidies are non-negative integers here, so "at most 1" means "in $\{0,1\}$".

17.6 Step 4: closing the case of one leftover item

Theorem 277. *If (A_1, A_2, A_3) is an EF partial allocation with exactly one leftover item e, some choice of which bundle receives e, some permutation of the resulting three bundles to the three agents, and some $p \in \{0,1\}^3$ gives a full EF allocation.*

Proof Put e into *any* bundle — say $Y_m := A_m \cup \{e\}$, with $Y_j := A_j$ for $j \neq m$; the choice of m is immaterial. By the discussion closing Section 17.5 it is enough to exhibit one assignment of total excess at most 1, and the identity assignment already is one.

For an agent $a \neq m$: their own original bundle is still their favourite, because

$$c_a(Y_a) \;=\; c_a(A_a) \;\leq\; c_a(A_j) \;\leq\; c_a(Y_j) \qquad \text{for every } j,$$

the first inequality being envy-freeness of the partial allocation and the second monotonicity ($Y_j \supseteq A_j$). Hence $c_a(Y_a) = \beta_a$ and $\varepsilon_a = 0$.

For the agent m: the single added item raises the cost by at most one unit, so for every j,

$$c_m(Y_m) \;=\; c_m(A_m \cup \{e\}) \;\leq\; c_m(A_m) + 1 \;\leq\; c_m(A_j) + 1 \;\leq\; c_m(Y_j) + 1,$$

using the unit-marginal property, then envy-freeness, then monotonicity. Taking the minimum over j gives $c_m(Y_m) \leq \beta_m + 1$, i.e. $\varepsilon_m \leq 1$.

The identity assignment therefore has total excess at most 1, so a minimum-cost assignment π has total excess at most 1 (Theorem 276). Being minimum-cost, (Y, π) is envy-freeable (Theorem 10), and its minimal subsidy is $p_i = \ell(i) \leq \sum_a \varepsilon_a \leq 1$ (Theorem 11 with Theorem 275). Subsidies are non-negative integers, so $p \in \{0,1\}^n$. □

Three things about this proof are worth stating explicitly, because they are what a scaling argument would need. It contains no search. It never uses $n = 3$ — it is written for, and valid at, every number of agents. And it does not need the "stuck" case distinction at all: it subsumes Theorem 272 rather than branching on it. Recording the consequence separately, since it is stronger than what Theorem 284 will need:

Corollary 278 (One leftover item, any number of agents)**.** *Let $(A_1, \ldots, A_n)$ be an envy-free partial allocation for any number n of agents with dichotomous costs, leaving exactly one item unallocated. Then a minimum-cost assignment of the augmented bundles admits a subsidy $p \in \{0,1\}^n$ making it envy-free, computable in polynomial time.*

17.7 Step 5: closing the case of two leftover items

This case needs one thing Theorem 277 did not: the fact that the partial allocation was produced by *a particular algorithm*. Used as a black box, Theorem 271 says only that at most two items are left. But its algorithm cannot stop with two items left at $n = 3$ unless the allocation it stops on has a very rigid shape, and that shape is exactly what closes the case.

Lemma 279 (What the algorithm leaves behind)**.** *Suppose the algorithm of Theorem 271, run on 3 agents, halts with an EF partial allocation (A_1, A_2, A_3) and two unallocated items $R = \{e_1, e_2\}$. Let G be the digraph on the agents with an arc $i \longrightarrow j$ $(i \neq j)$ exactly when $c_i(A_i) = c_i(A_j)$. Then:*

(a) G is strongly connected;

(b) $c_i(e \mid A_i) = 1$ for every agent i and every $e \in R$;

(c) $c_i(e \mid A_j) = 1$ for every arc $i \longrightarrow j$ of G and every $e \in R$.

Proof The algorithm halts only at its final line, having failed each of its three update rules in turn.

(b) is the failure of the first rule, which would otherwise hand a zero-marginal item to an agent who accepts it for free.

(a): the halting rule inspects a tail strongly connected component S of G and halts only when the number of unallocated items is smaller than $|S|$. Two items are unallocated, so $|S| \geq 3$, forcing $|S| = 3$; thus S is the whole agent set and G is strongly connected.

(c) is the failure of the second rule, which would otherwise rotate bundles around a directed cycle whenever some arc $i \longrightarrow j$ *lying on a directed cycle* has $c_i(e \mid A_j) = 0$. In a strongly connected digraph every arc lies on a directed cycle, so by (a) that restriction is vacuous and the condition applies to every arc of G. □

The single consequence we need is that, in this situation, *any* bundle carrying a leftover item is strictly worse for *everybody* than their own original bundle.

Lemma 280 (Augmented bundles are strictly expensive)**.** *Under Theorem 279, for every agent a, every j, and every $e \in R$,*

$$c_a(A_j \cup \{e\}) \;\geq\; c_a(A_a) + 1.$$

Proof Write $c_a(A_j \cup \{e\}) = c_a(A_j) + c_a(e \mid A_j)$. Envy-freeness of the partial allocation gives $c_a(A_j) \geq c_a(A_a)$. If that inequality is strict the claim follows, since costs are integers and marginals are non-negative. If it is an equality, then either $j = a$, and $c_a(e \mid A_a) = 1$ by Theorem 279(b); or $j \neq a$, in which case $a \longrightarrow j$ is an arc of G and $c_a(e \mid A_j) = 1$ by Theorem 279(c). Either way the marginal is 1. □

Theorem 281. *Let (A_1, A_2, A_3) be the EF partial allocation at which the algorithm of Theorem 271 halts with two unallocated items. Send the two items to two* distinct *bundles — any two — and assign the three resulting bundles to the three agents by any minimum-cost matching. The result is envy-free with a subsidy in $\{0, 1\}^3$.*

Proof Write Y_1, Y_2, Y_3 for the bundles after placement: two of them are *augmented* (one leftover item each) and one is *clean*. Put $L := \sum_a c_a(A_a)$, and for an assignment π put $g_a := c_a(Y_{\pi(a)}) - c_a(A_a)$, which is non-negative because $c_a(Y_{\pi(a)}) \geq c_a(A_{\pi(a)}) \geq c_a(A_a)$ by monotonicity and partial envy-freeness.

The minimum cost is exactly $L + 2$. For the lower bound, under any π the two agents holding augmented bundles have $g_a \geq 1$ by Theorem 280, and the third has $g_a \geq 0$, so the total is at least $L + 2$. For the upper bound, let u, v index the augmented bundles and hand each bundle to its original owner: agents u and v pay $c_u(A_u) + 1$ and $c_v(A_v) + 1$ by Theorem 279(b), and the remaining agent w pays $c_w(A_w)$, for a total of exactly $L + 2$.

Hence at a minimum-cost π the excesses are pinned. We have $\sum_a g_a = 2$ with $g \geq 1$ on each of the two augmented-bundle holders, so those two have $g = 1$ exactly and the agent c holding the clean bundle has $g_c = 0$.

Now read off the envy graph. Let a, b be the augmented-bundle holders. Recall $w(x, y) = c_x(Y_{\pi(x)}) - c_x(Y_{\pi(y)})$.

- $w(a, b) = \big(c_a(A_a) + 1\big) - c_a(Y_{\pi(b)}) \leq 0$: the first term because $g_a = 1$, and $c_a(Y_{\pi(b)}) \geq c_a(A_a) + 1$ by Theorem 280, the bundle $Y_{\pi(b)}$ being augmented. Symmetrically $w(b, a) \leq 0$.

- $w(a, c) = \big(c_a(A_a) + 1\big) - c_a(Y_{\pi(c)}) \leq 1$, since $Y_{\pi(c)}$ is clean and so costs a at least $c_a(A_a)$. Symmetrically $w(b, c) \leq 1$.

- $w(c, x) = c_c(A_c) - c_c(Y_{\pi(x)}) \leq 0$ for both x, since $g_c = 0$ gives the first term and every bundle costs c at least $c_c(A_c)$.

So every arc has weight at most 0 except the two arcs pointing *into* c, which have weight at most 1. A simple path enters c at most once, every arc leaving c has weight at most 0, and every other arc available to it has weight at most 0 as well. Hence no simple path has weight above 1.

The assignment is minimum-cost, so (Y, π) is envy-freeable (Theorem 10), and its minimal subsidy is $p_i = \ell(i) \leq 1$ (Theorem 11). Subsidies are non-negative integers, so $p \in \{0, 1\}^3$. □

Remark 282 (Why this does not follow from the black box)**.** Theorem 281 uses Theorem 271's *construction*, not merely its statement, and it has to. Among abstract cost tables satisfying only "EF partial allocation, two leftover items, every agent's marginal on their own bundle equal to 1", the conclusion genuinely fails for some placements: there the minimum-cost total excess reaches 2 and so does the longest path (Section 17.9). Such tables exist as tables; what Theorem 279 shows is that the algorithm never halts on one. Strong connectivity of G is what excludes them, and it is available only because two unallocated items is the *maximum* the algorithm may leave at $n = 3$ — which is precisely what forces its tail component to be the whole agent set.

Remark 283 (How far this reaches beyond $n = 3$)**.** The proof of Theorem 281 never uses $n = 3$ except through one fact: that the number of leftover items is one less than the number of agents, so that placing them in distinct bundles leaves exactly *one* clean bundle. Everything then goes through verbatim for any n: the minimum cost is $L + (n - 1)$, each of the $n - 1$ agents holding an augmented bundle has $g_a = 1$, the single clean-bundle holder has $g = 0$, arcs between augmented holders are non-positive, arcs into the clean holder are at most 1, and arcs out of it are non-positive — so again no simple path exceeds 1. This proves Conjecture 2 for every n on every instance where the algorithm of Theorem 271 leaves exactly $n - 1$ items, just as Theorem 278 does when it leaves exactly one.

What is *not* covered for $n \geq 4$ is the intermediate range $2 \leq |R| \leq n - 2$. Two things fail there at once. The halting rule gives only $|S| \geq |R| + 1$, so G need not be strongly connected on all agents and Theorem 280 loses its justification; and with two or more clean bundles a simple path can alternate — an arc of weight 1 into one clean holder, a non-positive arc out of it, then another arc of weight 1 into a second clean holder — so the path bound of 1 is no longer forced. Closing that range is what a full proof of Conjecture 2 now requires. At $n = 3$ the range is empty, which is why $n = 3$ closes here.

17.8 Putting it together

Theorem 284 (Conjecture 2 holds at $n = 3$)**.** *For every instance of 3 dichotomous cost functions on a finite item set M, there is an allocation $\mathcal{A}$ and a subsidy $p \in \{0, 1\}^3$ such that $(\mathcal{A}, p)$ is envy-free, computable in polynomial time.*

Proof Run the algorithm of Theorem 271 to get an EF partial allocation with at most 2 leftover items. If none are left, $p = 0$ finishes it. If one or two items are left, Theorem 277 or Theorem 281 (respectively) produces a full EF allocation with $p \in \{0, 1\}^3$. Every step is constructive and the extra search in Theorems 277 and 281 is over a bounded number of choices — at most 6 placements, 6 permutations, 8 subsidy vectors — that does not grow with $|M|$, so the whole procedure runs in polynomial time. □

17.9 Verification (not part of any proof)

This section records finite checks. *Nothing above depends on any of them*; every result in Sections 17.6 and 17.7 is proved outright. The checks are reported because they were how the results were found, and because one of them delimits the argument in a way worth knowing.

The enumeration is complete rather than sampled, by the following observation, which is what makes a bounded search decisive at all.

Lemma 285 (Saturation)**.** *Let D be a 3×3 table, D_{ij} the cost to agent i of candidate bundle j. Whether some permutation of bundles to agents and some $p \in \{0, 1\}^3$ make the allocation envy-free depends on D only through, for each row i and pair of columns j, k, whether $D_{ij} - D_{ik}$ is ≤ -1, exactly 0, exactly 1, or ≥ 2. Hence if a base matrix C has zero diagonal and off-diagonal entries in $\{0, 1, 2\}$, enumerating all such C covers every case an unbounded cost table could produce.*

Proof Agent i holding column j does not envy column k exactly when $D_{ij} - p_j \leq D_{ik} - p_k$, i.e. $D_{ij} - D_{ik} \leq p_j - p_k$, and $p_j - p_k$ ranges only over $\{-1, 0, 1\}$; the four cases follow. Subtracting C_{ii} from row i changes no within-row difference, so the diagonal may be taken to be zero, and entries in $\{0, 1, 2\}$ already realise differences across $\{-2, \ldots, 2\}$, hitting all four cases. □

With the two leftover items sent to distinct bundles, the doubly-stuck configuration is described by 6 base entries $C_{ij} \in \{0, 1, 2\}$ and 12 marginal bits $\mu_{i,e,j} = c_i(e \mid A_j)$, $i \neq j$, $e \in \{1, 2\}$, with $\mu_{i,e,i} = 1$ forced. Checking all $3^6 \times 2^{12} = 2{,}985{,}984$ combinations against all 6 ordered choices of receiving bundles, all 6 permutations and all 8 subsidy vectors finds a valid completion every time, with no exceptions (`update_12/prove_r2.py`, about 20 seconds). Separately, `update_12/verify_real.py` draws genuine random dichotomous cost triples rather than bounded abstract tables, isolates real doubly-stuck EF partial allocations (over 9,000 with two leftover items and over 30,000 with one, in 300 trials), and confirms the same completions there.

Two negative findings from the same enumeration are more informative than the positive one, and they are what Theorem 282 reports: over abstract cost tables — unconstrained by any algorithm — the minimum-cost total excess genuinely reaches 2 (`update_49/eps_only_r2.py`), and so does the longest path (`update_49/test_excess_r2.py`). This is why Theorem 281 cannot be proved from the statement of Theorem 271 alone and must appeal to its algorithm: the obstruction is real for tables in general and is removed only by Theorem 279. A further enumeration restricted to tables satisfying Theorem 279's conclusions (`update_49/check_proof_r2.py`) finds no failures, agreeing with the proof.

What this covers, and what it does not. Theorem 284 makes no assumption on the cost functions beyond dichotomy: not additive, not submodular, not related across agents in any way, and no bound on the number of items. Neither of the two theorems it rests on is inherently a 3-agent argument — Theorem 278 holds at every n when one item is left over, and Theorem 283 shows Theorem 281 holds at every n when $n-1$ are. What is missing for $n \geq 4$ is the intermediate range $2 \leq |R| \leq n-2$, and Theorem 283 gives the two independent reasons the present argument does not reach it. That range is empty at $n = 3$, which is exactly why $n = 3$ closes here and general n does not.

18 Approach 13: Unit Subsidies via Partial Allocations

Throughout this section $N = \{1, \ldots, n\}$ is a set of agents and M a finite set of indivisible chores. Each agent i has a dichotomous cost function $c_i : 2^M \to \mathbb{Z}_{\geq 0}$ in the sense of Definition 7: $c_i(\emptyset) = 0$ and $c_i(S \cup \{e\}) - c_i(S) \in \{0, 1\}$ for all $S \subseteq M$ and $e \in M \setminus S$. We write $c_i(e \mid S) := c_i(S \cup \{e\}) - c_i(S)$. Every such c_i is integer-valued and non-decreasing.

A *partial allocation* is a tuple $\mathcal{A} = (A_1, \ldots, A_n)$ of pairwise disjoint subsets of M; its *residue* is $R(\mathcal{A}) := M \setminus \bigcup_i A_i$, and $\mathcal{A}$ is an allocation when $R(\mathcal{A}) = \emptyset$. A partial allocation is *envy-free* if $c_i(A_i) \leq c_i(A_j)$ for all $i, j \in N$. For $p \in \mathbb{Z}^n_{\geq 0}$ the pair $(\mathcal{A}, p)$ is an *envy-free solution* if $c_i(A_i) - p_i \leq c_i(A_j) - p_j$ for all i, j.

We use two facts from [HS19], recorded as Theorem 10 and Theorem 11 in Section 2. In the cost formulation they read as follows. Given an allocation $\mathcal{A}$, its *envy graph* is the complete digraph on N with arc weights $w(i, j) := c_i(A_i) - c_i(A_j)$. First, $\mathcal{A}$ admits an envy-free solution for some $p \in \mathbb{Z}^n_{\geq 0}$ if and only if $\mathcal{A}$ minimises $\sum_i c_i(A_{\sigma(i)})$ over all permutations σ of its own bundles among the agents; equivalently, if and only if its envy graph has no positive-weight directed cycle. Second, when this holds, the pointwise-minimal such subsidy is $p_i = \ell_{\mathcal{A}}(i)$, the maximum weight of a simple directed path in the envy graph starting at i (the empty path being allowed, so $\ell_{\mathcal{A}}(i) \geq 0$).

Finally we use the following, which is Theorem 5.1 of [TWYZ23].

Theorem 286 ([TWYZ23])**.** *For every n and all dichotomous cost functions $c_1, \ldots, c_n$ there is a polynomial-time algorithm returning an envy-free partial allocation $\mathcal{A}$ with $|R(\mathcal{A})| \leq n - 1$.*

18.1 A potential for the assignment of bundles

Fix pairwise disjoint bundles $Y_1, \dots, Y_n$ with $\bigcup_j Y_j = M$, and let π be a permutation of N, agent a receiving $Y_{\pi(a)}$. Put

$$\beta_a \ := \ \min_{j\in N} c_a(Y_j), \qquad \varepsilon_a(\pi) \ := \ c_a(Y_{\pi(a)}) - \beta_a \ \geq \ 0,$$

and call $\Xi(\pi) := \sum_{a\in N} \varepsilon_a(\pi)$ the *total excess* of π.

Lemma 287. *π minimises $\sum_a c_a(Y_{\pi(a)})$ if and only if it minimises $\Xi(\pi)$.*

Proof Summing the definition of $\varepsilon_a(\pi)$ over a gives $\sum_a c_a(Y_{\pi(a)}) = \sum_a \beta_a + \Xi(\pi)$. The quantity $\sum_a \beta_a$ does not depend on π, so the two minimisation problems have the same minimisers. □

Lemma 288. *Let $\mathcal{A}$ be the allocation assigning $Y_{\pi(a)}$ to a. Every simple directed path in the envy graph of $\mathcal{A}$ has weight at most $\Xi(\pi)$. In particular $\ell_{\mathcal{A}}(a) \leq \Xi(\pi)$ for every a.*

Proof For any a, b,

$$w(a,b) = c_a(Y_{\pi(a)}) - c_a(Y_{\pi(b)}) \leq c_a(Y_{\pi(a)}) - \beta_a = \varepsilon_a(\pi),$$

because $\beta_a \leq c_a(Y_{\pi(b)})$ by definition of β_a. A simple path has at most one arc leaving each vertex, so its weight is at most $\sum_a \varepsilon_a(\pi) = \Xi(\pi)$. The empty path has weight $0 \leq \Xi(\pi)$. □

Lemma 289. *Suppose some permutation σ satisfies $\Xi(\sigma) \leq 1$, and let π minimise $\sum_a c_a(Y_{\pi(a)})$. Then the allocation assigning $Y_{\pi(a)}$ to a admits an envy-free solution with subsidy in $\{0,1\}^n$, and π is computable in polynomial time given a value oracle.*

Proof By Lemma 287, π minimises Ξ, so $\Xi(\pi) \leq \Xi(\sigma) \leq 1$. Being cost-minimising among reassignments of its own bundles, the allocation admits an envy-free solution by Theorem 10, and by Theorem 11 its minimal subsidy is $p_a = \ell(a)$, which lies in $[0, \Xi(\pi)] \subseteq [0,1]$ by Lemma 288. Each $\ell(a)$ is a difference of integers, hence $p \in \{0,1\}^n$. Computing π is an assignment problem on n agents and n bundles, solvable in time polynomial in n after n^2 oracle calls. □

18.2 Residues of size one

Theorem 290. *Let $\mathcal{A} = (A_1, \dots, A_n)$ be an envy-free partial allocation with $R(\mathcal{A}) = \{e\}$. For any $m \in N$, set $Y_m := A_m \cup \{e\}$ and $Y_j := A_j$ for $j \neq m$, and let π minimise $\sum_a c_a(Y_{\pi(a)})$. Then the allocation assigning $Y_{\pi(a)}$ to a admits an envy-free solution with subsidy in $\{0,1\}^n$.*

Proof By Lemma 289 it suffices to exhibit one permutation of total excess at most 1; we show the identity is such a permutation.

Let $a \neq m$. For every j,

$$c_a(Y_a) = c_a(A_a) \leq c_a(A_j) \leq c_a(Y_j),$$

the first inequality by envy-freeness of $\mathcal{A}$ and the second because $A_j \subseteq Y_j$ and c_a is non-decreasing. Hence $\beta_a = c_a(Y_a)$ and $\varepsilon_a(\mathrm{id}) = 0$.

For $a = m$ and every j,

$$c_m(Y_m) = c_m(A_m) + c_m(e \mid A_m) \le c_m(A_m) + 1 \le c_m(A_j) + 1 \le c_m(Y_j) + 1,$$

using in turn that marginals lie in $\{0, 1\}$, envy-freeness of $\mathcal{A}$, and monotonicity. Taking the minimum over j gives $c_m(Y_m) \le \beta_m + 1$, i.e. $\varepsilon_m(\mathrm{id}) \le 1$.

Therefore $\Xi(\mathrm{id}) \le 1$. □

18.3 Residues of maximum size

Theorem 290 uses only the envy-freeness of $\mathcal{A}$. The remaining case uses properties of the partial allocation returned by the algorithm of Theorem 286, which we now record. That algorithm maintains an envy-free partial allocation and the digraph

$$G \;=\; (N, E), \qquad (i, j) \in E \iff i \ne j \text{ and } c_i(A_i) = c_i(A_j), \tag{1}$$

and it iterates three rules: (R1) if some $e \in R(\mathcal{A})$ and some i satisfy $c_i(e \mid A_i) = 0$, give e to i; (R2) if some $e \in R(\mathcal{A})$ and some arc $(i, j) \in E$ lying on a directed cycle of G satisfy $c_i(e \mid A_j) = 0$, rotate the bundles along that cycle and give e to i; (R3) otherwise select a tail strongly connected component S of G — one with no arc of E leaving S — and, if $|R(\mathcal{A})| \ge |S|$, give one residual item to each agent of S. The three rules are attempted in this order and the algorithm loops while $R(\mathcal{A}) \ne \emptyset$; the only exit with $R(\mathcal{A}) \ne \emptyset$ is the failure of the test in (R3). Hence if the algorithm halts with $R(\mathcal{A}) \ne \emptyset$, then (R1) and (R2) do not apply at the halting state and $|R(\mathcal{A})| < |S|$ for the component S selected by (R3).

Lemma 291. *Let $n \ge 2$, let $\mathcal{A}$ be the partial allocation at which the algorithm of Theorem 286 halts, and suppose $|R(\mathcal{A})| = n - 1$. Then, with G as in (1):*

(a) G is strongly connected;

(b) $c_i(e \mid A_i) = 1$ for every $i \in N$ and every $e \in R(\mathcal{A})$;

(c) $c_i(e \mid A_j) = 1$ for every arc $(i, j) \in E$ and every $e \in R(\mathcal{A})$.

Proof Since $|R(\mathcal{A})| = n - 1 \ge 1$, the algorithm halted with $R(\mathcal{A}) \ne \emptyset$, so the description above applies.

(a) At the halting state the component S selected by (R3) satisfies $|R(\mathcal{A})| < |S|$, so $|S| > n - 1$, forcing $|S| = n$. Thus $S = N$, and S is a strongly connected subgraph of G by construction, so G is strongly connected.

(b) Rule (R1) does not apply, so $c_i(e \mid A_i) \ne 0$ for all i and all $e \in R(\mathcal{A})$; marginals lie in $\{0, 1\}$.

(c) Let $(i, j) \in E$. By (a) there is a directed path in G from j to i; choosing one with no repeated vertex and appending the arc (i, j) yields a directed cycle of G through (i, j). Since (R2) does not apply, $c_i(e \mid A_j) \ne 0$ for every $e \in R(\mathcal{A})$. □

Lemma 292. *In the situation of Lemma 291, for every $a, j \in N$ and every $e \in R(\mathcal{A})$,*

$$c_a(A_j \cup \{e\}) \;\ge\; c_a(A_a) + 1.$$

Proof Envy-freeness of $\mathcal{A}$ gives $c_a(A_j) \geq c_a(A_a)$, and both sides are integers. If $c_a(A_j) \geq c_a(A_a) + 1$ the claim follows from $c_a(A_j \cup \{e\}) \geq c_a(A_j)$. Otherwise $c_a(A_j) = c_a(A_a)$. If $j = a$ then $c_a(e \mid A_a) = 1$ by Lemma 291(b); if $j \neq a$ then $(a, j) \in E$ by (1) and $c_a(e \mid A_j) = 1$ by Lemma 291(c). In both cases $c_a(A_j \cup \{e\}) = c_a(A_a) + 1$. □

Theorem 293. *Let $n \geq 2$, let $\mathcal{A} = (A_1, \dots, A_n)$ be the partial allocation at which the algorithm of Theorem 286 halts, and suppose $R(\mathcal{A}) = \{e_1, \dots, e_{n-1}\}$ has exactly $n - 1$ elements. Choose distinct indices $m_1, \dots, m_{n-1}$ and set*

$$Y_{m_t} := A_{m_t} \cup \{e_t\} \quad (1 \leq t \leq n-1), \qquad Y_w := A_w \text{ for the remaining index } w.$$

Let π minimise $\sum_a c_a(Y_{\pi(a)})$. Then the allocation assigning $Y_{\pi(a)}$ to a admits an envy-free solution with subsidy in $\{0, 1\}^n$.

Proof Call $Y_{m_1}, \dots, Y_{m_{n-1}}$ *augmented* and Y_w *clean*. Put $L := \sum_a c_a(A_a)$ and, for a permutation σ,

$$g_a(\sigma) \;:=\; c_a(Y_{\sigma(a)}) - c_a(A_a).$$

Since $A_{\sigma(a)} \subseteq Y_{\sigma(a)}$ and $\mathcal{A}$ is envy-free, $c_a(Y_{\sigma(a)}) \geq c_a(A_{\sigma(a)}) \geq c_a(A_a)$, so $g_a(\sigma) \geq 0$, and $\sum_a c_a(Y_{\sigma(a)}) = L + \sum_a g_a(\sigma)$.

The minimum of $\sum_a c_a(Y_{\sigma(a)})$ equals $L + n - 1$. For the lower bound, fix σ. Exactly $n - 1$ agents receive augmented bundles, and each such agent a has $g_a(\sigma) \geq 1$ by Lemma 292; the remaining agent has $g_a(\sigma) \geq 0$. Hence $\sum_a g_a(\sigma) \geq n - 1$. For the upper bound, take $\sigma = \mathrm{id}$: for $t = 1, \dots, n-1$ agent m_t pays $c_{m_t}(A_{m_t}) + c_{m_t}(e_t \mid A_{m_t}) = c_{m_t}(A_{m_t}) + 1$ by Lemma 291(b), and agent w pays $c_w(A_w)$; the total is $L + n - 1$.

Consequently $g_a(\pi) = 1$ for each of the $n-1$ agents holding an augmented bundle, and $g_c(\pi) = 0$ for the agent $c := \pi^{-1}(w)$ holding the clean bundle. Indeed $\sum_a g_a(\pi) = n - 1$ by the previous paragraph, while the $n - 1$ agents holding augmented bundles contribute at least 1 each and $g_c(\pi) \geq 0$; so all these inequalities are equalities.

Arc weights. Write $w(x, y) = c_x(Y_{\pi(x)}) - c_x(Y_{\pi(y)})$ and let a, b denote agents holding augmented bundles.

- $w(a, b) \leq 0$. Indeed $c_a(Y_{\pi(a)}) = c_a(A_a) + 1$ since $g_a(\pi) = 1$, while $Y_{\pi(b)}$ is augmented, say $Y_{\pi(b)} = A_{\pi(b)} \cup \{e\}$, so $c_a(Y_{\pi(b)}) \geq c_a(A_a) + 1$ by Lemma 292.
- $w(a, c) \leq 1$. Indeed $c_a(Y_{\pi(c)}) = c_a(A_w) \geq c_a(A_a)$ by envy-freeness of $\mathcal{A}$, and $c_a(Y_{\pi(a)}) = c_a(A_a) + 1$.
- $w(c, y) \leq 0$ for every y. Indeed $c_c(Y_{\pi(c)}) = c_c(A_c)$ since $g_c(\pi) = 0$, while $c_c(Y_{\pi(y)}) \geq c_c(A_{\pi(y)}) \geq c_c(A_c)$ by monotonicity and envy-freeness of $\mathcal{A}$.

Thus every arc of the envy graph has weight at most 0, except arcs whose head is c, which have weight at most 1.

Path bound. Let P be a simple directed path. At most one arc of P has head c, since P visits c at most once; every other arc of P has weight at most 0. Hence P has weight at most 1. In particular the envy graph has no positive-weight cycle, so by Theorem 10 the allocation admits an envy-free solution, and by Theorem 11 its minimal subsidy satisfies $0 \leq p_a = \ell(a) \leq 1$ for every a. As $\ell(a)$ is an integer, $p \in \{0, 1\}^n$. □

18.4 Three agents

Corollary 294. *Conjecture 2 holds for $n \le 3$: for all dichotomous cost functions $c_1, \dots, c_n$ with $n \le 3$ there exist an allocation $\mathcal{A}$ and $p \in \{0,1\}^n$ such that $(\mathcal{A}, p)$ is an envy-free solution, and both are computable in polynomial time.*

Proof Run the algorithm of Theorem 286 to obtain an envy-free partial allocation $\mathcal{A}$ with $|R(\mathcal{A})| \le n - 1 \le 2$.

If $|R(\mathcal{A})| = 0$ then $\mathcal{A}$ is an allocation and $(\mathcal{A}, \mathbf{0})$ is an envy-free solution.

If $|R(\mathcal{A})| = 1$, apply Theorem 290.

If $|R(\mathcal{A})| = 2$ then $n = 3$ and $|R(\mathcal{A})| = n - 1$, so Theorem 293 applies.

In each case the construction is a minimum-cost assignment on at most three bundles preceded by a polynomial-time algorithm, so the total running time is polynomial. □

18.5 The range not covered

Theorem 290 and Theorem 293 hold for every n, at residue sizes 1 and $n - 1$ respectively; these coincide with the full range $\{0, 1, \dots, n-1\}$ permitted by Theorem 286 exactly when $n \le 3$. For $n \ge 4$ the residue sizes k with $2 \le k \le n - 2$ are not covered, and neither proof extends to them, for the following two reasons.

First, Lemma 291(a) fails. At a halting state with $|R(\mathcal{A})| = k$ the selected component satisfies only $|S| \ge k + 1$, so S may be a proper subset of N; arcs of E with tail outside S need not lie on any directed cycle, since S has no outgoing arcs. Consequently Lemma 292 is unavailable for agents outside S, and with it the lower bound $\sum_a g_a(\sigma) \ge k$ in the proof of Theorem 293.

Second, even granting that lower bound, the path bound fails. With k augmented bundles there are $n - k \ge 2$ clean bundles, so the envy graph may contain two distinct vertices $c_1 \ne c_2$ holding clean bundles. The argument of Theorem 293 then bounds by 1 every arc with head c_1 or c_2 and by 0 every other arc, which permits a simple path $a_1 \to c_1 \to a_2 \to c_2$ of weight up to 2: the arcs into c_1 and into c_2 each contribute at most 1, and the arc $c_1 \to a_2$ contributes at most 0, but nothing in the argument forces the two contributions of 1 to be non-simultaneous.

The two obstructions are independent: removing the first restores the bound $\sum_a g_a(\sigma) \ge k$ but leaves the alternating path of the second available, and removing the second leaves the lower bound unjustified. For $2 \le k \le n - 2$ it would suffice, for a given halting state, to exhibit either a placement of $R(\mathcal{A})$ into bundles and a permutation of total excess at most 1, in which case Lemma 289 applies directly; or a placement together with a minimum-cost permutation for which every simple path in the envy graph has weight at most 1, in which case Theorem 10 and Theorem 11 apply as at the end of the proof of Theorem 293. Establishing either for every halting state with $2 \le |R(\mathcal{A})| \le n - 2$ would, with Theorem 290 and Theorem 293, prove Conjecture 2 for all n.

19 Computational Evidence

All searches operate directly on the graph-theoretic form of Conjecture 2: enumerate allocations, build the envy graph, detect positive-weight cycles and compute $\ell_{\mathcal{A}}$ by a

Bellman–Ford iteration on maximum-weight walks, and record $\min_{\mathcal{A}} \max_i \ell_{\mathcal{A}}(i)$.

Experiment	Script	Result
Exhaustive: $n = m = 3$, all 9,880 instances up to agent symmetry	`search.py`	worst-case minimum per-agent subsidy = 1
Randomised: $n \in \{3,4,5\}$, $m \in \{4,5,6\}$, $\approx 3.5 \times 10^4$ instances	`fast.py`	no counterexample
Binary additive construction of Theorem 19, 2×10^5 instances, $n \leq 5$, $m \leq 7$	`binadd.py`	maximum per-agent subsidy ever observed = 1
The obstruction of Theorem 36	`deadend.py`	all three insertions require subsidy 2; full instance solvable with 0
Does *some* utilitarian-optimal allocation attain ≤ 1?	`msw.py`	yes, in all but one of $\approx 2.6 \times 10^4$ instances
Does *every* utilitarian-optimal allocation attain ≤ 1?	`msw.py`	no — fails in $\approx 2.5\%$ of instances at $n = 3, m = 4$, rising to $\approx 5\%$ at $m = 5$
The single exception	`mswcex.py`	refutes Conjecture 18 (Proposition 38)
Peel schedule and trap-state refusal on the witness of Theorem 36	`peel.py`	six legal peels reach the zero-subsidy allocation; the trap state has $\ell(1) = 2$
Full peel state space of that witness, $7^3 = 343$ states, both move types	`peel3.py`	175 legal, 166 good, **9 dead ends**, root good, all six reachable terminals are perfect matchings

and, for the approaches attempted after the replica transform,

Experiment	Script	Result
No dichotomous reweighting separates: all 990^3 triples on the instance of Theorem 93, plus the separable P-family on three instances	`dupsep.py`	**0** separating triples (Theorem 94)
Type-ordered peel schedules, witness plus 890 random instances, $n \in \{3,4\}$, $m \le 4$	`typeorder.py`, `stress.py`	every instance admitting a legal schedule admits a type-ordered one; arg max recipient rule fails (Proposition 91)
Double-refinement selection rule, ≈ 1050 random instances	`rules.py`	no failure — but see the next row
The same rule on the instance of Proposition 38	`checkD.py`	**fails**: selected allocation needs 2, another needs 0 (Remark 41)
Iterated minimum-marginal perfect matching, non-additive costs	`rules.py`	fails on 13/250 at $n = 3, m = 4$ and 10/120 at $n = 3, m = 5$
Uniformly balanced families: exhaustive $n = m = 3$, then ≈ 1900 random instances up to $n = 5$	`routeA.py`	no failure, and the implication chain verified on every instance
The same, hunted adversarially over structured families	`routeA_adv.py`	10 **failures** at $n = 3, m = 4$, plus the certified-hard set-splitting instance (Proposition 100)
Arc weights of the good allocations on that counterexample	`routeA_cex.py`	all 19 have an arc ≤ -2; minimum -3 (Proposition 101)
Conjecture 2 itself, hunted adversarially over non-additive instances: exhaustive structured triples at $m = 3, 4$ plus $\approx 8.4 \times 10^3$ endpoint-constant instances to $n = 5$	`twotier.py`	**no counterexample** (Remark 295)
How often a subsidy is needed at all	`twotier.py`	exactly envy-free in 98.6% of 11,884 instances (Remark 296)
Structure of the paid set on the 164 instances that need one	`hardcases.py`	$\min \lvert S\rvert \le 3$, always inside $\arg\max_i c_i(A_i)$ (Remark 109)
The $n = 3$ characterisation, against the true longest-path computation	`n3check.py`	0 mismatches over 5.8×10^5 (instance, allocation) pairs (Proposition 104)

and, for the construct-and-repair algorithm of Section 11,

Experiment	Script	Result
Item-by-item insertion on the transformed goods instance, greedily steered	`guidedR3.py`, `guidedR3_test.py`	1,635/9,880 (fixed order), 440/9,880 (best of 6)
Full backtracking over every legal insertion execution on the smallest failure	`guidedR3_full.py`	spread 2 is the best reachable; spread 0 is unreachable
Independent, algorithm-free confirmation that spread 0 is achievable there	`verify_reach_gap.py`	confirmed — a genuine reachability obstruction
IMWPM warm start alone, no repair	`imwpm_raw.py`	exhaustive $n = m = 3$ and all hard instances pass; $\approx 18\%$ of larger random instances fail
Swap-based repair from a round-robin start alone	`targetGbal_local.py`	9,880/9,880 exhaustive; 357,746/357,760 structured triples at $m = 5$
The 14 residual local optima, classified	`classify_failures.py`	all 14 are local-search artefacts — exhaustive balanced search finds spread 0 on every one
The same 14, retried with shuffled restarts	`restart_check.py`	14/14 recovered, at most 4 restarts each
The combined algorithm: IMWPM warm start, swap repair, restart on failure	`final_algorithm.py`	0 **failures across** 389,215 **instances** — exhaustive $n = m = 3$, exhaustive structured triples to $m = 5$, every named hard instance, and randomised sweeps to $n = 8, m = 10$
The $n = 2$ completeness theorem and its $O(m\tau)$ constructive walk	`n2proof_check.py`	0 failures over 521,310 instances — exhaustive $m \leq 4$ (490,545 pairs at $m = 4$), random $m \in \{5, 6, 7\}$; walk never exceeded 2 evaluations (Theorem 118, Corollary 121)

The exhaustive row is the only unconditional evidence for Conjecture 2. The dichotomous functions on m items number $2, 6, 38, 990$ for $m = 1, 2, 3, 4$. Agents are interchangeable, so an instance is a multiset of three of them, and at $m = 3$ there are $\binom{38+3-1}{3} = \binom{40}{3} = 9{,}880$ of these — feasible to enumerate, and already out of reach at $m = 4$.

The last three rows are the ones that changed the shape of the problem. The utilitarian-optimal set almost always contains a witness, but not always, and one exception is enough: Proposition 38 is built from the single instance in which it did not.

19.1 Why the randomised evidence is weak

The count of $\approx 3.5 \times 10^4$ instances without a counterexample overstates the support for Conjecture 2, and the reason is structural rather than a matter of sample size.

The generator builds a dichotomous function by visiting subsets in order of increasing size and, at each subset S, computing the interval $[\ell, h]$ of values consistent with what has already been fixed — namely $\ell = \max_{g \in S} c(S \setminus \{g\})$ and $h = \min_{g \in S} c(S \setminus \{g\}) + 1$ — then choosing an endpoint by an unbiased coin whenever $\ell \neq h$. Independent fair coins put almost all of the mass on functions that wander in the middle of the feasible region. Structured extremal functions, which take the same endpoint at every subset, are exponentially unlikely: on m items there are $2^m - 1$ nonempty subsets and a function that requires a prescribed endpoint at each of them is sampled with probability $2^{-(2^m-1)}$, which is $1/128$ at $m = 3$ and about 3×10^{-5} at $m = 4$, per agent.

This matters because the two instances that carry the results of this paper are of exactly that kind. The obstruction of Theorem 36 uses $c_1(S) = \max(0, |S| - 1)$ alongside $c_2 = c_3 = |S|$, all three of which are endpoint-constant; drawing that triple at $m = 3$ has probability $2^{-21} \approx 5 \times 10^{-7}$, so a run of 3.5×10^4 samples would be expected to produce it about once in every sixty runs. Both instances were found by hand or by a targeted search, not by the sampler. The sampler has, in effect, never looked in the region where the interesting behaviour lives.

Remark 295 (The experiment worth running)**.** If Conjecture 2 is false, the counterexample is in the corner the current generator cannot reach, and no amount of additional uniform sampling will find it. An adversarial generator — biased towards endpoint-constant functions, towards supermodular costs, towards agents whose free chores overlap heavily, and towards instances in which the utilitarian optimum is unique — is the single highest-value experiment outstanding against the *conjecture*.

It has now been built (`routeA_adv.py`, `twotier.py`) and it works: it refuted the invariant of Section 10 in a single run, on an instance that $\approx 1.2 \times 10^4$ uniformly sampled instances had missed. Pointed at Conjecture 2 itself it finds nothing. Since binary additive instances are settled by [LMS26], a counterexample must be non-additive and the search was restricted accordingly: exhaustive over all structured non-additive triples at $m = 3$ (120 instances) and $m = 4$ (5,984), then $\approx 8.4 \times 10^3$ endpoint-constant instances up to $n = 5$, $m = 6$. **No instance without a good allocation was found.** This is the first evidence for the conjecture produced by a method demonstrated to find counterexamples on this problem, and it is a genuine update: the accumulating failures of Sections 6, 7, 9 and 10 are evidence about the *methods*, not about the statement.

Remark 296 (Where the difficulty is concentrated)**.** The same sweep records how often a subsidy is needed at all. Across 11,884 adversarial non-additive instances an *exactly* envy-free allocation — zero subsidy — exists in 98.6% of them; only 164 require any subsidy, and among those the minimum number of paid agents is 1 or 2 at $n = 3$ and never exceeds 3 at $n = 4$.

Two cautions. The pool is adversarially structured rather than representative, so the percentage describes that pool and is not a claim about the model. And the residual 1.4% is not negligible in the way the number suggests: deciding whether an exactly envy-free allocation exists is NP-complete already for binary additive chores [BSV21], so those instances are hard for a reason and they are precisely the ones any construction must handle.

19.2 The peel-frame search

The highest-value experiment against the *approach* of Section 8 is different, and it is cheap. Reuse the `gen.py/search.py` harness over the full $n = 3$, $m = 3$ dichotomous family — all 9,880 instances up to agent symmetry — and for each instance run the fixed-point computation of `peel3.py`: is the root good, and how many dead ends are there? An instance with a bad root refutes Conjecture 54 and kills the peel approach.

It would *not*, by Remark 55, refute Conjecture 2: a good terminal state can in principle exist while every schedule reaching it passes through an illegal intermediate state. The two searches therefore test different things and both are worth running — the allocation-space search above tests the conjecture, and the peel-frame search tests whether this route to it survives.

Two follow-ups, in order. First, test the hypothesis of Remark 53 across that family: is every dead end a state already committed to an unbalanced terminal? If so, formulate the balance rule precisely and check whether greedy-under-that-rule reaches a terminal without backtracking. Second, try candidate rules in the order (a) balance-first, peel so that a balanced terminal remains reachable; (b) spectator-free peels first by Lemma 49, ties broken by balance; (c) $\arg\max_i \ell(i)$ with a balance tie-break — noting that (c) alone already fails on the witness.

19.3 Reproducing the results

Script	Purpose
`gen.py`	enumerate all dichotomous functions on m items
`search.py`	exhaustive search, arguments `m n`
`fast.py`	randomised counterexample hunt, `m n T seed`
`msw.py`	utilitarian-optimal test, `m n T seed`
`tiebreak.py`	selection rules of Section 7.1
`localsearch.py`	local minima of single-transfer search
`deadend.py`	Theorem 36
`binadd.py`	Theorem 19
`mswcex.py`	Proposition 38
`peel.py`	the schedule and trap-state refusal of Section 8.4
`peel3.py`	the state-space table of Proposition 52
`dupsep.py`	Theorems 93 and 94; `dupsep.py pfamily` for the P-family sweeps
`typeorder.py`	type-ordered schedules on the witness and small families
`stress.py`	randomised type-ordering sweep, seeded
`rules.py`	the refinement rules and the matching lift
`checkD.py`	Remark 41

and, continuing,

Script	Purpose
`routeA.py`	Proposition 99; exhaustive $n = m = 3$ and randomised sweeps
`routeA_adv.py`	adversarial hunt; finds Proposition 100
`routeA_cex.py`	Propositions 100 and 101
`twotier.py`	Remarks 295 and 296
`hardcases.py`	Remark 109
`n3check.py`	Proposition 104 and Corollary 105
`monotone_check.py`	Remark 44: monotonicity alone does not suffice
`targetG.py`	Conjecture 111, exhaustive $n = m = 3$
`targetG_adv.py`	Target G against the Approach 5 adversary
`guidedR3.py`, `guidedR3_test.py`	guided-insertion greedy; Section 11.3
`guidedR3_full.py`	the insertion reachability obstruction
`verify_reach_gap.py`	algorithm-free confirmation of that gap
`targetGbal.py`	Definition 113, exhaustive over balanced partitions
`targetGbal_local.py`	the swap-based repair, round-robin start
`imwpm_raw.py`	the IMWPM warm start alone
`targetGbal_stress.py`	the stress sweep that found the 14 local optima
`classify_failures.py`	classifies the 14 as search artefacts
`restart_check.py`	confirms all 14 recovered by restarts
`final_algorithm.py`	the combined algorithm; the 389,215-instance sweep
`n2proof_check.py`	Theorem 118 and Corollary 121, exhaustive $m \leq 4$ plus random $m \leq 7$
`balanced_msw.py`, `tiebreak_balanced.py`, `ruleD_adv.py`	the min-cost-balanced rule and its tie-breaks
`structural.py`	Theorem 23, Corollary 24, two-type sweep

The instance of Proposition 38 was produced by `msw.py 4 3 3000 11` at trial 2460 and is re-derived independently by `mswcex.py`, which re-checks that all three cost functions are negative dichotomous, that the utilitarian optimum is unique, and that a zero-subsidy allocation exists outside it.

20 Open Threads: Status Register

Every thread this investigation has raised is listed once below, with a status. The vocabulary is fixed:

PROVED a theorem of this report.

REFUTED a counterexample or machine-verified failure exists.

WITHDRAWN asserted here, then falsified by later work here.

OPEN precisely stated, unrefuted, and load-bearing.

UNEXPLORED stated, never tested.

SUPERSEDED absorbed into a later, stronger formulation.

EXCLUDED deliberately out of scope.

The live chain

Conjecture 2 currently reduces as follows; every implication is proved, so the chain is only as strong as its weakest open link.

$$\text{Conj. 2} \Longleftarrow \text{Target G} \Longleftarrow \text{conj:warmstart} \Longleftarrow \text{conj:imwpm-bound} + \text{conj:last-rung}$$

thread	where	status
p^* has a zero coordinate; Conj. 2 is one clause	Prop. 13	PROVED
Conjecture 2 for $n = 2$, all m	Thm. 118	PROVED
Conjecture 2 for $m \le n$	Cor. 24	PROVED
Target G $\Rightarrow$ Conj. 2	Prop. 112	PROVED
$q_i = \max_j[d(i,j) + \lvert A_j \rvert]$	Lem. 133	PROVED
IMWPM = round-wise minimum total marginal	Lem. 135	PROVED
Round dominance, and its corollary	Lem. 137	PROVED
$w_{\mathcal{A}}(i,j) = (\kappa_i - \kappa_j) - E_{ij}$	Lem. 139	PROVED
Peel reachability, full $n = m = 3$ family: no bad root	Obs. 56	VERIFIED
Balance $\Rightarrow$ live (characterises the dead ends)	Obs. 57	VERIFIED
Balance rule $\Rightarrow$ Conj. 54 $\Rightarrow$ Conj. 2	Prop. 61	PROVED
IMWPM warm start has q-spread ≤ 2	Conj. 128	OPEN
Spread 2 repairs in one move	Conj. 129	OPEN
Repair from the warm start needs no restarts	Conj. 125	OPEN
Balance-rule maintenance	Conj. 59	OPEN — best candidate

Closed threads

thread	where	status
Item-by-item insertion (chores, peel, and goods frames)	Thm. 36	REFUTED
Utilitarian-optimal allocations, and every refinement	Cor. 39	REFUTED
Encoding coverage into the numbers	§9	REFUTED
Uniform balance (all arcs ≥ -1)	Prop. 100	REFUTED
Leximax objective; all 12 tie-break chains	Prop. 164, §13	REFUTED
Greedy insertion never stuck (NS), and its balanced form (NSB)	Prop. 160	REFUTED
Descent potentials P_1, $(\max \ell, \max \lvert B \rvert)$, $(\sum \ell, \sum \lvert B \rvert^2)$	§12	REFUTED
All five local rules for the descent witness	§12.6	REFUTED
The u-spread window at $n \geq 3$	§11.9	REFUTED
Dichotomous analogue of [BDN+20, Thm. 1.3]	Prop. 131	REFUTED
IMWPM produces balanced bundles	Prop. 132	REFUTED
Transplanting [LMS26]'s telescoping	Prop. 147	REFUTED
Closing-arc bound as a route	Prop. 144	REFUTED
Parity of the good set	Prop. 163	REFUTED
Any natural distribution concentrating on good allocations	Prop. 164	REFUTED
spread(κ) ≤ 2 and bounded path defect	Rem. 142	WITHDRAWN
Induction on the depth of the cost functions (the chain/layer route)	retired, git `8be6e54`	WITHDRAWN
Descent lemma route (strictly stronger than needed)	Rem. 155	SUPERSEDED
Balance lemma (a weakening of Target G-bal)	Rem. 158	SUPERSEDED
Choose the paid set first / build the two tiers first	Prop. 15	SUPERSEDED

Never tested

thread	where	status
Type-ordered peel search (a further restriction)	Conj. 90	UNEXPLORED
Peel converse: does Conj. 2 imply reachability?	Rem. 55	UNEXPLORED
Agent induction carrying a *path*-based invariant	§10	UNEXPLORED
At most two distinct cost functions among the n agents	—	UNEXPLORED
Exchange *paths* rather than single transfers	—	UNEXPLORED
Balanced minimum-cost allocation minimising envious pairs	—	UNEXPLORED
A non-uniform weighting concentrating on good allocations	§13.5	UNEXPLORED
The m-threshold: $\ell_{\mathcal{A}} = 2$ occurs only at small m	Rem. 142	UNEXPLORED
Hard adversarial search for a counterexample	§19.1	UNEXPLORED

Out of scope

Binary submodular / matroid-rank valuations, and the milestone ladder through them, are EXCLUDED: the dichotomous *goods* corner is already settled in [BKNS22] by matroid machinery, and retreating to that hypothesis would re-prove a known result rather than the open one — see Remark 148 for why submodularity is exactly what this problem lacks. The mixed goods-and-chores model of [KMS+24] is likewise EXCLUDED as a different problem.

What the closed column means

The refutations above are not independent. Grouped by the technique they kill: constructive routes fail because the witness exists but no local rule finds it (§12.6, §13.4); decompositional routes fail because every two-term split of $\ell_{\mathcal{A}}$ has both parts unbounded while the whole stays small (Rem. 143, four instances); transplanted routes fail because additivity is load-bearing in [BDN+20] and submodularity in [LMS26] (§11.12, §11.17); and non-local routes fail for want of a distribution or an identity (§13.5). Every named technique class has now been tried. That is a statement about where a proof will not come from, not evidence that Conjecture 2 is false — it survives every test at every size reached, now to $m = 12$.

21 Conclusion and Future Work

We have not settled Conjecture 2 in general, but Section 11 now carries a construct-and-repair algorithm together with a proof of most of what one would want from it. Four properties are theorems for every n: the algorithm's outputs are *certified*, so it never lies (Theorem 116); it always terminates, in polynomial time $O(n^3m^3\tau + n^4m^3)$ (Theorem 117); its search space is genuinely the unordered balanced partitions (Lemma 115); and for *two agents* it is *complete*

(Theorem 118) — every two-agent dichotomous instance admits a balanced split of q-spread at most 1, found in $O(m\tau)$ time with no restarts. That last result even strengthens the known $n = 2$ chore case, adding a balancedness guarantee the complementation route of Remark 9 does not provide.

For $n \geq 3$ what is missing is completeness alone, and it is backed by **zero observed failures across** 389,215 **test instances**, including every hard instance in this project. So the headline is a proved algorithm whose only unproved property is that it always succeeds — a single, sharply stated combinatorial claim (Conjecture 123), with a proved base case and a named proof technique to generalise.

Getting there required ruling out three strategies a reader would reasonably try first: item-by-item insertion, the strategy that works in [BKNS22] (Theorem 36, and again in the pure goods coordinates of Section 11.3); restriction to utilitarian-optimal allocations, together with *every* refinement of that set (Proposition 38, Corollary 39); and encoding the coverage constraint into the numbers, at either the valuation or the algorithm level (Section 9). Alongside those sit a small set of positive results — binary additive costs (Theorem 19, now subsumed by [LMS26]), identical costs (Theorem 21), two agents — and structural tools that say precisely how the chore problem differs from its goods mirror.

The lesson that ties the whole investigation together. Every insertion-based strategy failed for the same reason, in three different coordinate systems: chores directly (Section 6), the replica/peel frame (Section 8), and even the pure goods frame (Section 11.3). Building an allocation one item at a time fixes decisions before their consequences are visible, and by the time a bad grouping shows itself, insertion has no way to undo it — only permutation between agents, never a genuine re-split of a bundle's contents, is available. The algorithm that finally worked abandoned that template entirely: *construct* a whole candidate partition directly, then *repair* it by moves that can freely undo an earlier bad grouping. That is the one idea from this project worth carrying into any future attempt on a similar problem.

The three negative results are more useful together than separately, because they fail in the same way. Insertion fixes an owner before the consequences are visible; a selection rule fixes a criterion before knowing which allocation it will pick out; a reweighting fixes a penalty before knowing who will hold the duplicate. Each is *a rule fixed ex ante against a quantity determined ex post* (Section 9.1). What survives is procedures that decide late — which is exactly the design of the peel frame, where ownership is settled only by the last move touching a chore.

Both negative results are failures of *timing*, and the constraint they jointly impose is that *a correct algorithm must be free to move already-allocated chores between bundles, and free to give up total cost while doing so*. The replica transform of Section 8 is the response: Theorem 43 turns the instance into a goods instance with an identical envy graph, Corollary 45 reduces the conjecture to the single constraint that no agent be relieved of a chore twice, and reading that dynamically defers every ownership decision to the last move touching a chore, so there is nothing to re-open. This is currently the most promising route. It is not known to work — Proposition 52 shows the state space has genuine dead ends, and Remark 55 notes that the reduction is sufficient but not known to be complete — and what is missing is a peel rule that provably never enters a deadlock.

Every construction here that succeeds does so by exhibiting a per-agent potential ϕ

with $w_{\mathcal{A}}(i, j) \le \phi_i - \phi_j$, which telescopes along paths. Finding such a potential for general negative dichotomous costs, or proving none exists, remains the problem; the balance signal of Remark 53 is the current best guess at its shape.

Directions. Every thread this project has raised, with its status, is registered once in Section 20; the former enumerated list here duplicated it and had gone stale. The live chain is the one recorded at the head of that register: Conjecture 2 follows from Target G, which follows from Conjecture 125, which follows from Conjectures 128 and 129. Those two are the only open links, and both are statements about the output of a single construction rather than about all allocations.

Where this would sit. Conjecture 2 is the fourth corner of the square in Section 1: additive goods [BDN+20], dichotomous goods [BKNS22] and additive chores [LMS26] all give one dollar per agent and $n-1$ in total, and dichotomous chores is what remains. For valuations that are dichotomous but not additive the best published bound is still that of [KMS+24], $n-1$ per agent and $n(n-1)/2$ in total from any EF1 allocation, so the conjecture would improve the total by a factor of n — the same factor [BKNS22] gains over [BDN+20] on the goods side.

Two cautions follow from the literature check. First, Theorem 19 is already subsumed by [LMS26] (Remark 20); what survives of it is the mechanism, not the statement. Second, [LMS26] appeared in July 2026, which is a reminder that the surrounding corners are being filled in quickly and that the search should be repeated before circulation rather than treated as done.

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